ACPC Notes — Contents

1 Contest Playbook 1
- 1.1 ACPC, Kattis, and Team Strategy 1
- 1.2 Template, Debugging, Stress Testing 2
- 1.3 When You're Stuck 3
- 1.4 ACPC 2016–2024: Every Problem → Technique 6

2 Core Techniques 9
- 2.1 Binary Search, Ternary Search, Parametric Search 9
- 2.2 Prefix Sums, Difference Arrays, Sweep Line, Contribution 10
- 2.3 Two Pointers, Monotonic Stack, Monotonic Deque 11
- 2.4 Greedy and Exchange Arguments 12
- 2.5 Complete Search, Meet in the Middle, Square Root, Randomization 13
- 2.6 Bit Manipulation and Bitset 15

3 Data Structures 16
- 3.1 Disjoint Set Union 16
- 3.2 Fenwick Tree (BIT) 17
- 3.3 Segment Tree 18
- 3.4 Sparse Table, Ordered Set, Treap, Mo's, Small-to-Large 19

4 Dynamic Programming 22
- 4.1 DP Design and Classic Patterns 22
- 4.2 Bitmask DP, Digit DP, SOS, Broken Profile 23
- 4.3 Tree DP and Rerooting 25
- 4.4 DP Optimisations 26
- 4.5 Probability, Expected Value, Games 28

5 Graphs and Trees 30
- 5.1 Shortest Paths and Graph Search 30
- 5.2 MST, DAGs, Functional Graphs, Permutations 31
- 5.3 SCC, 2-SAT, Bridges, Articulation Points, Euler Paths 33
- 5.4 Trees: LCA, Euler Tour, HLD, Centroid Decomposition 34

6 Flows, Matching, Assignment 37
- 6.1 Max Flow, Min Cut, and Modelling 37
- 6.2 Bipartite Matching and Covers 38
- 6.3 Min-Cost Flow and the Assignment Problem 40

7 Math 42
- 7.1 Modular Arithmetic, GCD, CRT, Binomials 42
- 7.2 Primes, Factorisation, Divisors, Multiplicative Functions 43
- 7.3 Counting 45
- 7.3 distinct objects = (1/|G|) Σ_{g ∈ G} fix(g). For colourings, fix(g) = k^{c(g)} with c(g) = number of cycles of g on positions. 46
- 7.4 Matrices, Linear Recurrences, Gaussian Elimination, XOR Basis 46
- 7.5 FFT / NTT (Convolution) 48

8 Strings 50
- 8.1 Hashing, Prefix Function (KMP), Z-Function 50
- 8.2 Palindromes, Rotations, Tries, Aho–Corasick, Suffix Structures 51

9 Geometry 54
- 9.1 Geometry Primitives 54
- 9.2 Convex Hull, Calipers, Closest Pair, Circles, Integration 55

1. Contest Playbook

1.1 ACPC, Kattis, and Team Strategy

How to use these notes. Stuck on a problem → go to *When You're Stuck* (§1.3): constraint → complexity table, then keyword → technique table, which points to the section with code. Every C++ block compiles as-is after the template (§1.2) with `#include <bits/stdc++.h>`, `using namespace std;`, `using ll = long long;`. Blocks marked **[stress-tested]** were checked against brute force; the rest compile and were hand-verified, so still test on samples.

1.1.1 What ACPC looks like (from all sets 2016–2024 on Kattis)

- 8–15 problems per set. Kattis difficulty (1–10 scale) spread is very consistent:
- **~35% easy (1.2–2.9):** implementation, careful reading, simulation. Solve these *fast and without WA* — penalty time decides ranks among teams with equal solves.
- **~35% medium (3.0–5.0):** one standard idea: binary search on answer + greedy, monotonic stack, Dijkstra/BFS with a twist, tree DP, counting mod p, interval DP, grid flood fill, offline DSU.
- **~30% hard (5.1–8.3):** probability/expected-value DP, flows with modelling (min flow with lower bounds, assignment), FFT, number theory (factoring, gcd tricks), interactive problems with tiny query budgets, geometry + search, string canonical forms, nasty interval DP.
- Expert/CM level: easy+medium ≈ CF Div2 A–C; hard ≈ Div2 D–F / Div1 C–D. Hard ACPC problems are usually *one known technique plus modelling* rather than research-level tricks — recognizing the model is the whole game. See §1.4 for every past problem mapped to its technique.
- Themes recur: Kattis "Medium/Hard" problems from ACPC very often involve **probability**, **flows/matching**, **Dijkstra on augmented state**, **binary search on the answer**, and **number theory**.

1.1.2 Kattis specifics

- Language: C++ compiled with GCC, `-O2`, gnu++17 or newer, 64-bit. `__int128`, `__builtin_*`, `<bits/stdc++.h>`, `__gnu_pbds` all work.
- Time limit is **per test file**; there may be dozens of files. Memory is usually 1024 MB (check the problem page).
- Verdicts stop at the first failing test file — you don't learn which case failed. A WA on test 1 means the sample itself fails (or format is wrong).
- Output: trailing whitespace/newline differences are ignored. Floating answers are accepted within the stated absolute/relative error; print with `fixed << setprecision(10)`.
- **Interactive problems** (ACPC 2024 had two): flush after every query (`cout << ... << endl` or `cout.flush()`), never read past what the judge sends, terminate immediately after the final answer. See §10.1.
- Multiple test cases inside one file (`T` given): **reset all global state**.
- Python: big integers are trivial in Python; consider it for one-off big-number problems (e.g. ACPC 2017 *Divide by 100* has numbers with 10^6 digits — just string manipulation though).
- Recursion: Kattis has a generous stack, but DFS depth 10^6 in C++ is still risky; prefer iterative for path graphs with $n \geq 10^6$.

1.1.3 Team strategy (2–3 people, one machine)

1. **First 15 min:** split the problem set; everyone reads, labels each problem `easy / medium / hard / unknown` and writes one-line summary + constraints on paper. Read *all* problems — ACPC difficulty is not ordered by letter.
2. **Keyboard discipline:** only one person types; others think on paper. Never debug at the keyboard while someone with a ready solution waits — print/write the code and debug on paper.
3. Watch the scoreboard: problems solved by many teams are easier than they look. If 10 teams solved X and you think it's hard, you're missing something simple.
4. **Before submitting:** run all samples, a tiny edge case ($n = 1$, all equal, max values), and check output format. One WA costs 20 minutes of penalty.
5. **After WA:** reread the statement *first* (most WAs are misreads), then check §1.3 bug list, then write a brute force and compare on random small inputs.
6. **Last hour:** stop starting new hard problems unless the idea is complete; finish/debug what's closest. Freeze-time submissions still count.
7. Two people can pair on the hardest problem: one explains the problem from scratch to the other — this exposes wrong assumptions.

1.1.4 Precision and number sizes (memorize)

type	max	notes
`int`	$2.1 \cdot 10^9$	`a*b` overflows when both $\sim 5 \cdot 10^4$
`long long`	$9.2 \cdot 10^{18}$	product of two 10^9 values fits; two 10^{10} doesn't
`__int128`	$1.7 \cdot 10^{38}$	multiply two `ll` safely; no I/O, write a printer
`double`	15–16 digits	integers exact up to $2^{53} \approx 9 \cdot 10^{15}$
`long double`	18–19 digits	80-bit on x86 GCC; slower but helps precision

- Operations/second: $\sim 3 \cdot 10^8$ simple ops in 1 s with `-O2`; `set/map` $\sim 10^7$ ops/s; `unordered_map` $\sim 3 \cdot 10^7$ with reserve; `cin` with sync off $\sim 10^7$ ints/s.
- Memory: `int` array of 10^8 = 400 MB. `vector<vector<int>>` with 10^6 small vectors has big overhead — use CSR / flat arrays.

1.2 Template, Debugging, Stress Testing

1.2.1 Main template

```cpp
#include <bits/stdc++.h>
using namespace std;
using ll = long long;
using ull = unsigned long long;
using ld = long double;
using pii = pair<int, int>;
using pll = pair<ll, ll>;
#define all(x) begin(x), end(x)
#define sz(x) (int)(x).size()
#define rep(i, a, b) for (int i = (a); i < (b); ++i)
#ifdef LOCAL
#define dbg(...) cerr << "[" << #__VA_ARGS__ << "] = ", _dbg(__VA_ARGS__)
template <class T> void _dbg(const T& x) { cerr << x << endl; }
template <class T, class... R> void _dbg(const T& x, const R&... r) {
    cerr << x << ", "; _dbg(r...);
}
#else
#define dbg(...)
#endif
const ll INF = 4e18;  // careful: INF + INF overflows. Use 1e18 if you add two.
const int MOD = 1e9 + 7;

void solve() {
}

int main() {
    cin.tie(nullptr)->sync_with_stdio(false);
    int T = 1;
    // cin >> T;
    while (T--) solve();
}
```

Compile locally: `g++ -std=gnu++17 -O2 -Wall -Wextra -DLOCAL -fsanitize=address,undefined -g a.cpp`. Sanitizers catch out-of-bounds and signed overflow immediately — use them whenever you get RE/WA.

1.2.2 Useful snippets

```cpp
// print __int128
void print128(__int128 x) {
    if (x < 0) { putchar('-'); x = -x; }
    if (x > 9) print128(x / 10);
    putchar('0' + (int)(x % 10));
}
// floor/ceil division that work for negatives
ll fdiv(ll a, ll b) { return a / b - ((a % b != 0) && ((a < 0) != (b < 0))); }
ll cdiv(ll a, ll b) { return a / b + ((a % b != 0) && ((a < 0) == (b < 0))); }
// exact integer sqrt (floor)
ll isqrt(ll n) { ll r = sqrtl((ld)n); while (r * r > n) --r; while ((r + 1) * (r + 1) <= n) ++r; return r; }
// random
mt19937_64 rng(chrono::steady_clock::now().time_since_epoch().count());
ll rnd(ll l, ll r) { return uniform_int_distribution<ll>(l, r)(rng); }
// anti-hack hash for unordered_map<ll, V, chash>
```

```cpp
struct chash {
    static ull sm(ull x) { x += 0x9e3779b97f4a7c15; x = (x ^ (x >> 30)) * 0xbf58476d1ce4e5b9;
        x = (x ^ (x >> 27)) * 0x94d049bb133111eb; return x ^ (x >> 31); }
    size_t operator()(ull x) const { static const ull R = rng(); return sm(x + R); }
};
// hash a pair: use key = (ll)a << 32 | (unsigned)b
```

- Read until EOF: `while (cin >> x)`. Read a line: `getline(cin, s)`; after `cin >> n` call `cin.ignore()` (or `getline` once) before the first `getline`.
- Output many lines: use `'\n'`, never `endl` (except interactive).
- Doubles: `cout << fixed << setprecision(10) << x;` — avoid printing `-0.000000` (add `+ 0.0` or check `abs(x) < eps`).
- Sorting with comparator must be a *strict weak ordering* (`<`, never `<=`), otherwise RE/UB.
- `lower_bound(all(v), x)` → first `>= x`; `upper_bound` → first `> x`. Count of `x` in sorted v: `upper - lower`.
- `set.lower_bound(x)` is O(log n); `lower_bound(all(set), x)` is **O(n)**. Same for multiset/map.
- `multiset.erase(x)` removes **all** copies of x; use `ms.erase(ms.find(x))`.
- `priority_queue` is a max-heap; min-heap: `priority_queue<T, vector<T>, greater<T>>`.
- `accumulate(all(v), 0LL)` — the `0LL` matters.
- `(int)v.size() - 1` — `v.size() - 1` is unsigned and wraps when empty.
- `1 << k` is `int`; use `1LL << k` for k ≥ 31. `__builtin_popcountll` for `ll`.
- `memset` only for 0 / -1 / `0x3f` (int ≈ 1.06e9, sum of two still fits) .

1.2.3 Stress testing (do this after the first unexplained WA)

```bash
# gen.py prints a random small test; brute.cpp is the obviously-correct slow solution
g++ -O2 -o sol sol.cpp && g++ -O2 -o brute brute.cpp
for i in $(seq 1 1000); do
  python3 gen.py $i > in.txt
  ./sol < in.txt > o1.txt; ./brute < in.txt > o2.txt
  if ! cmp -s o1.txt o2.txt; then echo "DIFF on seed $i"; cat in.txt; break; fi
done
```

```python
# gen.py
import random, sys
random.seed(int(sys.argv[1]))
n = random.randint(1, 6)
print(n)
print(*[random.randint(1, 10) for _ in range(n)])
```

Keep tests *tiny* (n ≤ 6, values ≤ 10): the smallest failing case is the easiest to debug by hand.

1.3 When You're Stuck

1.3.1 Constraints → target complexity

Budget ≈ $3 \cdot 10^8$ simple operations per second (less with maps/sets, recursion, cache misses).

n up to	target	typical techniques
8–11	O(n!·n)	all permutations (`next_permutation`), brute force
15–20	$O(2^n \cdot n)$, $O(3^n)$ at 15	bitmask DP, subset enumeration, submask DP
35–45	$O(2^{n/2})$	meet in the middle
50–100	$O(n^4)$, $O(n^3 \log)$	Floyd, flows, small Hungarian, DP with 3–4 dims
300–500	$O(n^3)$	interval DP, Floyd, Gaussian elimination, Hungarian
2000–5000	$O(n^2)$	2D DP, all pairs, $O(n^2)$ with bitset /32 for 10^4–10^5
10^5–$2 \cdot 10^5$	O(n log n), $O(n \log^2 n)$, O(n√n)	sort, BIT/segtree, binary search, Dijkstra, Mo's
10^6	O(n), O(n log n) small const	linear scans, sieve, prefix sums, two pointers, KMP/Z
10^9–10^{18}	O(log n), O(√n), O(cube-root)	math formula, binary search, matrix power, digit DP, factorization

- **Sum of n over tests ≤ X** → per-test complexity can use X. Otherwise multiply by T!
- Two small parameters (e.g. n ≤ 10^5, k ≤ 20) → O(n·k) or $O(n \cdot 2^k)$ DP is intended (ACPC 2022 *City Numbers*: k ≤ 20 colours on a tree).

- A weirdly specific bound is a hint: `n ≤ 15` (bitmask/interval brute), `values ≤ 200` (enumerate the value — ACPC 2018 *Dog Trouble*), `prime factors < 10^6` (trial division works), `p < 10^10, s ≤ 10^5` (s^2 can exceed p).
- Output modulo a prime → counting, combinatorics, DP; answer would be huge otherwise.
- Output a real number with 1e-6 → binary/ternary search on reals, geometry, probability.

1.3.2 Keyword → technique

if the problem says / looks like ...	try	where
minimize the maximum / maximize the minimum	binary search on answer + greedy/BFS check	§2.1
"find the smallest X such that feasible" and feasibility is monotone	binary search on X	§2.1
maximize ratio Σa/Σb	binary search λ: is max Σ(a − λb) ≥ 0 (Dinkelbach)	§2.1
unimodal function of a real variable	ternary search / golden section	§2.1
k-th smallest of something implicit	binary search on value, count ≤ mid	§2.1
subarray sums/xors, count subarrays with property	prefix sums + hashmap / two pointers	§2.2, §2.3
range add many times, query once	difference array	§2.2
sum over all pairs/subarrays of f	contribution technique: count how often each element contributes	§2.2
nearest greater/smaller, "visible", histogram	monotonic stack	§2.3
sliding window min/max	monotonic deque	§2.3
schedule jobs with deadlines/durations	sort by deadline + heap; exchange argument	§2.4
choose k things to maximize something, n large	greedy + heap; check with exchange argument	§2.4
n ≤ 40, subset sums	meet in the middle	§2.5
connectivity under edge deletions, offline	reverse time → DSU additions	§3.1
point update + range query	Fenwick / segment tree	§3.2–3.3
range update + range query	lazy segment tree	§3.3
static range min/gcd	sparse table	§3.4
k-th element in dynamic set, order of key	PBDS ordered set / BIT walk	§3.4
many range queries, offline, no updates, hard to merge	Mo's algorithm	§3.4
count / optimize over sequences, choices at each step	DP; define state = what you need to know about the past	§4.1
subsequence counting (e.g. palindromic subsequences)	interval DP with inclusion–exclusion	§4.1
interleaving two strings, edit distance, LCS	2D DP	§4.1
merge adjacent / parenthesize / split interval	interval DP $O(n^3)$; keep max AND min if signs	§4.1
n ≤ 20 items, choose order/assignment	bitmask DP	§4.2
count numbers in [L, R] with digit property	digit DP	§4.2
tree, choose labels/colours per node with neighbour constraint	tree DP (keep best & second best)	§4.3
answer for every root	rerooting DP	§4.3
dp[i] = min_j (dp[j] + b_j·a_i)	convex hull trick / Li Chao	§4.4
dp[k][i] = min_j dp[k−1][j] + cost(j,i), cost "Monge"	divide & conquer opt / Knuth	§4.4
expected value, probability of winning, optimal stopping	linearity of expectation; DP over states backward	§4.5
turns repeat in a cycle with probabilities	geometric series / solve linear equations	§4.5
two players, optimal play, who wins	win/lose DP, Sprague–Grundy, find pattern by brute force	§4.5
shortest path, unweighted / weights 0–1 / nonneg	BFS / 0-1 BFS / Dijkstra	§5.1
shortest path with extra condition (fuel, horse, key, parity)	Dijkstra on (node, state)	§5.1
distance to nearest of many sources	multi-source BFS/Dijkstra	§5.1
path maximizing the minimum edge/vertex value	binary search + BFS, or max spanning tree, or modified Dijkstra	§5.1
which sums are representable by coins (unbounded)	Dijkstra on residues mod smallest coin	§5.1
minimum ratio spanning tree	binary search λ + MST	§5.2
dependencies / ordering	topological sort, DAG DP	§5.2
each node has one outgoing edge	functional graph: cycles + trees, binary lifting	§5.2
boolean constraints "a or b"	2-SAT	§5.3

if the problem says / looks like ...	try	where
edges whose removal disconnects	bridges / articulation points	§5.3
use every edge exactly once	Euler path (Hierholzer)	§5.3
tree path queries, ancestors, distances	LCA / binary lifting / Euler tour + BIT / HLD	§5.4
count paths in tree with property	centroid decomposition / small-to-large	§5.4
assign workers to jobs one-to-one, min cost	Hungarian / min-cost flow	§6.3
max pairs with compatibility	bipartite matching	§6.2
min #paths covering all vertices of a DAG	n − max matching (split graph)	§6.2
min #paths covering all **edges** of a DAG	min flow with lower bounds = Dilworth on edges	§6.1
cut with costs, "choose set with dependencies, max profit"	min cut / project selection	§6.1
grid with dominoes / chessboard parity	bipartite matching on black/white cells	§6.2
gcd, lcm, divisibility, a·x + b·y = c	ext. gcd, CRT	§7.1–7.2
sum of divisors, factorization of big n	sieve / trial division up to √n / Pollard rho	§7.2
count arrangements, multisets, paths	stars & bars, Catalan, inclusion–exclusion, Burnside	§7.3
linear recurrence with huge n	matrix exponentiation	§7.4
system of linear (mod 2) equations, max xor subset	Gaussian elimination / xor basis	§7.4
count pairs with sum = k for all k, pattern matching counts	FFT / NTT convolution	§7.5
string equality of substrings fast	polynomial hashing	§8.1
pattern occurrences / borders / periods	KMP prefix function, Z-function	§8.1
rotations / cyclic shifts equal	minimal rotation (Booth) as canonical form	§8.2
longest palindrome substring	Manacher	§8.2
many patterns in a text	Aho–Corasick	§8.2
distinct substrings, LCP of suffixes	suffix array + LCP	§8.2
polygon area, orientation, lattice points	shoelace, cross product, Pick	§9.1–9.2
smallest enclosing / outer boundary of points	convex hull	§9.2
interactive: judge answers queries	binary search / information-theoretic planning / gcd tricks	§10.1
evaluate expressions, mini-language	recursive descent parser	§10.2

1.3.3 Stuck-on-an-idea checklist

1. **Re-read the statement.** Underline every constraint. Is something guaranteed (distinct values, connected, sorted, a < b)? That guarantee is usually *used*.
2. **Solve tiny cases by hand** (n = 1, 2, 3) and **write a brute force**; print answers for n = 1..10 and look for a pattern (then search OEIS-style in your head: powers of 2, Catalan, Fibonacci, n^2, C(2n,n)).
3. **What is the answer monotone in?** If "answer ≥ x is feasible" is monotone → binary search. Fixing the answer often turns optimization into a yes/no greedy.
4. **Fix one thing and iterate over it:** fix the maximum element, the left endpoint, the last chosen item, the root, the value of one variable (e.g. enumerate the max time t ≤ 200 then solve the rest).
5. **Sort** the input (by start, end, value, ratio). Many greedy/DP problems become easy after sorting by the right key; find the key with an exchange argument.
6. **Reverse** something: process time backward (deletions → additions), reverse the graph, think about the complement ("at least one" = total − "none").
7. **Change what you iterate over:** over values instead of indices, over divisors instead of multiples (harmonic sum n log n), over the answer instead of the input.
8. **Model as a graph:** states = nodes, moves = edges. Grid + extra state (direction, keys, used power-up) → BFS/Dijkstra on product graph.
9. **Model as a flow/matching** when there are "pairings", "assign each X to one Y", "each used at most once", "minimum number of chains/paths".
10. **Linearity of expectation:** E[sum] = Σ E[indicator]. Never compute the full distribution if only an expectation is asked.
11. **Bound the search:** the answer is small (≤ 60 when doubling, ≤ log n), the number of distinct values is small (≤ √(2n) distinct lengths if Σ = n), only O(√n) distinct `n / i` values.
12. **Look for invariants** (parity, sum mod k, xor) in games and constructive problems.

13. Think about the **special structure:** tree (unique paths, DP), DAG (topo order), bipartite, planar grid, permutation (cycles), convex polygon ($O(n^3)$ DP over vertices).
14. Still stuck after 20–30 min with no progress → switch problems, come back later.

1.3.4 WA / RE / TLE checklist

- **Overflow**: `int * int`, sums of 10^5 values of 10^9, `1 << 40`, `INF + x`. When in doubt, `ll` everywhere.
- **Multi-test state** not reset (globals, adjacency lists, visited arrays, DSU).
- **Off-by-one**: inclusive vs exclusive ranges; 0- vs 1-indexed input; array size `n` vs `n + 1`.
- **Edge cases**: n = 1, empty set, all equal, all negative, answer 0, disconnected graph, self-loops, multi-edges, duplicate points, collinear points.
- **Reading**: multiple values per line, strings with spaces (`getline`), trailing spaces, Windows `\r`.
- **Output**: exact format/capitalisation ("Yes" vs "YES"), precision, one answer per line, `-0.0`.
- **Negative modulo**: `((a % m) + m) % m`. Subtraction in modular arithmetic: `(a - b + MOD) % MOD`.
- **Doubles compared with `==`**; use eps or exact integer arithmetic (cross products in `ll`).
- **Uninitialised** local variables / arrays (globals are zeroed, locals aren't).
- **Recursion depth** (10^6 deep DFS), **stack arrays** too big (`int a[10^7]` inside `main` → use global/vector).
- **TLE**: `endl` flushing, `map` where an array works, passing `vector` by value, O(n) `lower_bound` on `set`, `string +=` in a loop creating copies (`s = s + c` is O(n)), recomputing in loops, too many `long double`.
- **Dijkstra**: skip stale heap entries (`if (d > dist[u]) continue;`), else worst case is quadratic.
- **Comparator** not strict weak ordering → crash in `sort`.
- **Wrong algorithm**: greedy without proof → stress test against brute force.

1.4 ACPC 2016–2024: Every Problem → Technique

Kattis difficulty in brackets. Technique = the intended core idea (my analysis of the statements; `?` = likely but unverified). Use this to calibrate: what does a 4.0 vs 6.5 ACPC problem look like?

year	problem [diff]	core technique
2016	Mirror Images [1.6], Soylent [1.7], Bits [2.2]	implementation
	Wood Cutting [2.5]	sort ascending (shortest processing time first), prefix sums
	Dacey the Dice [3.3]	BFS on (cell, die orientation) — 24 orientations
	Flower Garden [3.6]	prefix of Manhattan distances + sieve for primes
	Palindromic Naming [4.1]	count palindromic subsequences, interval DP $O(n^2)$ mod p (§4.1)
	City Destruction [5.6]	linear DP over buildings with state "was neighbour destroyed before me"
	Modelling Problems [6.8]	parse a mini-language, brute force over unknown 0/1 values
2017	Musical Scales [1.6], Tetration [1.7]	implementation / math ($x = a^{1/a}$)
	Quantum Superposition [2.2]	DAG DP: set of reachable path lengths (bitset), combine two sets
	Anthony and Diablo [2.3]	geometry: max area for perimeter is a circle
	Divide by 100 [3.2]	big-number string manipulation
	Concentration [4.8]	probability / game simulation reasoning ?
	Lane Switching [5.5]	maximize the minimum (bottleneck path) over intervals: Dijkstra/BFS ?
	Maximal Sequences [6.2]	offline by start index; ordered set of next occurrences per value; check first m+1
	Matchings [7.5]	2-D pattern matching counts via FFT convolution, one FFT per colour (§7.5)
2018	Early Winter [2.0], Eating Out [2.5]	implementation / pigeonhole counting
	Arachnophobia [4.0]	multi-source Dijkstra from spiders + binary search D + Dijkstra s→t
	Dog Trouble [5.8]	enumerate max time t (≤ 200), assignment (Hungarian) maximizing Σt_i (§6.3)
	PUBNite [5.8]	geometry + binary/ternary search on time

year	problem [diff]	core technique
	Exploding Kittens [6.4]	simulation, skipping irrelevant cards ?
	Homework [6.5]	string interleaving DP O(len1·len2) (bitset if tight)
	Trimming Convex Polygons [6.9]	DP over convex polygon vertices $O(n^3)$ (fix first kept vertex, triangle fan)
2020	Test Drive [1.9], Laptop Stickers [1.9], Warring Scoring [2.8]	implementation
	Path Crossings [2.9]	hashing (x,y,t) / sort events
	Straights [2.9]	greedy on value counts
	Listen To Your Boss [3.9]	tree ancestor queries: Euler tour tin/tout (§5.4)
	Wormhole Extreme!!! [4.4]	kinematics; minimum initial velocity via v^2 recursion / binary search
	Hired Help [5.5]	binary search #employees + greedy deadline check
	Monochromatic Minesweeper [6.2]	constructive / search with pruning ?
	Password Rotation [6.3]	canonical form = min rotation of s and reverse(s) (Booth), hash set (§8.2)
	Wrapping Trees [7.3]	fractal → convex hull of the self-similar set; geometry
	Best Investing [7.8]	math / greedy over deposit–withdraw schedule ?
2021	Finding An A [1.2], GCVWR [1.4], Square Peg [1.5], Simone [1.8], Colour Wars [2.3]	implementation
	Snakes and Masters [2.9]	Fibonacci DP mod 10^6
	Bitmask [3.1]	string/bit matching with DP ?
	Menu Updates [3.3]	maintain smallest available number: set/heap
	Associative Exponents [3.4]	compare $a^{b\hat{}c}$ vs $(a^b)^c \Leftrightarrow b^c$ vs b·c, careful with overflow
	Ghostbusters 3 [3.8]	non-crossing assignment: sort + greedy/DP ?
	Wonky Pizza [4.2]	polar area $\int r^2/2\, d\theta$ closed form + binary search the angle
	Beep Code [4.6]	bit decomposition of a sum of square waves; reason level by level
	Balls and Bins [5.1]	optimal-stopping expected-value DP over (#bins with 1, #bins with 2) (§4.5)
	Swapping Places [8.1]	constructive: permutation routing on spanning tree, process leaves first
	Compression [8.2]	DP over block boundaries with careful cost precomputation
2022	Undead or Alive [1.3], Ordinals [1.8], Count Doubles [1.8]	implementation
	Contingency Planning [2.1]	$\Sigma_k\ n!/(n-k)!$ with cap
	Simple Factoring [2.4]	discriminant $b^2 - 4ac$ is a perfect square
	Maximum Fix [3.4]	each element votes for the one shift that fixes it; count array
	Supply Routes [3.5]	offline: reverse edge deletions → DSU unions (§3.1)
	Fight to Survive [3.8]	greedy / simulation ?
	City Numbers [4.4]	tree DP dp[v][colour], keep best & second best child value (§4.3)
	Long Walk [5.5]	implicit graph $i \leftrightarrow 2i \bmod 2^n$: bit manipulation, LCA-like on binary strings
	Braaaains [5.8]	Frobenius/coin reachability: Dijkstra on residues mod smallest size (§5.1)
	Tricky Factoring [6.7]	factor a, c (primes $< 10^6$), enumerate divisor pairs, count distinct b = dg + ef
	Maximal Parentheticals [8.0]	interval DP over expression tracking max and min (signs flip)
	Escape Routes [8.3]	min #paths covering all DAG edges = min flow with lower bounds (§6.1)
2023	Buildin' Fences [1.5]	implementation
	Flying Straight [3.0]	trigonometry (arc vs tangent)

year	problem [diff]	core technique
	Buildin' Roads [3.3]	minimum ratio spanning tree: binary search λ + MST (§2.1, §5.2)
	OBD Codes [3.9]	pattern matching on event streams ?
	Counting Cacti [4.0]	grid parsing + flood fill components
	Outlaw Numbers [4.0]	aliquot sequence: sum of proper divisors via factorization, ≤ 32 steps
	Wrangler Rating [4.5]	careful simulation of rules
	Circuit Game [5.2]	graph connectivity under switch removal + budget ?
	Dice Tables [5.2]	counting multisets (stars and bars) over overlapping face ranges, mod 10^9+9
	Desert Strike [5.4]	lattice counting of a rasterized rotated square (math)
	High Noon Language Showdown [5.5]	brute force over n with exact integer ceil(log2)
	Temporal Turmoil [6.7]	clock-hand alignment math with exact fractions
	Horse Snatchers [6.8]	binary search horse speed + Dijkstra on (node, has horse)
	Gunslingers [7.8]	probability DP over survivor sets with cyclic turns (geometric series)
2024	Snowfall [1.6], Melting Snow [1.7], Handheld Fan [2.2]	implementation
	Wrapping Snowballs [3.5]	?
	Alpine Agility [3.8]	grid DP (moves S/E/SE)
	Slippery Floor [3.8]	BFS where a move slides until hitting an obstacle
	Topographic Isolation [4.0]	monotonic stack: nearest $\geq$ on each side (§2.3)
	A Little to the Right [4.2]	count distinct sorted orders among properties with all-distinct values
	Hockey Fans [4.4]	binary search on D + greedy left-to-right placement
	Uniform Variation [4.7]	brute force: groups of all-different are ≤ 4; enumerate small subsets
	Snow Way Out [6.7]	interactive, 6 comparisons on 10×10 grid: offline minimax search for a strategy
	Four Questions [7.1]	interactive: $s^2 \bmod p \Rightarrow p$ divides $s^2 - r$; gcd of such values (§10.1)
	Speedy Slopes [7.2]	shortest path with speed state on grid ?

Patterns to take away

- Binary search on the answer appears almost every year (Arachnophobia, Hired Help, Hockey Fans, Horse Snatchers, Buildin' Roads).
- Dijkstra variants: multi-source, state-augmented, residue graph (Braaaains).
- Probability/expected value DP appears most years at 5–8 difficulty.
- One flow/matching problem every year or two (Dog Trouble, Escape Routes).
- Interactive problems appeared in 2024 (two of them) — read §10.1 before the contest.
- Number theory with "all prime factors $< 10^6$" or big-number twists.

2. Core Techniques

2.1 Binary Search, Ternary Search, Parametric Search

2.1.1 Integer binary search on the answer

Use when `ok(x)` is monotone: `false false ... false true true ...`. Write the invariant, not a formula you half-remember: **`lo` is always bad, `hi` is always good.**

```
// smallest x in (lo, hi] with ok(x) true; requires ok(hi) true, ok(lo) false (lo may be a sentinel)
template <class F> ll firstTrue(ll lo, ll hi, F ok) {
    while (hi - lo > 1) {
        ll mid = lo + (hi - lo) / 2;
        (ok(mid) ? hi : lo) = mid;
    }
    return hi;
}
// largest good x for "true ... true false ... false": lastTrue = firstTrue of !ok, minus 1
```

- Bounds: pick `lo` = something surely infeasible (often `-1` or `0`), `hi` = surely feasible (`1e18`, `2e9`). Check that `ok` never overflows at `hi` (e.g. `mid * cnt` with `mid = 1e18` → use `__int128` or early exit).
- **Maximize the minimum** (e.g. Hockey Fans, ACPC 2024: largest D such that M non-overlapping windows of length S all have values ≥ D): `ok(D)` = greedy scan left to right taking a window as soon as the last S values are all ≥ D. Monotone because a smaller D is easier.
- **Minimize the maximum** (split array into k parts minimizing max sum; minimize max edge on a path): `ok(X)` = greedy "how many parts do I need with cap X" ≤ k.
- **k-th smallest of an implicit set** (k-th smallest pairwise sum/difference, k-th fraction): binary search value v, count elements ≤ v (two pointers on sorted arrays is usually O(n)).
- **Binary search + graph** (Arachnophobia, Horse Snatchers): fix the parameter, then run BFS/Dijkstra on the allowed subgraph and check if the target is reachable within the time limit. Complexity × log(range).
- Searching over sorted *candidate* values (e.g. distinct edge weights) instead of the whole range keeps the answer exact and iterations ≤ log n.

2.1.2 Real-valued binary search

```
// sketch: ok(mid) = your feasibility test
double lo = 0, hi = 1e9;
for (int it = 0; it < 100; ++it) {   // 100 halvings: 2^-100 relative; never loop on hi-lo>eps
    double mid = (lo + hi) / 2;
    if (ok(mid)) hi = mid; else lo = mid;
}
```

Fixed iteration count avoids infinite loops caused by precision. 60 iterations are enough for 1e-9 relative on [0, 1e9]; cost is usually negligible.

2.1.3 Ternary search (unimodal functions)

```
// maximum of strictly unimodal f on [lo, hi] (reals)
template <class F> double ternaryMax(double lo, double hi, F f) {
    for (int it = 0; it < 200; ++it) {
        double m1 = lo + (hi - lo) / 3, m2 = hi - (hi - lo) / 3;
        if (f(m1) < f(m2)) lo = m1; else hi = m2;
    }
    return (lo + hi) / 2;
}
// integers: shrink while hi - lo > 2, then check lo..hi directly
template <class F> ll ternaryMaxInt(ll lo, ll hi, F f) {
    while (hi - lo > 2) {
        ll m1 = lo + (hi - lo) / 3, m2 = hi - (hi - lo) / 3;
        if (f(m1) < f(m2)) lo = m1 + 1; else hi = m2;
    }
    ll best = lo;
    for (ll x = lo; x <= hi; ++x) if (f(x) > f(best)) best = x;
    return best;
}
```

- Integer version fails with plateaus (f(m1) == f(m2) not at the max). If f can be flat, binary search on the sign of `f(x+1) - f(x)` instead (valid when f is concave: differences are non-increasing).

- Convex functions: sums of convex functions are convex (e.g. $\Sigma|x - a_i|$, Σ dist(P(t), Q_i) for linear motion). Distance between two points moving linearly over time is convex in t → ternary search on time.
- Nested ternary search works for functions convex in 2 variables (e.g. Fermat point, smallest enclosing circle centre) but is slow: 100×100 evaluations.

2.1.4 Parametric search / fractional programming

Maximize $\Sigma a_i / \Sigma b_i$ over feasible sets S ($b > 0$). The answer $\geq \lambda$ iff $\max_S \Sigma(a_i - \lambda b_i) \geq 0$. So binary search λ and solve the *unweighted* problem with weights $a_i - \lambda b_i$:

- Choose exactly k items: sort by $a_i - \lambda b_i$, take top k.
- Spanning tree minimizing Σcost/Σgain (ACPC 2023 *Buildin' Roads*): min ratio ⇔ binary search λ, MST with weights cost $- \lambda \cdot$gain, check total ≤ 0.
- Cycle with maximum mean (Σw/len): binary search λ, look for positive cycle with weights $w - \lambda$ (Bellman–Ford).

Dinkelbach converges faster: start $\lambda = 0$; repeat: $S = \text{argmax } \Sigma(a - \lambda b)$; $\lambda = \Sigma_S a / \Sigma_S b$; until no improvement (usually < 10 iterations).

2.1.5 Binary lifting search ("walk down powers of two")

When you can test "is the answer $\geq$ pos + 2^k" incrementally (Fenwick tree prefix walk, binary lifting on ancestors, sparse table), descend over k from high to low instead of a separate binary search — saves a log factor. See BIT `lower_bound` (§3.2) and LCA (§5.4).

2.1.6 Parallel binary search (offline, many queries)

Q queries each need "first time t when condition holds" where the structure evolves over time (e.g. first time u,v connected after edge additions; first time a sum reaches a threshold). Do log T rounds: in each round sort queries by their current `mid`, replay the events once from the start, and answer all `ok(mid)` checks along the way. Cost: O((T + Q) log T × cost of one event).

2.2 Prefix Sums, Difference Arrays, Sweep Line, Contribution

2.2.1 Prefix sums (1D/2D) and difference arrays

```
// 1D: p[i] = a[0] + ... + a[i-1]; sum of a[l..r] = p[r+1] - p[l]
// 2D: P[i][j] = sum of a[0..i-1][0..j-1]
struct Pref2D {
    vector<vector<ll>> P;
    Pref2D(const vector<vector<ll>>& a) {
        int n = a.size(), m = a[0].size();
        P.assign(n + 1, vector<ll>(m + 1, 0));
        rep(i, 0, n) rep(j, 0, m) P[i + 1][j + 1] = a[i][j] + P[i][j + 1] + P[i + 1][j] - P[i][j];
    }
    ll query(int r1, int c1, int r2, int c2) {  // inclusive rectangle
        return P[r2 + 1][c2 + 1] - P[r1][c2 + 1] - P[r2 + 1][c1] + P[r1][c1];
    }
};
```

- **Difference array:** add v to a[l..r] ⇒ `d[l] += v; d[r+1] -= v;` then prefix-sum d once. 2D: `d[r1][c1] += v; d[r1][c2+1] -= v; d[r2+1][c1] -= v; d[r2+1][c2+1] += v`.
- Adding an arithmetic progression to ranges: apply difference twice (second-order difference).
- **Count subarrays with sum = k:** iterate prefix p, `ans += cnt[p - k]; cnt[p]++` (hashmap; start with `cnt[0] = 1`). Same for xor (`p ^ k`), and for "sum divisible by m" (`cnt[p mod m]`).
- **Longest subarray with sum ≤ k (non-negative values)**: two pointers. With negatives: prefix + monotonic stack / sorted prefix minima + binary search.
- **Prefix of a function over a cycle** (rotations, circular arrays): double the array, take windows of length n.

2.2.2 Coordinate compression

```
// sketch: vals = every coordinate that matters
vector<ll> xs = vals;                       // all coordinates that matter
sort(all(xs)); xs.erase(unique(all(xs)), xs.end());
auto id = [&](ll x) { return int(lower_bound(all(xs), x) - xs.begin()); };
```

For intervals on a compressed axis, remember that compressed index i represents the *point* xs[i]; the gap between xs[i] and xs[i+1] has length `xs[i+1] - xs[i]` (needed for union-of-intervals length / area).

2.2.3 Sweep line

Turn geometry/intervals into **events sorted by coordinate**, process with a data structure.

- **Max overlapping intervals:** events (l, +1), (r, −1); sort; for closed intervals put +1 before −1 at equal coordinate.
- **Union length of intervals:** sort by l, merge greedily.
- **Area of union of rectangles:** sweep x; segment tree over compressed y storing "count covered > 0 length" (classic: node stores `cnt` and `covered len`, no push-down needed).
- **Pairs of intersecting segments / closest pair:** sweep with `set` ordered by current y (rarely needed at ACPC).
- **Offline queries:** sort queries together with updates by coordinate/time; answer each query when the sweep reaches it (e.g. "count points with x ≤ X and y ≤ Y" → sort by x, BIT on y).
- **Events at same coordinate:** decide the order explicitly (insertions before queries? removals first?) — this is the #1 sweep bug.

```
// sketch: max number of simultaneously active closed intervals [l, r]
vector<pair<ll,int>> ev;
for (auto [l, r] : iv) { ev.push_back({l, -1}); ev.push_back({r, +1}); } // -1 = open sorts first
sort(all(ev));
int cur = 0, best = 0;
for (auto [x, t] : ev) { cur -= t; best = max(best, cur); }
```

2.2.4 Contribution technique (sum over all subsets/subarrays/pairs)

Instead of enumerating structures, ask: **for each element, in how many structures does it contribute, and how much?**

- Σ over all subarrays of sum = Σ a_i · (i+1)·(n−i) (0-indexed).
- Σ over all subarrays of max: a_i is the max of subarrays with l ∈ (L_i, i], r ∈ [i, R_i) where L_i = previous index with value ≥ a_i (strict on one side, non-strict on the other to avoid double counting ties), R_i = next index with value > a_i. Contribution a_i·(i − L_i)·(R_i − i). Use monotonic stack (§2.3).
- Σ over pairs |a_i − a_j|: sort; a_i contributes `a_i·i − (sum of previous)`.
- Σ over all subsets of max: sort; a_i is the max in 2^i subsets.
- Σ of path lengths over all pairs in a tree: edge (parent, v) is used by sz[v]·(n − sz[v]) pairs.
- Expected value versions: same with probabilities (linearity of expectation, §4.5).

2.2.5 Offline tricks summary

- Answer queries in a convenient order (sorted by right endpoint, by threshold, by time) and restore original order at the end — store the query index.
- Reverse time for deletions (DSU can't delete, but can add).
- Divide & conquer over time / CDQ: split operations, handle "left half updates → right half queries" with a sort + BIT.

2.3 Two Pointers, Monotonic Stack, Monotonic Deque

2.3.1 Two pointers / sliding window

Valid when, for each left end `l`, the best/maximal right end `r(l)` is non-decreasing in `l` (window property is monotone: shrinking a valid window keeps it valid).

```
// longest window with at most k distinct values
int longestKDistinct(const vector<int>& a, int k) {
    unordered_map<int, int> cnt;
    int best = 0;
    for (int l = 0, r = 0; r < sz(a); ++r) {
        cnt[a[r]]++;
        while (sz(cnt) > k) { if (--cnt[a[l]] == 0) cnt.erase(a[l]); ++l; }
        best = max(best, r - l + 1);
    }
    return best;
}
```

- Count windows satisfying a monotone property: for each r, all l' ≥ l(r) work → `ans += r - l + 1`.
- "Exactly k" = atMost(k) − atMost(k−1).
- Two sorted arrays: pairs with a_i + b_j ≤ X — one pointer forward on a, one backward on b.
- **Does not work with negative numbers for sums** (window sum isn't monotone) — use prefix sums + sorted structure instead.

2.3.2 Monotonic stack: nearest greater / smaller

```cpp
// prv[i] = largest j < i with a[j] >= a[i] (or -1); nxt[i] = smallest j > i with a[j] > a[i] (or n)
// (>= on one side, > on the other: ties are counted exactly once in "subarrays where a[i] is max")
void nearestGreater(const vector<ll>& a, vector<int>& prv, vector<int>& nxt) {
    int n = sz(a); prv.assign(n, -1); nxt.assign(n, n);
    vector<int> st;
    rep(i, 0, n) {
        while (!st.empty() && a[st.back()] < a[i]) { nxt[st.back()] = i; st.pop_back(); }
        prv[i] = st.empty() ? -1 : st.back();
        st.push_back(i);
    }
}
```

Uses: largest rectangle in histogram (nearest smaller on both sides, area $h_i \cdot (R - L - 1)$); sum of subarray maxima/minima (contribution, §2.2); ACPC 2024 *Topographic Isolation* (for each peak, nearest index on the left and right with height ≥ peak: answer = min distance); visibility ("how many people can i see"); next greater element in a circular array (iterate 2n).

Maximal all-ones rectangle in a 0/1 grid: for each row keep `h[j]` = consecutive ones above; run histogram per row. O(nm).

2.3.3 Monotonic deque: sliding window min/max

```cpp
// mins[i] = min(a[i-k+1..i]) for i >= k-1
vector<ll> slidingMin(const vector<ll>& a, int k) {
    deque<int> dq; vector<ll> res;
    rep(i, 0, sz(a)) {
        while (!dq.empty() && a[dq.back()] >= a[i]) dq.pop_back();
        dq.push_back(i);
        if (dq.front() <= i - k) dq.pop_front();
        if (i >= k - 1) res.push_back(a[dq.front()]);
    }
    return res;
}
```

- **DP optimisation:** `dp[i] = min_{i-k ≤ j < i} dp[j] + c_i` → keep deque of j by dp value. Also `dp[i] = max_{j} (dp[j] - f(j)) + g(i)` with a sliding range of j.
- 2D sliding window min: apply 1D on rows, then on columns of the result.
- Shortest subarray with sum ≥ K with negative numbers: deque over prefix sums (increasing), pop front while `p[i] - p[front] ≥ K`.

2.3.4 Other amortised patterns

- **Stack simulation**: bracket matching, "remove adjacent equal pairs", evaluating expressions (shunting-yard).
- **Next pointer / "jump" with path compression**: "find next unvisited index ≥ i" — DSU where `nxt[i] = i+1` after visiting i; O(α) per query. Great for "paint ranges, each cell painted once" problems.
- **Small-to-large merging**: merge smaller set into larger → O(n log n) total insertions (§3.4).

2.4 Greedy and Exchange Arguments

A greedy is only as good as its proof. Standard proof: take an optimal solution that differs from greedy at the first position; **swap** to agree with greedy without making it worse. If you can't prove it, stress test against a brute force (§1.2) before submitting.

2.4.1 Finding the sort key by exchange

Two adjacent items i, j in an order; compare cost(…i j…) vs cost(…j i…). The difference only involves i and j ⇒ sort by that comparison.

problem	key
minimise Σ completion times (1 machine)	ascending processing time (SPT) — ACPC 2016 *Wood Cutting*
minimise $\Sigma w_i \cdot C_i$	ascending t_i / w_i, i.e. compare `t_i*w_j < t_j*w_i` (integers!)
minimise max lateness	ascending deadline (EDF)
concatenate strings to get smallest string	`a + b < b + a`
tasks with requirement r_i and cost c_i (need ≥ r to start, lose c)	descending $r_i - c_i$
two machines, each job on A then B (Johnson)	jobs with a<b first by a ascending, then others by b descending

Always compare with cross-multiplication in integers instead of dividing.

2.4.2 Interval greedies

- **Max number of disjoint intervals:** sort by right end, take if start > last end.
- **Min points to stab all intervals:** sort by right end; if interval not stabbed, put a point at its right end.
- **Min intervals to cover [0, L]:** sort by left; repeatedly among intervals with left ≤ covered, take the one with max right.
- **Partition intervals into min number of "rooms":** sort by start, min-heap of end times; answer = max overlap.
- **Assign intervals to k machines maximizing count:** sort by end, multiset of machine free times; assign to the machine with the latest free time ≤ start.

2.4.3 Deadline scheduling

Unit jobs with deadlines, maximise #jobs (or profit): sort by profit desc, place each job in the latest free slot ≤ its deadline (DSU "next free slot" trick, §3.1). Or: sort by deadline, push profits into a min-heap, pop the smallest whenever heap size > deadline.

Jobs with durations and deadlines, maximise count (Moore–Hodgson):

```
// jobs (deadline, duration); maximum number completed on time
int maxOnTime(vector<pair<ll,ll>> jobs) {
    sort(all(jobs));
    priority_queue<ll> pq; ll t = 0;
    for (auto [d, p] : jobs) {
        pq.push(p); t += p;
        if (t > d) { t -= pq.top(); pq.pop(); }   // drop the longest job so far
    }
    return sz(pq);
}
```

Feasibility check with deadlines (ACPC 2020 *Hired Help*: can we form k groups of K unit jobs?): a set of unit jobs is schedulable on c machines iff for every t, #jobs with deadline ≤ t is ≤ c·t. Combine with binary search on k and pick the jobs with the *latest* deadlines.

2.4.4 Heap greedies

- **Merge costs (Huffman):** always merge two smallest; total cost Σ.
- **Regret / "undo" greedy:** buy low sell high with one transaction per day: min-heap; for each price p: if `top < p` then `profit += p - top`, pop, and push p (this copy lets the sale be "undone" later); then always push p. Same idea: choose k non-adjacent elements with max sum — take max, replace it with (left + right − it) node, linked list.
- **k largest sums / k best combinations:** heap of states with "next" transitions (e.g. pairs (i, j) of two sorted arrays, push (i+1, j) and (i, j+1) with a visited set).

2.4.5 Other common greedy facts

- To maximise Σ a_i·b_{π(i)}: sort both same direction (rearrangement inequality); to minimise: opposite directions.
- Median minimises Σ|x − a_i|; mean minimises Σ(x − a_i)².
- Making an array non-decreasing with ±1 ops (min cost) → slope trick / median heap.
- Lexicographically smallest answer: build character by character, at each step try the smallest choice and check feasibility of completing (feasibility check is often a simple counting argument).
- Coin change greedy only works for canonical systems (1,5,10,25…); for arbitrary coins use DP.
- Pair the largest with the smallest (boats/teams with capacity): sort + two pointers.

2.5 Complete Search, Meet in the Middle, Square Root, Randomization

2.5.1 Brute force done right

- Enumerate subsets: `for (int m = 0; m < (1 << n); ++m)`. Permutations: sort, then `do { } while (next_permutation(all(p)));` (n ≤ 10: 3.6M perms).
- **Backtracking with pruning:** order choices (most constrained first), prune with an optimistic bound (current + best possible remaining ≤ best found ⇒ stop), memoize repeated states.
- **Iterate over the small dimension:** if one of the parameters is ≤ 20–200 (number of colours, max value, number of properties), loop over it explicitly.
- **Search the answer space instead of the input space** when the answer is small (e.g. answer ≤ 60 because values double).
- Before brute forcing, count states × transitions and compare with $3 \cdot 10^8$.

2.5.2 Meet in the middle

Split n ≤ 40 items into halves, enumerate $2^{n/2}$ each, combine with sort + two pointers/binary search/hash map.

```
// number of subsets with sum <= S (n <= 40)
ll countSubsets(const vector<ll>& a, ll S) {
    int n = sz(a), h = n / 2;
    auto gen = [&](int l, int r) {
        vector<ll> s = {0};
        for (int i = l; i < r; ++i) {
            int k = sz(s);
            rep(j, 0, k) s.push_back(s[j] + a[i]);
        }
        sort(all(s)); return s;
    };
    vector<ll> L = gen(0, h), R = gen(h, n);
    ll cnt = 0; int j = sz(R) - 1;
    for (ll x : L) {                        // L ascending -> pointer in R only moves down
        while (j >= 0 && x + R[j] > S) --j;
        cnt += j + 1;
    }
    return cnt;
}
```

Also: 4-sum = pair sums of two halves; bidirectional BFS on huge implicit graphs (puzzle states) — expand from both ends, stop when frontiers meet.

2.5.3 Square root decomposition

- **Blocks of size B ≈ √n:** each block stores an aggregate (+ lazy tag). Update/query touch O(n/B) full blocks + O(B) loose elements. Handy when the operation is weird (e.g. "add x to range, count elements < y in range": keep each block sorted → O(√n log n)).
- **Heavy/light by frequency:** values occurring > √n times are "heavy" (at most √n of them) → precompute for heavy, brute force for light.
- **Σ of sizes = n ⇒ at most O(√n) distinct sizes.** (Knapsack with Σ w = n has O(√n) distinct weights → bounded knapsack per distinct weight.)
- **Divisor/quotient blocks:** `n / i` takes O(√n) distinct values; iterate blocks:

```
// sum_{i=1}^{n} f(n / i) grouped: for each block [l, r], n / i == q
ll sumQuotients(ll n) {
    ll res = 0;
    for (ll l = 1, r; l <= n; l = r + 1) {
        ll q = n / l; r = n / q;
        res += (r - l + 1) * q;     // replace q by f(q), (r-l+1) by the weight of [l, r]
    }
    return res;
}
```

- **Mo's algorithm** for offline range queries: §3.4.
- **Rebuild periodically:** buffer updates; every √q updates rebuild the static structure; queries check the static structure + the buffer.

2.5.4 Randomization

- **Random shuffle** before inserting into structures that degrade on sorted input (treap keys, BST, hull insertion).
- **Random sampling** to find a majority/frequent element: an element occupying ≥ 1/k of the array is hit with probability ≥ 1/k per sample; 30·k samples → failure ≈ e^{-30}.
- **Hashing with random base/mod** (§8.1); **xor-hashing of sets/multisets**: assign each value a random 64-bit number; set equality ⇔ xor (or sum) of hashes equal. Subarray is a permutation of 1..k ⇔ sum of hashes of subarray = prefix-hash of {1..k}.
- **Random restarts / simulated annealing** for optimization problems with no clear structure (last resort; tune by time limit with `clock()`).
- `mt19937_64 rng(seed)`; never `rand()` (max 32767 on some platforms).

2.5.5 Iterative deepening / search with memo

- Game/puzzle with small depth: IDDFS or BFS over states stored in hash map (encode state into a 64-bit integer).
- **Memoized recursion over states** = DP. If the state space is unclear, write the recursive brute force, then add a `map<State, Answer>` and see if it's fast enough.

2.6 Bit Manipulation and Bitset

2.6.1 Tricks

expression	meaning
`x & -x`	lowest set bit
`x & (x - 1)`	clear lowest set bit; `== 0` ⇔ x is a power of two (x > 0)
`__builtin_popcountll(x)`	number of set bits
`__builtin_ctzll(x)` / `__builtin_clzll(x)`	trailing / leading zeros (undefined for 0!)
`63 - __builtin_clzll(x)`	floor(log2 x) (`__lg(x)` also works)
`x >> k & 1`	k-th bit
`x ^ (1LL << k)`	toggle k-th bit
`(1LL << n) - 1`	mask of n ones
`a + b == (a ^ b) + 2 * (a & b)`	sum decomposes into xor + carries
`a ^ b ^ b == a`	xor cancels; prefix xor gives range xor

```cpp
// iterate all submasks of mask (descending), including 0: total over all masks is 3^n
void subs(int mask) {
    for (int s = mask; ; s = (s - 1) & mask) {
        // use s
        if (s == 0) break;
    }
}
// iterate all supersets of mask within n bits
void supers(int mask, int n) {
    for (int s = mask; s < (1 << n); s = (s + 1) | mask) { /* use s */ }
}
// next mask with the same popcount (Gosper's hack)
ll nextSamePop(ll x) { ll c = x & -x, r = x + c; return (((r ^ x) >> 2) / c) | r; }
```

2.6.2 XOR facts

- XOR of 0..n: `n % 4 == 0 ? n : n % 4 == 1 ? 1 : n % 4 == 2 ? n + 1 : 0`.
- Maximum xor of two numbers in a set: binary trie, greedily go to the opposite bit (O(30) per query).
- Maximum xor of any subset / k-th xor / "is x representable": **linear basis** (§7.4).
- Per-bit independence: many problems on sums of xors/ands/ors split into 30 independent problems, one per bit (count pairs with bit k set in a_i ^ a_j = cnt0·cnt1).
- `a | b` and `a & b`: `a + b = (a | b) + (a & b)`.

2.6.3 Bitset

`bitset<N>` operations are N/64 word ops — 64× speedup for boolean DP.

```cpp
// subset-sum reachability up to MAXS: O(n * MAXS / 64)
const int MAXS = 100001;
bitset<MAXS> reachable(const vector<int>& w) {
    bitset<MAXS> b; b[0] = 1;
    for (int x : w) b |= b << x;
    return b;
}
```

- Transitive closure / reachability in a DAG: `reach[v] |= reach[u]` in reverse topo order: O(n·m/64).
- Counting triangles: for each edge (u,v): `(adj[u] & adj[v]).count()`.
- Sets of reachable path lengths in a DAG (ACPC 2017 *Quantum Superposition*): `len[v] |= len[u] << 1`.
- O(n^2) DP where the transition is an OR/AND over a row (LCS-like boolean, interleaving check like ACPC 2018 *Homework*) can often be done row-by-row with shifts.
- `_Find_first()`, `_Find_next(i)` iterate set bits fast. Size must be a compile-time constant; for dynamic sizes pick the max or template over sizes.

2.6.4 Gray code

`g = i ^ (i >> 1)`: consecutive values differ in one bit — enumerate subsets toggling one element at a time (useful when add/remove of an item is O(1) but recomputation is expensive).

3. Data Structures

3.1 Disjoint Set Union

```
struct DSU {
    vector<int> p;                      // p[x] < 0: x is root, size = -p[x]
    DSU(int n) : p(n, -1) {}
    int find(int x) { return p[x] < 0 ? x : p[x] = find(p[x]); }
    bool same(int a, int b) { return find(a) == find(b); }
    int size(int x) { return -p[find(x)]; }
    bool unite(int a, int b) {
        a = find(a), b = find(b);
        if (a == b) return false;
        if (p[a] > p[b]) swap(a, b);     // union by size
        p[a] += p[b]; p[b] = a;
        return true;
    }
};
```

Recursion depth is O(log n) thanks to union by size, so `find` is safe even for $n = 10^6$.

3.1.1 Uses

- Connected components online under edge additions; Kruskal MST (§5.2); cycle detection in undirected graphs (`unite` returns false).
- **Offline deletions** (ACPC 2022 *Supply Routes*): read all operations, start from the final graph (all never-deleted edges), process operations in reverse: a deletion becomes an addition, queries are answered in reverse and output reversed.
- **"Next free slot" pointer jumping:** `nxt = DSU` where after using slot i you link $i \to i+1$; `find(i)` gives the first free slot $\geq i$. Paint-the-range problems become $O(n\,\alpha)$.
- Track per-component aggregates (sum, min, count of something) in arrays indexed by root; merge them in `unite`.
- Bipartiteness / parity constraints online: DSU with parity (below).
- Process edges in sorted order of weight to answer "max edge on path / min bottleneck" queries offline (Kruskal reconstruction tree idea: when a union happens at weight w, all pairs between the two components have bottleneck w).

3.1.2 DSU with parity / potential (weighted DSU)

Maintains `d[x]` = value(x) − value(root) for relations like "x and y differ by w" or parity "x and y have different colours".

```
struct WDSU {
    vector<int> p; vector<ll> d;        // d[x] = val[x] - val[parent chain root]
    WDSU(int n) : p(n, -1), d(n, 0) {}
    int find(int x) {
        if (p[x] < 0) return x;
        int r = find(p[x]);
        d[x] += d[p[x]]; p[x] = r;      // d[p[x]] is already relative to r
        return r;
    }
    // add constraint val[b] - val[a] = w; returns false on contradiction
    bool unite(int a, int b, ll w) {
        int ra = find(a), rb = find(b);
        if (ra == rb) return d[b] - d[a] == w;
        if (p[ra] > p[rb]) { swap(ra, rb); swap(a, b); w = -w; }
        // attach rb under ra: val[rb] - val[ra] = d[a] + w - d[b]
        p[ra] += p[rb]; p[rb] = ra; d[rb] = d[a] + w - d[b];
        return true;
    }
    ll diff(int a, int b) { find(a); find(b); return d[b] - d[a]; }  // assumes same set
};
```

For parity, use w ∈ {0,1} and compare `(d[b] - d[a]) & 1` (or replace + by xor).

3.1.3 DSU with rollback (for offline dynamic connectivity / divide & conquer)

No path compression; `undo()` pops the last union. Each `find` is O(log n).

```
struct RollbackDSU {
    vector<int> p; vector<pair<int,int>> st;   // (attached root, old parent size)
    int comps;
    RollbackDSU(int n) : p(n, -1), comps(n) {}
    int find(int x) { while (p[x] >= 0) x = p[x]; return x; }
```

```
    bool unite(int a, int b) {
        a = find(a), b = find(b);
        if (a == b) { st.push_back({-1, -1}); return false; }
        if (p[a] > p[b]) swap(a, b);
        st.push_back({b, p[b]});
        p[a] += p[b]; p[b] = a; --comps;
        return true;
    }
    void undo() {
        auto [b, sb] = st.back(); st.pop_back();
        if (b < 0) return;
        int a = p[b]; p[b] = sb; p[a] -= sb; ++comps;
    }
};
```

Offline dynamic connectivity: each edge is alive during a time interval; insert the interval into a segment tree over time (O(log T) nodes); DFS the segment tree, `unite` on entry, `undo` on exit, answer queries at leaves. O(m log T log n).

3.2 Fenwick Tree (BIT)

Point update, prefix query, O(log n), tiny constant. Works for any **invertible** operation (sum, xor) for range queries; for max/min only prefix queries with non-decreasing updates.

```
// [stress-tested] 0-indexed interface, internally 1-indexed
struct BIT {
    int n; vector<ll> t;
    BIT(int n) : n(n), t(n + 1, 0) {}
    void add(int i, ll v) { for (++i; i <= n; i += i & -i) t[i] += v; }
    ll pref(int i) { ll s = 0; for (; i > 0; i -= i & -i) s += t[i]; return s; }  // sum [0, i)
    ll sum(int l, int r) { return pref(r) - pref(l); }                            // sum [l, r)
    // smallest idx such that pref(idx + 1) >= k (all values must be >= 0); returns n if none
    int lower_bound(ll k) {
        int pos = 0;
        for (int pw = 1 << __lg(max(n, 1)); pw; pw >>= 1)
            if (pos + pw <= n && t[pos + pw] < k) { pos += pw; k -= t[pos]; }
        return pos;
    }
};
```

- **Order statistics:** BIT over compressed values with counts; `lower_bound(k)` = k-th smallest (1-based k). Insert/erase = `add(v, ±1)`.
- **Count inversions:** iterate from left, `ans += i - pref(a[i] + 1)` (number of previous elements > a[i]), then `add(a[i], 1)`. Values compressed to 0..n−1.
- **Count of distinct values in ranges (offline):** sort queries by r; when processing position i, `add(i, +1)` and `add(last[a[i]], -1)`; answer = sum(l, r+1).
- **2D points dominance count:** sort by x, BIT on y (sweep, §2.2).

3.2.1 Range update, point query / range update, range query

```
// range add, point query: store differences
struct BITRangeAdd {
    BIT b; BITRangeAdd(int n) : b(n + 1) {}
    void add(int l, int r, ll v) { b.add(l, v); b.add(r + 1, -v); }   // a[l..r] += v (inclusive)
    ll get(int i) { return b.pref(i + 1); }
};
// range add, range sum: two BITs
struct BITRangeRange {
    BIT b1, b2; BITRangeRange(int n) : b1(n + 1), b2(n + 1) {}
    void add(int l, int r, ll v) {                                    // inclusive
        b1.add(l, v); b1.add(r + 1, -v);
        b2.add(l, v * l); b2.add(r + 1, -v * (r + 1));
    }
    ll pref(int i) { return b1.pref(i) * i - b2.pref(i); }            // sum [0, i)
    ll sum(int l, int r) { return pref(r + 1) - pref(l); }            // inclusive
};
```

3.2.2 2D BIT

```
struct BIT2D {   // point add, rectangle-prefix sum; n*m memory
    int n, m; vector<vector<ll>> t;
    BIT2D(int n, int m) : n(n), m(m), t(n + 1, vector<ll>(m + 1)) {}
    void add(int x, int y, ll v) {
        for (int i = x + 1; i <= n; i += i & -i)
```

```
            for (int j = y + 1; j <= m; j += j & -j) t[i][j] += v;
    }
    ll pref(int x, int y) {  // sum over [0,x) x [0,y)
        ll s = 0;
        for (int i = x; i > 0; i -= i & -i)
            for (int j = y; j > 0; j -= j & -j) s += t[i][j];
        return s;
    }
};
```

If coordinates are large and points known in advance: compress, or do the offline sweep with 1D BIT instead.

3.3 Segment Tree

3.3.1 Iterative (point update, range query) — fastest to type

```
// [stress-tested] change T / ID / op; op must be associative (need not be commutative)
struct Seg {
    using T = ll;
    static constexpr T ID = LLONG_MIN;
    static T op(T a, T b) { return max(a, b); }
    int n; vector<T> t;
    Seg(int n) : n(n), t(2 * n, ID) {}
    void set(int i, T v) {
        for (t[i += n] = v; i > 1; i >>= 1) t[i >> 1] = op(t[i & ~1], t[i | 1]);
    }
    T query(int l, int r) {                // [l, r)
        T ra = ID, rb = ID;
        for (l += n, r += n; l < r; l >>= 1, r >>= 1) {
            if (l & 1) ra = op(ra, t[l++]);
            if (r & 1) rb = op(t[--r], rb);
        }
        return op(ra, rb);
    }
};
```

3.3.2 Lazy propagation: range assign + range add, query sum/min/max

```
// [stress-tested] 0-indexed, inclusive [l, r]. Delete what you don't need.
struct LazySeg {
    struct Node { ll sum = 0, mn = INF, mx = -INF; };
    static Node merge(const Node& a, const Node& b) {
        return {a.sum + b.sum, min(a.mn, b.mn), max(a.mx, b.mx)};
    }
    int n; vector<Node> t; vector<ll> add, setv; vector<char> hasSet;
    LazySeg(const vector<ll>& a) : n(sz(a)), t(4 * n), add(4 * n), setv(4 * n), hasSet(4 * n) {
        build(1, 0, n - 1, a);
    }
    void build(int x, int l, int r, const vector<ll>& a) {
        if (l == r) { t[x] = {a[l], a[l], a[l]}; return; }
        int m = (l + r) / 2;
        build(2 * x, l, m, a); build(2 * x + 1, m + 1, r, a);
        t[x] = merge(t[2 * x], t[2 * x + 1]);
    }
    void applySet(int x, int l, int r, ll v) {
        t[x] = {v * (r - l + 1), v, v}; hasSet[x] = 1; setv[x] = v; add[x] = 0;
    }
    void applyAdd(int x, int l, int r, ll v) {
        t[x].sum += v * (r - l + 1); t[x].mn += v; t[x].mx += v;
        if (hasSet[x]) setv[x] += v; else add[x] += v;
    }
    void push(int x, int l, int r) {
        int m = (l + r) / 2;
        if (hasSet[x]) {
            applySet(2 * x, l, m, setv[x]); applySet(2 * x + 1, m + 1, r, setv[x]); hasSet[x] = 0;
        }
        if (add[x]) {
            applyAdd(2 * x, l, m, add[x]); applyAdd(2 * x + 1, m + 1, r, add[x]); add[x] = 0;
        }
    }
    // type 0 = add v, type 1 = assign v
    void update(int ql, int qr, ll v, int type) { update(1, 0, n - 1, ql, qr, v, type); }
    void update(int x, int l, int r, int ql, int qr, ll v, int type) {
        if (qr < l || r < ql) return;
        if (ql <= l && r <= qr) { type ? applySet(x, l, r, v) : applyAdd(x, l, r, v); return; }
        push(x, l, r);
```

```
        int m = (l + r) / 2;
        update(2 * x, l, m, ql, qr, v, type); update(2 * x + 1, m + 1, r, ql, qr, v, type);
        t[x] = merge(t[2 * x], t[2 * x + 1]);
    }
    Node query(int ql, int qr) { return query(1, 0, n - 1, ql, qr); }
    Node query(int x, int l, int r, int ql, int qr) {
        if (qr < l || r < ql) return Node();
        if (ql <= l && r <= qr) return t[x];
        push(x, l, r);
        int m = (l + r) / 2;
        return merge(query(2 * x, l, m, ql, qr), query(2 * x + 1, m + 1, r, ql, qr));
    }
    // first index i >= ql with a[i] >= v, or -1  (O(log^2 n) worst, usually ~log n)
    int firstAtLeast(int ql, ll v) { return firstAtLeast(1, 0, n - 1, ql, v); }
    int firstAtLeast(int x, int l, int r, int ql, ll v) {
        if (r < ql || t[x].mx < v) return -1;
        if (l == r) return l;
        push(x, l, r);
        int m = (l + r) / 2;
        int res = firstAtLeast(2 * x, l, m, ql, v);
        return res != -1 ? res : firstAtLeast(2 * x + 1, m + 1, r, ql, v);
    }
};
```

How to design a new lazy segment tree: decide (1) the node summary, (2) the tag, (3) how a tag changes a node summary *using only the summary and the segment length*, (4) how two tags compose (new tag applied after old). If (3) is impossible, the problem needs a different summary (e.g. store sum of squares to support range add with sum-of-squares query: $\Sigma(x+v)^2 = \Sigma x^2 + 2v\Sigma x + \text{len}\cdot v^2$).

3.3.3 Useful node designs

query	node stores	merge
max subarray sum	`sum, pref, suf, best`	`best = max(a.best, b.best, a.suf + b.pref)`, `pref = max(a.pref, a.sum + b.pref)`
count of minimum	`mn, cnt`	take smaller, add counts on tie
longest run of 1s	`len, pref1, suf1, best`	like max subarray
bracket sequence validity	`open, close` unmatched	`m = min(a.open, b.close)`; `open = a.open + b.open - m`, `close = a.close + b.close - m`
polynomial hash of range	`hash, len`	`a.hash · B^len_b + b.hash`
matrix product (DP transitions along array)	2×2 / 3×3 matrix	matrix multiply (order matters)
area of union of rectangles (sweep)	`cnt, coveredLen`	no push: `covered = cnt > 0 ? len : left + right`

3.3.4 Variations (when you need them)

- **Segment tree on values** (counts per value, compressed): k-th smallest by descending (go left if left count ≥ k).
- **Merge-sort tree**: each node stores the sorted subarray → count elements ≤ x in [l, r] in $O(\log^2 n)$. Memory n log n.
- **Dynamic / implicit segment tree**: create children on demand (`vector<array<int,2>> ch`) when coordinates are up to 10^9 and online. Offline → compress instead.
- **Persistent segment tree**: every update copies O(log n) nodes; version i = prefix i of array → k-th smallest in range via `root[r] - root[l-1]`.
- **Segment tree over time** for offline dynamic connectivity (§3.1).
- Recursion is fine for $n \le 10^6$; size arrays `4n`.

3.4 Sparse Table, Ordered Set, Treap, Mo's, Small-to-Large

3.4.1 Sparse table (static RMQ in O(1))

Works for idempotent operations (min, max, gcd, and, or). O(n log n) build.

```
struct Sparse {
    vector<vector<ll>> t;
    static ll op(ll a, ll b) { return min(a, b); }
    Sparse(const vector<ll>& a) {
        int n = sz(a), K = __lg(max(n, 1)) + 1;
```

```
        t.assign(K, a);
        rep(k, 1, K) rep(i, 0, n - (1 << k) + 1)
            t[k][i] = op(t[k - 1][i], t[k - 1][i + (1 << (k - 1))]);
    }
    ll query(int l, int r) {            // inclusive, l <= r
        int k = __lg(r - l + 1);
        return op(t[k][l], t[k][r - (1 << k) + 1]);
    }
};
```

For non-idempotent ops (sum, product) use prefix sums / disjoint sparse table / segment tree. Range GCD with sparse table is O(log) per query due to gcd cost.

3.4.2 Ordered set (PBDS)

```
#include <ext/pb_ds/assoc_container.hpp>
#include <ext/pb_ds/tree_policy.hpp>
using namespace __gnu_pbds;
template <class T>
using ordered_set = tree<T, null_type, less<T>, rb_tree_tag, tree_order_statistics_node_update>;
// os.find_by_order(k) -> iterator to k-th smallest (0-based); os.order_of_key(x) -> #elements < x
// multiset: store pair<value, unique_id>; order_of_key({x, -1}) = #elements < x
```

3.4.3 Implicit treap (sequence with split/merge, reverse, range sum)

Use for: cut a block and paste elsewhere, reverse ranges, insert/delete at position — all O(log n).

```
// [stress-tested]
mt19937 trng(91);
struct Treap {
    struct N { int l = 0, r = 0, sz = 0; unsigned pr = 0; ll val = 0, sum = 0; bool rev = false; };
    vector<N> t;
    Treap() { t.push_back(N()); }       // node 0 = null
    int make(ll v) { N x; x.sz = 1; x.pr = trng(); x.val = x.sum = v; t.push_back(x); return sz(t) - 1; }
    void pull(int x) {
        t[x].sz = 1 + t[t[x].l].sz + t[t[x].r].sz;
        t[x].sum = t[x].val + t[t[x].l].sum + t[t[x].r].sum;
    }
    void push(int x) {
        if (x && t[x].rev) {
            swap(t[x].l, t[x].r);
            if (t[x].l) t[t[x].l].rev ^= 1;
            if (t[x].r) t[t[x].r].rev ^= 1;
            t[x].rev = false;
        }
    }
    // first k elements -> a, rest -> b
    void split(int x, int k, int& a, int& b) {
        if (!x) { a = b = 0; return; }
        push(x);
        if (t[t[x].l].sz < k) { split(t[x].r, k - t[t[x].l].sz - 1, t[x].r, b); a = x; }
        else { split(t[x].l, k, a, t[x].l); b = x; }
        pull(x);
    }
    int merge(int a, int b) {
        if (!a || !b) return a ^ b;
        if (t[a].pr > t[b].pr) { push(a); t[a].r = merge(t[a].r, b); pull(a); return a; }
        push(b); t[b].l = merge(a, t[b].l); pull(b); return b;
    }
    // helpers on root (0-indexed, inclusive ranges)
    void insert(int& root, int pos, ll v) { int a, b; split(root, pos, a, b); root = merge(merge(a, make(v)), b); }
    void erase(int& root, int pos) { int a, b, c; split(root, pos, a, b); split(b, 1, b, c); root = merge(a, c); }
    void reverse(int& root, int l, int r) {
        int a, b, c; split(root, l, a, b); split(b, r - l + 1, b, c);
        t[b].rev ^= 1; root = merge(merge(a, b), c);
    }
    ll rangeSum(int& root, int l, int r) {
        int a, b, c; split(root, l, a, b); split(b, r - l + 1, b, c);
        ll s = t[b].sum; root = merge(merge(a, b), c); return s;
    }
    void collect(int x, vector<ll>& out) {   // in-order
        if (!x) return; push(x); collect(t[x].l, out); out.push_back(t[x].val); collect(t[x].r, out);
    }
};
```

Keyed treap (ordered by value) = same with split by key `t[x].val < k`. Add other lazy tags (range add) exactly like the segment tree: apply to node + mark children in `push`.

3.4.4 Mo's algorithm (offline range queries, O((n + q)√n))

Needs O(1) (or O(log)) add/remove of one element at either end. Example: number of distinct values in [l, r].

```
vector<int> mo(const vector<int>& a, const vector<pair<int,int>>& qs) {  // inclusive queries
    int n = sz(a), q = sz(qs), B = max(1, (int)(n / sqrt(max(q, 1))));
    vector<int> ord(q); iota(all(ord), 0);
    sort(all(ord), [&](int i, int j) {
        int bi = qs[i].first / B, bj = qs[j].first / B;
        if (bi != bj) return bi < bj;
        return (bi & 1) ? qs[i].second > qs[j].second : qs[i].second < qs[j].second;
    });
    vector<int> cnt(*max_element(all(a)) + 1), res(q);   // compress a first if values are big
    int cur = 0, L = 0, R = -1;
    auto add = [&](int i) { if (cnt[a[i]]++ == 0) ++cur; };
    auto rem = [&](int i) { if (--cnt[a[i]] == 0) --cur; };
    for (int id : ord) {
        auto [l, r] = qs[id];
        while (R < r) add(++R);
        while (L > l) add(--L);
        while (R > r) rem(R--);
        while (L < l) rem(L++);
        res[id] = cur;
    }
    return res;
}
```

Expand before shrinking (as above) so the window is never "negative". Mo on trees: run on the Euler tour (ET of size 2n, LCA handled separately).

3.4.5 Small-to-large merging

Merging sets of children into the parent: always insert the smaller into the larger (`if (sz(A) < sz(B)) swap(A, B);`). Each element moves O(log n) times → O(n log² n) with `set/map`, O(n log n) with hash/vectors. Classic use: for every subtree, number of distinct colours / most frequent colour / answer queries about subtree.

```
// distinct colours in every subtree; g = children lists, col = colours
void dfsMerge(int v, const vector<vector<int>>& g, const vector<int>& col,
              vector<set<int>>& S, vector<int>& ans) {
    S[v].insert(col[v]);
    for (int c : g[v]) {
        dfsMerge(c, g, col, S, ans);
        if (sz(S[c]) > sz(S[v])) swap(S[c], S[v]);     // O(1) swap of std::set
        for (int x : S[c]) S[v].insert(x);
        S[c].clear();
    }
    ans[v] = sz(S[v]);
}
```

3.4.6 Other standard-library power tools

- `multiset` as a priority queue with deletion; `prev(ms.end())` = max.
- `map<ll, ll>` as an "interval map" (Chtholly / ODT): store maximal constant segments `[l → value]`; range assign = split at l and r+1, erase between, insert one — amortised O(log n) with random assignments.
- Set of disjoint intervals (merge on insert): `map<l, r>`, find the first interval with `prev(upper_bound(l))`, absorb overlaps.
- `deque` supports O(1) push/pop both ends + random access.
- `bitset` §2.6; `__int128` for 128-bit keys.

4. Dynamic Programming

4.1 DP Design and Classic Patterns

4.1.1 Designing a DP

1. **State** = the minimum information about the past that determines the future. Write it as a sentence: "dp[i][j] = max score using the first i items with j capacity used".
2. **Transition**: what is the last decision? (last item taken or not, where the last segment starts, which colour the last node got.)
3. **Order**: compute states after everything they depend on (increasing i; increasing interval length; post-order on a tree; topological order on a DAG). If there are cycles in the dependency, it's a shortest path / system of equations, not a DP.
4. **Count states × transitions** — reduce with prefix sums/min, monotonic deque, data structures, or a smarter state.
5. Base cases, answer location, modulo, overflow, ±INF for impossible states (use `INF/2` so additions don't overflow).

Tricks: **memoized recursion** is the fastest way to a correct DP when the order is unclear. **Rolling arrays** (`dp[i & 1]`) cut memory. **Sort first** if items can be processed in any order (by deadline, by weight) — then a knapsack-like DP over the sorted order is valid. To **reconstruct** the solution, store the argmax choice per state or re-derive by checking which transition achieves the optimum.

4.1.2 Knapsack family

```cpp
// 0/1 knapsack: max value with capacity W  (iterate capacity DOWN)
ll knap01(const vector<int>& w, const vector<ll>& v, int W) {
    vector<ll> dp(W + 1, 0);
    rep(i, 0, sz(w)) for (int c = W; c >= w[i]; --c) dp[c] = max(dp[c], dp[c - w[i]] + v[i]);
    return dp[W];
}
// unbounded: iterate capacity UP.  count ways to make sum (order-independent): items outer loop
ll countWays(const vector<int>& coins, int S, ll mod) {
    vector<ll> dp(S + 1, 0); dp[0] = 1;
    for (int c : coins) rep(s, c, S + 1) dp[s] = (dp[s] + dp[s - c]) % mod;
    return dp[S];   // swap loops (sum outer, coins inner) -> counts ordered sequences
}
```

- **Bounded** (item i available k_i times): split k into 1, 2, 4, …, remainder → O(W Σ log k_i). Or monotonic deque per residue class mod w → O(nW).
- **Values small, weights huge:** dp[value] = min weight.
- **Subset sum reachability**: bitset (§2.6). Σ w = n ⇒ ≤ √(2n) distinct weights.
- Min items to reach exact sum: dp[s] = min(dp[s − c] + 1).
- **Huge sums with few coin types (Frobenius / reachability for all large X):** Dijkstra on residues modulo the smallest coin (§5.1, ACPC 2022 *Braaaains*).

4.1.3 Sequences

```cpp
// LIS length, O(n log n). For non-decreasing use upper_bound.
int lis(const vector<ll>& a) {
    vector<ll> d;                        // d[k] = min tail of an increasing subsequence of length k+1
    for (ll x : a) {
        auto it = lower_bound(all(d), x);
        if (it == d.end()) d.push_back(x); else *it = x;
    }
    return sz(d);
}
```

To reconstruct LIS: store for each i the position `pos[i]` it took in d and `prev[i]` = last index placed at `pos[i]-1`. Count of LIS: BIT over values storing (best length, count).

- **LCS** O(nm): `dp[i][j] = s[i-1]==t[j-1] ? dp[i-1][j-1]+1 : max(dp[i-1][j], dp[i][j-1])`.
- **Edit distance**: `dp[i][j] = min(dp[i-1][j]+1, dp[i][j-1]+1, dp[i-1][j-1] + (s[i-1]!=t[j-1]))`.
- **Interleaving** s = merge of s1, s2 keeping order (ACPC 2018 *Homework*): `ok[i][j]` = first i+j chars of s are an interleaving of s1[0..i), s2[0..j): `ok[i][j] = (ok[i-1][j] && s1[i-1]==s[i+j-1]) || (ok[i][j-1] && s2[j-1]==s[i+j-1])`. $25 \cdot 10^6$ booleans is fine; use `vector<char>` or bitset rows.
- **Count distinct subsequences**: `dp[i] = 2·dp[i-1] − dp[last[c]-1]` where last[c] = previous occurrence of the same char.
- Max subarray sum (Kadane): `cur = max(x, cur + x)`.

4.1.4 Interval DP

`dp[l][r]` from smaller intervals; iterate `len` from 1 to n. $O(n^3)$ with a split point, $O(n^2)$ without.

```
// number of palindromic subsequences (as index sets, non-empty) mod p — ACPC 2016 Palindromic Naming
ll countPalSubseq(const string& s, ll mod) {
    int n = sz(s);
    vector<vector<ll>> dp(n, vector<ll>(n, 0));
    for (int l = n - 1; l >= 0; --l) {
        dp[l][l] = 1;
        rep(r, l + 1, n) {
            ll in = (l + 1 <= r - 1) ? dp[l + 1][r - 1] : 0;
            dp[l][r] = (dp[l + 1][r] + dp[l][r - 1] - in + mod) % mod;  // inclusion–exclusion
            if (s[l] == s[r]) dp[l][r] = (dp[l][r] + in + 1) % mod;     // pair s[l], s[r] around any inner one or no
ne
        }
    }
    return dp[0][n - 1];
}
```

- **Split point**: `dp[l][r] = min over l ≤ k < r of dp[l][k] + dp[k+1][r] + cost(l, r)` (matrix chain, merging stones, optimal BST; Knuth optimisation §4.4 if cost is quadrangle-monotone).
- **Expressions with + and −, choose parenthesisation to maximise** (ACPC 2022 *Maximal Parentheticals*): keep **both max and min** of every sub-expression, since `a - (b)` is maximised by minimising b. With multiplication by negatives also keep both.
- **Remove/merge adjacent elements** ("zuma", "remove all with cost"): dp over [l, r] plus an extra dimension for "how many equal elements are glued to the right end".
- **Convex polygon DP** (triangulations, ACPC 2018 *Trimming Convex Polygons*): vertices 0..n−1 in order; choosing a sub-polygon = choosing a subsequence of vertices; its area is a fan of triangles from the first chosen vertex. dp over (first kept vertex fixed, last kept vertex) with transitions adding one triangle: $O(n^3)$.
- **Circular** intervals: double the array and take windows of length n, or fix the element at position 0.

4.1.5 Grid, DAG, partition DPs

- Grid paths with obstacles: `dp[i][j] = dp[i-1][j] + dp[i][j-1]`. Count paths avoiding k special cells (n, m large): sort cells, inclusion–exclusion with C(dx+dy, dx) — $O(k^2)$.
- Grid with moves S/E/SE and accumulating state (ACPC 2024 *Alpine Agility*): row-major order works because all moves go forward.
- DAG longest path / count paths: topological order (§5.2).
- **Partition into segments**: `dp[i] = min_j dp[j] + cost(j, i)`; with k segments add a dimension → D&C / CHT optimisations (§4.4) or the λ-trick.
- **State "was my neighbour already processed"**: when the order of operations on a line matters only locally (destroying buildings that damage neighbours, ACPC 2016 *City Destruction*), dp over position i with a flag for whether i−1 is destroyed before or after i.

4.1.6 Counting DPs (mod p)

- dp over (position, last value) with prefix sums over last value → $O(n \cdot V)$.
- "Number of sequences with sum S": dp over sum; with bounded parts use prefix sums to make each step $O(1)$.
- Avoid double counting: define a **canonical** construction (e.g. always extend from the leftmost undecided position, or count by "first time the event happens").
- When the DP counts objects up to symmetry, use Burnside (§7.3) instead.

4.2 Bitmask DP, Digit DP, SOS, Broken Profile

4.2.1 Bitmask DP

Assignment / ordering of n ≤ 20 items. `dp[mask]` = best for having used the set `mask`; the next item index is often `popcount(mask)`.

```
// TSP / Hamiltonian path: dp[mask][v] = min cost path visiting mask, ending at v. O(2^n n^2)
ll tsp(const vector<vector<ll>>& d) {      // returns min Hamiltonian cycle from 0
    int n = sz(d);
    vector<vector<ll>> dp(1 << n, vector<ll>(n, INF));
    dp[1][0] = 0;
    rep(mask, 1, 1 << n) rep(v, 0, n) {
        if (!(mask >> v & 1) || dp[mask][v] >= INF) continue;
        rep(u, 0, n) if (!(mask >> u & 1))
```

```
            dp[mask | 1 << u][u] = min(dp[mask | 1 << u][u], dp[mask][v] + d[v][u]);
    }
    ll best = INF;
    rep(v, 1, n) best = min(best, dp[(1 << n) - 1][v] + d[v][0]);
    return n == 1 ? 0 : best;
}
```

- **Assignment** (n workers ≤ 20 to n jobs): `dp[mask]` over assigned jobs, worker = popcount(mask). $O(2^n \cdot n)$. For n > 20 use Hungarian (§6.3).
- **Partition into groups** (min #groups each with weight ≤ W): `dp[mask] = (groups, current fill)` pairs, compare lexicographically. $O(2^n \cdot n)$.
- **Over submasks**: `dp[mask] = min over sub ⊆ mask of dp[mask ^ sub] + cost[sub]` → $O(3^n)$; n ≤ 15–16.
- Memory: `2^20 × 20` ints = 80 MB; prefer `int` and drop dimensions.
- Bitmask over a **small dimension** of a grid (rows ≤ 10): dp[column][mask of current column].

4.2.2 SOS DP (sum over subsets)

`F[mask] = Σ_{sub ⊆ mask} A[sub]` in $O(n \cdot 2^n)$:

```
void sos(vector<ll>& F, int n) {           // in place: F starts as A
    rep(i, 0, n) rep(mask, 0, 1 << n) if (mask >> i & 1) F[mask] += F[mask ^ (1 << i)];
}
// superset sums: condition !(mask >> i & 1), F[mask] += F[mask | 1 << i]
// inverse (Möbius): same loops with -=
```

Uses: count pairs with `a_i & a_j == 0` (count submasks of the complement), `a_i | a_j ⊆ x`, number of elements that are submasks of each x. AND/OR convolution = SOS transform, multiply pointwise, inverse transform.

4.2.3 Digit DP

Count integers in [0, N] with a property depending on digits. Answer for [L, R] = f(R) − f(L − 1).

```
// count x in [0, N] whose digit sum is divisible by K (K <= 100)
ll countDigitSum(ll N, int K) {
    if (N < 0) return 0;
    string s = to_string(N); int n = sz(s);
    // memo[pos][rem] for non-tight states only (tight path is unique, recomputed)
    vector<vector<ll>> memo(n, vector<ll>(K, -1));
    function<ll(int, int, bool)> go = [&](int pos, int rem, bool tight) -> ll {
        if (pos == n) return rem == 0;
        if (!tight && memo[pos][rem] != -1) return memo[pos][rem];
        int lim = tight ? s[pos] - '0' : 9;
        ll res = 0;
        rep(d, 0, lim + 1) res += go(pos + 1, (rem + d) % K, tight && d == lim);
        if (!tight) memo[pos][rem] = res;
        return res;
    };
    return go(0, 0, true);
}
```

Extra state flags as needed: `started` (to handle leading zeros, e.g. "no two adjacent equal digits"), `last digit`, `count of nonzero digits`, `rem mod m`. For "k-th number with property": compute counts per prefix choice and descend digit by digit.

4.2.4 Broken profile DP (tilings)

Count domino tilings of an n × m grid (n ≤ 10): process cells in row-major order; mask = which of the next m cells are already covered.

```
ll dominoTilings(int n, int m, ll mod) {          // m = width (small)
    vector<ll> dp(1 << m, 0); dp[0] = 1;
    rep(i, 0, n) rep(j, 0, m) {
        vector<ll> nd(1 << m, 0);
        rep(mask, 0, 1 << m) if (dp[mask]) {
            ll v = dp[mask];
            if (mask >> j & 1) { nd[mask ^ (1 << j)] = (nd[mask ^ (1 << j)] + v) % mod; continue; } // already covere
d
            nd[mask | (1 << j)] = (nd[mask | (1 << j)] + v) % mod;                  // vertical domino down
            if (j + 1 < m && !(mask >> (j + 1) & 1))                                // horizontal domino right
                nd[mask | (1 << (j + 1))] = (nd[mask | (1 << (j + 1))] + v) % mod;
        }
        dp = nd;
    }
```

```
    return dp[0];
}
```

Bit j of the mask refers to cell (i, j) when processing row i (set = covered by a vertical domino from above or a horizontal domino from the left). The same skeleton handles "place objects with local constraints" problems (e.g. minesweeper-like boards: keep the last two rows in the state when constraints reach distance 1).

4.3 Tree DP and Rerooting

Root the tree (usually at 0), compute children before parents. For n up to 10^6 use an **iterative** order: BFS/DFS order array, then process it in reverse.

```
// parent + order (pre-order). Process `for (int i = n-1; i >= 0; --i) v = order[i]` for bottom-up.
void rootTree(const vector<vector<int>>& g, int root, vector<int>& par, vector<int>& order) {
    int n = sz(g); par.assign(n, -1); order.clear(); order.reserve(n);
    vector<int> st = {root}; par[root] = root;
    while (!st.empty()) {
        int v = st.back(); st.pop_back(); order.push_back(v);
        for (int u : g[v]) if (u != par[v]) { par[u] = v; st.push_back(u); }
    }
    par[root] = -1;
}
```

4.3.1 Standard subtree DPs

- **Max independent set** (weights w): `in[v] = w_v + Σ out[c]`, `out[v] = Σ max(in[c], out[c])`.
- **Min vertex cover** = n − max independent set (trees are bipartite); **max matching** greedy from leaves: match a leaf with its parent if both free.
- **Diameter**: for each v keep the two longest downward paths `d1 ≥ d2`; diameter = max(d1 + d2). (Or BFS twice for unweighted/positive weights.)
- **Subtree sizes / sums**, **counting pairs at distance k**: small-to-large or centroid (§5.4).

Colouring with k colours minimising cost (ACPC 2022 *City Numbers* pattern)

`dp[v][c]` = min cost of subtree of v when v gets colour c = cost(c) + Σ_children min_{c' ≠ c} dp[child][c']. The inner min over c' ≠ c is O(1) with the child's **best and second best**:

```
// colours 1..k, cost of colour c = c. Returns min total cost (k >= 2 always feasible for a tree)
ll minColouring(const vector<vector<int>>& g, int k) {
    int n = sz(g); vector<int> par, order; rootTree(g, 0, par, order);
    vector<vector<ll>> dp(n, vector<ll>(k + 1, 0));
    for (int i = n - 1; i >= 0; --i) {
        int v = order[i];
        rep(c, 1, k + 1) dp[v][c] += c;
        if (par[v] < 0) continue;
        ll b1 = INF, b2 = INF; int bc = -1;          // best, second best over colours of v
        rep(c, 1, k + 1) {
            if (dp[v][c] < b1) { b2 = b1; b1 = dp[v][c]; bc = c; }
            else if (dp[v][c] < b2) b2 = dp[v][c];
        }
        rep(c, 1, k + 1) dp[par[v]][c] += (c == bc ? b2 : b1);
    }
    return *min_element(dp[0].begin() + 1, dp[0].end());
}
```

(Fact: an optimal colouring never needs more than O(log n) colours, so `k = min(k, 20)` is safe; with k = 1 and n > 1 it's impossible.)

4.3.2 Tree knapsack (merge children with size bounds) — O(n²) total

`dp[v][j]` = best value choosing j nodes in subtree of v. Merging child c into v: loop `j ≤ size(v so far)`, `t ≤ size(c)`. Bounding loops by current sizes makes the total O(n²) (each pair of nodes is "merged" once at their LCA). With a cap K on j: O(nK).

```
// sketch of merge step: cur = dp of v so far (size sv), ch = dp of child (size sc)
vector<ll> nd(sv + sc + 1, -INF);
rep(a, 0, sv + 1) if (cur[a] > -INF) rep(b, 0, sc + 1) if (ch[b] > -INF)
    nd[a + b] = max(nd[a + b], cur[a] + ch[b]);
cur = nd; sv += sc;
```

4.3.3 Rerooting (answer for every root)

Two passes: `down[v]` (subtree answer) bottom-up, then `up[v]` (answer of the part outside v's subtree) top-down; combine. Requires: the contribution of a child to its parent can be computed from the child's value, and the combination is "removable" (sum) or handled with prefix/suffix aggregates (max).

```
// sum of distances from every vertex to all others, O(n)
vector<ll> sumDistAll(const vector<vector<int>>& g) {
    int n = sz(g); vector<int> par, order; rootTree(g, 0, par, order);
    vector<ll> sub(n, 1), down(n, 0), ans(n, 0);
    for (int i = n - 1; i > 0; --i) { int v = order[i]; sub[par[v]] += sub[v]; down[par[v]] += down[v] + sub[v]; }
    ans[0] = down[0];
    rep(i, 1, n) { int v = order[i]; ans[v] = ans[par[v]] - sub[v] + (n - sub[v]); }
    return ans;
}
```

Generic rerooting with max (e.g. farthest node from every vertex): for each node compute children contributions `f(child)`, prefix-max and suffix-max over the children list, so the value "excluding child c" is `max(prefix[i-1], suffix[i+1], up-contribution of v)`.

4.3.4 Other tree techniques (pointers)

- Euler tour (tin/tout) turns subtree into a contiguous range → BIT/segment tree on subtree queries (§5.4).
- "Is u an ancestor of v": `tin[u] ≤ tin[v] && tout[v] ≤ tout[u]` (ACPC 2020 *Listen To Your Boss*).
- DP on paths/LCA: binary lifting tables store aggregates (min/max edge) along 2^k jumps.
- Tree isomorphism: canonical hashing of sorted child hashes (AHU) from the centre(s).

4.4 DP Optimisations

recurrence shape	technique	complexity
`dp[i] = min_j dp[j] + f(j) + g(i)` with j in a sliding window	monotonic deque (§2.3)	O(n)
`dp[i] = min_j (a_j · x_i + b_j)` (+ terms of i)	convex hull trick / Li Chao	O(n log n)
`dp[k][i] = min_{j<i} dp[k-1][j] + C(j, i)`, opt(i) monotone	divide & conquer	O(k n log n)
`dp[i][j] = min_{i≤k<j} dp[i][k] + dp[k+1][j] + C(i, j)`	Knuth (opt[i][j−1] ≤ opt[i][j] ≤ opt[i+1][j])	O(n²)
"exactly k groups" and answer convex in k	Aliens trick (λ-penalty, binary search)	O(n log n log C)
`dp[x]` is convex piecewise linear in a value x	slope trick (heaps of breakpoints)	O(n log n)

Monge / quadrangle inequality `C(a, c) + C(b, d) ≤ C(a, d) + C(b, c)` for a ≤ b ≤ c ≤ d implies monotone opt (both D&C and Knuth). Typical Monge costs: (sum of segment)², sum of |x − median| over a segment, number of pairs inside a segment. When unsure, stress-test opt monotonicity with a brute force.

4.4.1 Convex hull trick (any insertion/query order)

```
// [stress-tested] max of k*x + m. For min: add(-k, -m) and negate query.
struct Line {
    mutable ll k, m, p;
    bool operator<(const Line& o) const { return k < o.k; }
    bool operator<(ll x) const { return p < x; }
};
struct LineContainer : multiset<Line, less<>> {
    static const ll inf = LLONG_MAX;
    ll div(ll a, ll b) { return a / b - ((a ^ b) < 0 && a % b); }  // floor
    bool isect(iterator x, iterator y) {
        if (y == end()) return x->p = inf, 0;
        if (x->k == y->k) x->p = x->m > y->m ? inf : -inf;
        else x->p = div(y->m - x->m, x->k - y->k);
        return x->p >= y->p;
    }
    void add(ll k, ll m) {
        auto z = insert({k, m, 0}), y = z++, x = y;
        while (isect(y, z)) z = erase(z);
        if (x != begin() && isect(--x, y)) isect(x, y = erase(y));
        while ((y = x) != begin() && (--x)->p >= y->p) isect(x, erase(y));
    }
    ll query(ll x) {
        auto l = *lower_bound(x);
        return l.k * x + l.m;
    }
};
```

Watch overflow: $|k \cdot x|$ must fit in `ll`. Example: `dp[i] = min_j dp[j] + (p_i − p_j)²` = `p_i² + min_j (−2p_j·p_i + dp[j] + p_j²)` → line with slope `−2p_j`, intercept `dp[j] + p_j²`, query at `p_i`.

4.4.2 Li Chao tree (min, over a known set of query x's)

Also supports adding a line only on an x-range (descend to O(log n) nodes and insert there).

```
// [stress-tested]
struct LiChao {
    struct L { ll k, m; ll eval(ll x) const { return k * x + m; } };
    vector<ll> xs; vector<L> t; int n;
    LiChao(vector<ll> pts) : xs(pts) {
        sort(all(xs)); xs.erase(unique(all(xs)), xs.end());
        n = sz(xs); t.assign(4 * n, {0, INF});
    }
    void add(ll k, ll m) { add({k, m}, 1, 0, n - 1); }
    void add(L nw, int x, int l, int r) {
        int mid = (l + r) / 2;
        bool lef = nw.eval(xs[l]) < t[x].eval(xs[l]);
        bool md = nw.eval(xs[mid]) < t[x].eval(xs[mid]);
        if (md) swap(t[x], nw);
        if (l == r) return;
        if (lef != md) add(nw, 2 * x, l, mid); else add(nw, 2 * x + 1, mid + 1, r);
    }
    ll query(ll qx) {                    // qx must be one of pts
        int i = lower_bound(all(xs), qx) - xs.begin(), x = 1, l = 0, r = n - 1;
        ll res = INF;
        while (true) {
            res = min(res, t[x].eval(qx));
            if (l == r) return res;
            int mid = (l + r) / 2;
            if (i <= mid) { x = 2 * x; r = mid; } else { x = 2 * x + 1; l = mid + 1; }
        }
    }
};
```

The empty line `{0, INF}` evaluates to INF; don't add lines with huge slopes where `k*x` overflows.

4.4.3 Divide & conquer optimisation

```
// [stress-tested] split a[0..n) into exactly K contiguous groups minimising sum of cost(j, i) for
// group = a[j..i). cost must satisfy the quadrangle inequality.
ll partitionDC(int n, int K, const function<ll(int, int)>& cost) {
    vector<ll> prv(n + 1, INF), cur(n + 1, INF);
    prv[0] = 0;
    function<void(int, int, int, int)> solve = [&](int l, int r, int optl, int optr) {
        if (l > r) return;
        int mid = (l + r) / 2, opt = optl;
        ll best = INF;
        for (int j = optl; j <= min(mid - 1, optr); ++j)
            if (prv[j] < INF && prv[j] + cost(j, mid) < best) { best = prv[j] + cost(j, mid); opt = j; }
        cur[mid] = best;
        solve(l, mid - 1, optl, opt); solve(mid + 1, r, opt, optr);
    };
    rep(k, 0, K) { fill(all(cur), INF); solve(1, n, 0, n - 1); swap(prv, cur); }
    return prv[n];
}
```

4.4.4 Knuth optimisation (interval DP)

```
// sketch: dp[i][j] over closed intervals, C(i, j) Monge and monotone on inclusion
for (int i = 0; i < n; ++i) { dp[i][i] = base(i); opt[i][i] = i; }
for (int len = 2; len <= n; ++len)
    for (int i = 0; i + len - 1 < n; ++i) {
        int j = i + len - 1; dp[i][j] = INF;
        for (int k = opt[i][j - 1]; k <= min(j - 1, opt[i + 1][j]); ++k) {
            ll v = dp[i][k] + dp[k + 1][j] + C(i, j);
            if (v < dp[i][j]) { dp[i][j] = v; opt[i][j] = k; }
        }
    }
```

4.4.5 Aliens trick (Lagrangian relaxation)

If `f(k)` = optimal cost with exactly k groups/items is **convex** in k, then for a penalty λ solve the unconstrained problem where every group costs an extra λ: `g(λ) = min_k f(k) + λk`. Track the number of groups used (minimum count among ties). Binary

search λ (integer) so that the count is ≤ k with smallest λ; answer `f(k) = g(λ) − λ·k`. Turns O(n·k) states into O(n) per λ.

4.4.6 Slope trick

Maintain a convex piecewise-linear function by the multiset of its slope-change points. Classic: minimum Σ|a_i − b_i| to make b non-decreasing:

```
ll minCostNonDecreasing(const vector<ll>& a) {
    priority_queue<ll> pq; ll ans = 0;
    for (ll x : a) {
        pq.push(x);
        if (pq.top() > x) { ans += pq.top() - x; pq.pop(); pq.push(x); }
    }
    return ans;   // strictly increasing: apply to a_i - i
}
```

4.5 Probability, Expected Value, Games

ACPC has a probability/EV problem at difficulty 5–8 most years (Balls and Bins, Gunslingers, Concentration). They are almost always **DP over a small state space**, solved backward from terminal states.

4.5.1 Core facts

- **Linearity:** E[X + Y] = E[X] + E[Y] always (even dependent). Write the quantity as a sum of indicators: E[#fixed points of random permutation] = n·(1/n) = 1; E[#inversions] = n(n−1)/4.
- E[X] for X ≥ 0 integer: Σ_{k≥1} P(X ≥ k). E[max] via P(max ≤ k) = Π P(X_i ≤ k).
- **Geometric:** success prob p per trial ⇒ expected trials 1/p. **Coupon collector:** n·H_n = n(1 + 1/2 + … + 1/n).
- Independent events: P(A ∩ B) = P(A)P(B). Conditional: P(A | B) = P(A ∩ B)/P(B). Bayes: P(A | B) = P(B | A)P(A)/P(B).
- **Gambler's ruin** (fair): start at a, absorb at 0 or N ⇒ P(reach N) = a/N, expected steps a(N − a). Biased with ratio r = q/p: P(reach N) = $(1 - r^a)/(1 - r^N)$.
- Sum of probabilities over all outcomes must be 1 — assert it in your DP to catch bugs.
- Random permutation: element i ends before j with prob 1/2; i is the max of a random set of size k with prob 1/k.

4.5.2 Expected value DP

`E[s] = cost(s) + Σ_t P(s → t)·E[t]`, terminal states have E = 0. Compute in an order where t is done before s (e.g. "number of remaining items" decreases).

- **Self-loops** (state can stay the same with prob p): E = (c + Σ_{t ≠ s} P(t)E[t]) / (1 − p).
- **Optimal stopping / decisions** (ACPC 2021 *Balls and Bins*): `V[s] = max(value if stop now, Σ P(t)·(reward + V[t]))`. The state must capture everything that affects future probabilities — e.g. (#bins with 1 ball, #bins with 2 balls); the number of balls thrown is then determined.
- **Forward probability DP** when you need the distribution: `P[t] += P[s]·p(s → t)`; answer = Σ over final states.
- **Probability of winning with optimal play:** `W[s] = max over moves of Σ P(outcome)·W[next]` (or 1 − opponent's W when turns alternate).
- **Cycles in the state graph**:
- A *cyclic chain* (turns rotate among players until something happens): unroll. If `x_1 = a_1 + b_1 x_2, …, x_k = a_k + b_k x_1` then `x_1 = (a_1 + b_1 a_2 + b_1 b_2 a_3 + … ) / (1 − b_1 b_2 ⋯ b_k)`, then back-substitute. (Gunslingers: within a fixed set of survivors, misses rotate the turn; hits go to a smaller set, already solved.)
- General cycles: Gaussian elimination on the linear system (§7.4), O(n³) for n ≤ ~500.
- Infinite processes with decreasing probability: iterate until change < 1e-12 (value iteration) when the system is contractive.

```
// x[i] = a[i] + b[i] * x[(i+1) % k]  (cyclic chain), returns all x. Requires prod(b) != 1.
vector<double> solveCycle(const vector<double>& a, const vector<double>& b) {
    int k = sz(a); double num = 0, mul = 1;
    rep(i, 0, k) { num += mul * a[i]; mul *= b[i]; }
    vector<double> x(k);
    x[0] = num / (1 - mul);
    for (int i = k - 1; i >= 1; --i) x[i] = a[i] + b[i] * x[(i + 1) % k];
    return x;
}
```

Expected value modulo p

"Output E as P·Q⁻¹ mod 998244353": do the same DP with modular arithmetic; every division by an integer d becomes multiplication by `inv(d)` (§7.1). Probabilities like 1/n become `inv(n)`. `1/(1 - p)` becomes `inv((1 - p + MOD) % MOD)`.

Precision with doubles

Use `double` (or `long double`) and print 9+ decimals. Avoid computing tiny probabilities as products of thousands of terms (underflow → use logs). Subtracting nearly equal numbers loses precision — reformulate `(1 - (1 - p)^n` via `-expm1(n * log1p(-p))`).

4.5.3 Combinatorial games

Win/lose DP on a DAG of positions: a position is winning iff some move leads to a losing position; terminal positions are losing (normal play: player unable to move loses). Compute by memoized recursion.

Games with cycles (draws possible) — retrograde BFS: start from terminal positions; a position becomes LOSE when all its successors are WIN (count remaining successors), WIN when any successor is LOSE; positions never labelled are DRAW.

```cpp
// g = moves (position -> next positions). res: 1 = win, -1 = lose, 0 = draw (for player to move)
vector<int> retrograde(const vector<vector<int>>& g) {
    int n = sz(g); vector<vector<int>> rg(n); vector<int> deg(n), res(n, 0);
    rep(v, 0, n) { deg[v] = sz(g[v]); for (int u : g[v]) rg[u].push_back(v); }
    queue<int> q;
    rep(v, 0, n) if (deg[v] == 0) { res[v] = -1; q.push(v); }
    while (!q.empty()) {
        int u = q.front(); q.pop();
        for (int v : rg[u]) if (res[v] == 0) {
            if (res[u] == -1) { res[v] = 1; q.push(v); }
            else if (--deg[v] == 0) { res[v] = -1; q.push(v); }
        }
    }
    return res;
}
```

Sprague–Grundy: for a sum of independent impartial games, `grundy(position) = mex{grundy(next)}`; the whole game is losing iff the xor of the components' Grundy values is 0.

- **Nim:** Grundy of a heap of size x is x → first player wins iff xor $\neq 0$.
- **Subtraction game** (remove 1..k): g(x) = x mod (k+1).
- **Splitting games** (a heap splits into two): g = mex over splits of g(a) xor g(b).
- **Staircase Nim:** only odd-indexed steps matter (xor of piles on odd steps).
- **Misère Nim:** same as Nim unless all heaps ≤ 1, then first player wins iff the number of heaps of size 1 is even.
- **Grundy values often become periodic**: compute the first few hundred by brute force and look for the period.

Two-player optimisation games (both maximise own score, take from ends of an array): `dp[l][r]` = best (my score − opponent's score) for the player to move on [l, r] = max(a_l − dp[l+1][r], a_r − dp[l][r−1]).

Strategy-stealing / symmetry: first player often wins by mirroring (central symmetry on a board, pairing moves). When stuck, brute force small n and print the win/lose table — a pattern (parity, mod k, powers of 2) usually appears.

5. Graphs and Trees

5.1 Shortest Paths and Graph Search

situation	algorithm	complexity
unweighted	BFS	O(n + m)
weights 0/1 (or 0/k)	0-1 BFS with deque	O(n + m)
non-negative weights	Dijkstra	O(m log n)
negative weights, detect negative cycle	Bellman–Ford	O(nm)
all pairs, n ≤ 500	Floyd–Warshall	$O(n^3)$
DAG (any weights)	DP in topological order	O(n + m)

5.1.1 BFS on grids (template)

```
const int DR[] = {-1, 1, 0, 0}, DC[] = {0, 0, -1, 1};   // add diagonals for 8-neighbour
vector<vector<int>> gridBFS(const vector<string>& g, vector<pii> sources) {
    int R = sz(g), C = sz(g[0]);
    vector<vector<int>> d(R, vector<int>(C, -1));
    queue<pii> q;
    for (auto [r, c] : sources) { d[r][c] = 0; q.push({r, c}); }   // multi-source
    while (!q.empty()) {
        auto [r, c] = q.front(); q.pop();
        rep(k, 0, 4) {
            int nr = r + DR[k], nc = c + DC[k];
            if (nr < 0 || nr >= R || nc < 0 || nc >= C || g[nr][nc] == '#' || d[nr][nc] != -1) continue;
            d[nr][nc] = d[r][c] + 1; q.push({nr, nc});
        }
    }
    return d;
}
```

- **State-space BFS:** node = (cell, extra state): direction, die orientation (ACPC 2016 *Dacey the Dice*), collected keys bitmask, remaining fuel. Encode as one integer `((r * C + c) * S + s)`.
- **Sliding moves** (ice: move until hitting an obstacle, ACPC 2024 *Slippery Floor*): neighbours of a cell are the 4 stopping cells; precompute or walk (each walk is O(R + C)).
- **Flood fill / count components**: BFS from every unvisited cell.

5.1.2 0-1 BFS

```
// sketch: adj[u] = {(v, w)} with w in {0, 1}
deque<int> dq; dist.assign(n, INF); dist[s] = 0; dq.push_back(s);
while (!dq.empty()) {
    int u = dq.front(); dq.pop_front();
    for (auto [v, w] : adj[u]) if (dist[u] + w < dist[v]) {
        dist[v] = dist[u] + w;
        w ? dq.push_back(v) : dq.push_front(v);
    }
}
```

Typical: "minimum number of walls to break", "minimum number of direction changes" (state = (cell, direction), continuing straight costs 0, turning costs 1).

5.1.3 Dijkstra

```
// adj[u] = {(v, w)}, w >= 0. par gives the shortest path tree.
vector<ll> dijkstra(const vector<vector<pair<int, ll>>>& adj, const vector<int>& srcs, vector<int>* par = nullptr) {
    int n = sz(adj); vector<ll> d(n, INF);
    if (par) par->assign(n, -1);
    priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>> pq;
    for (int s : srcs) { d[s] = 0; pq.push({0, s}); }
    while (!pq.empty()) {
        auto [du, u] = pq.top(); pq.pop();
        if (du != d[u]) continue;                       // stale entry
        for (auto [v, w] : adj[u]) if (du + w < d[v]) {
            d[v] = du + w; pq.push({d[v], v});
            if (par) (*par)[v] = u;
        }
    }
```

```
    return d;
}
```

- Dense graph ($n \le 5000$, $m \approx n^2$): $O(n^2)$ Dijkstra with a linear scan for the minimum is faster.
- **Count shortest paths:** when `du + w == d[v]` add `cnt[u]` to `cnt[v]` (process in pop order).
- **Edge (u,v,w) lies on some shortest s–t path** iff `ds[u] + w + dt[v] == ds[t]` (run from s and from t on the reversed graph).
- **Product graph Dijkstra** (ACPC 2023 *Horse Snatchers*: state = (town, still has horse)); (node, number of coupons used $\le$ k); (node, time mod k). Size n·S nodes.
- **Maximise the minimum along the path** (bottleneck/widest path): Dijkstra with `max-heap` on `min(cap so far, w)`, or Kruskal in decreasing order until s, t connected, or binary search + BFS.
- **Binary search + Dijkstra** (ACPC 2018 *Arachnophobia*): multi-source Dijkstra from all spiders gives `spiderDist[v]`; binary search D, run Dijkstra s→t using only vertices with `spiderDist ≥ D`, check `dist ≤ T`.
- Vertex weights: put the weight on incoming edges. Undirected: add both directions.
- Shortest path with "time-dependent" edges that are FIFO (arriving later never lets you leave earlier): Dijkstra still works with arrival-time relaxation.

Dijkstra on residues: which totals are representable (coin problem)

Coins c_1..c_k (unbounded), smallest coin `a`. `d[r]` = smallest representable total ≡ r (mod a). Then X is representable ⇔ `d[X mod a] ≤ X`. Largest non-representable = max(d[r]) − a (if gcd = 1). ACPC 2022 *Braaaains*.

```
vector<ll> residueDist(const vector<ll>& coins) {
    ll a = *min_element(all(coins));
    vector<ll> d(a, INF); d[0] = 0;
    priority_queue<pair<ll, int>, vector<pair<ll, int>>, greater<>> pq; pq.push({0, 0});
    while (!pq.empty()) {
        auto [du, r] = pq.top(); pq.pop();
        if (du != d[r]) continue;
        for (ll c : coins) { int nr = (r + c) % a; if (du + c < d[nr]) { d[nr] = du + c; pq.push({d[nr], nr}); } }
    }
    return d;   // X representable iff d[X % a] <= X
}
```

5.1.4 Bellman–Ford and negative cycles

```
// edges (u, v, w). Returns dist; dist = -INF for vertices reachable from a negative cycle.
vector<ll> bellmanFord(int n, const vector<array<ll, 3>>& edges, int s) {
    vector<ll> d(n, INF); d[s] = 0;
    rep(it, 0, n) for (auto [u, v, w] : edges)
        if (d[u] < INF && d[u] + w < d[v]) d[v] = (it == n - 1 ? -INF : d[u] + w);
    rep(it, 0, n) for (auto [u, v, w] : edges)          // propagate -INF
        if (d[u] == -INF && d[v] != -INF) d[v] = -INF;
    return d;
}
```

Marking only in the n-th round misses some affected vertices, so the second loop propagates `-INF` through everything reachable. **Difference constraints** `x_v - x_u ≤ w` → edge u → v with weight w; add a source with 0-edges to all; feasible iff no negative cycle; `x = dist` is a solution (maximal for each variable with x ≤ 0).

5.1.5 Floyd–Warshall

```
// d[i][j] = edge weight or INF, d[i][i] = 0. Negative cycle through i iff d[i][i] < 0 afterwards.
void floyd(vector<vector<ll>>& d) {
    int n = sz(d);
    rep(k, 0, n) rep(i, 0, n) if (d[i][k] < INF) rep(j, 0, n)
        if (d[k][j] < INF) d[i][j] = min(d[i][j], d[i][k] + d[k][j]);
}
```

Variants (same triple loop): transitive closure (`reach[i][j] |= reach[i][k] && reach[k][j]`, or bitset rows), minimax path (`max` instead of `+`, `min` outer), counting paths in a DAG, shortest path when vertices are "unlocked" over time (process k in order of unlock time and answer queries in between).

5.2 MST, DAGs, Functional Graphs, Permutations

5.2.1 Minimum spanning tree

```
// Kruskal: edges (w, u, v). Returns total weight (n-1 edges if connected). Needs DSU (§3.1).
struct DSUk { vector<int> p; DSUk(int n) : p(n, -1) {}
```

```
    int f(int x) { return p[x] < 0 ? x : p[x] = f(p[x]); }
    bool u(int a, int b) { a = f(a); b = f(b); if (a == b) return 0; if (p[a] > p[b]) swap(a, b); p[a] += p[b]; p[b]
= a; return 1; } };
ll kruskal(int n, vector<tuple<ll, int, int>> edges, vector<tuple<ll, int, int>>* used = nullptr) {
    sort(all(edges)); DSUk d(n); ll tot = 0;
    for (auto& [w, a, b] : edges) if (d.u(a, b)) { tot += w; if (used) used->push_back({w, a, b}); }
    return tot;
}
```

- **Dense graphs** (complete graph on points, n ≤ 5000): Prim O(n²) with an array, no heap.
- **Cycle property:** the heaviest edge on any cycle is not needed. **Cut property:** the lightest edge across any cut is in some MST.
- MST minimises the **maximum** edge on paths too: the minimax path between u and v is the path in the MST. Max spanning tree for maximin paths.
- Second-best MST: for each non-tree edge (u, v, w), replace the max tree edge on path(u, v) (binary lifting with max, §5.4).
- **Min ratio spanning tree** Σcost/Σgain (ACPC 2023 *Buildin' Roads*): binary search λ, MST on `cost - λ·gain`, feasible iff total ≤ 0 (§2.1). Doubles, 60–100 iterations.
- Steiner tree on k ≤ 10 terminals: dp[mask][v] + Dijkstra per mask ($O(3^k n + 2^k m \log n)$).
- Count spanning trees: Kirchhoff (determinant of Laplacian minor, §7.4).

5.2.2 Topological sort and DAG DP

```
// Kahn. Returns empty vector if there is a cycle. Use a min-heap instead of queue for lexicographically smallest ord
er.
vector<int> topo(const vector<vector<int>>& g) {
    int n = sz(g); vector<int> indeg(n), order;
    rep(u, 0, n) for (int v : g[u]) indeg[v]++;
    queue<int> q; rep(u, 0, n) if (!indeg[u]) q.push(u);
    while (!q.empty()) {
        int u = q.front(); q.pop(); order.push_back(u);
        for (int v : g[u]) if (--indeg[v] == 0) q.push(v);
    }
    return sz(order) == n ? order : vector<int>();
}
```

- Longest path / count paths / probability of reaching: DP in topo order. Number of paths can be huge → mod or doubles.
- Unique topological order ⇔ queue size is always 1 ⇔ Hamiltonian path exists in the DAG.
- **Min path cover** of a DAG (vertices) = n − max matching (§6.2); **with path overlap allowed** = same on the transitive closure. **Cover all edges** (ACPC 2022 *Escape Routes*): min flow with lower bounds (§6.1).
- **Dilworth:** minimum number of chains covering a poset = size of the largest antichain. For 2D dominance (pairs sorted by x, chains need y non-decreasing): min chains = longest strictly decreasing subsequence of y.
- Directed graph with cycles → condense SCCs first (§5.3), then DAG DP.

5.2.3 Functional graphs (each node has exactly one outgoing edge)

Structure: each component is one cycle with in-trees hanging off it.

```
// cycId[v] = id of the cycle containing v (-1 if v is on a tail); cycLen[id] = its length
void functionalCycles(const vector<int>& f, vector<int>& cycId, vector<int>& cycLen) {
    int n = sz(f); vector<int> state(n, 0); cycId.assign(n, -1); cycLen.clear();
    rep(s, 0, n) if (!state[s]) {
        vector<int> path; int v = s;
        while (!state[v]) { state[v] = 1; path.push_back(v); v = f[v]; }   // 1 = on current path
        if (state[v] == 1) {                                               // found a new cycle
            int id = sz(cycLen), len = 0;
            for (int i = sz(path) - 1; ; --i) { cycId[path[i]] = id; ++len; if (path[i] == v) break; }
            cycLen.push_back(len);
        }
        for (int x : path) state[x] = 2;
    }
}
```

- **k-th successor** for huge k: binary lifting `up[j][v] = up[j-1][up[j-1][v]]`, O(n log k) build, O(log k) query. Or walk to the cycle and use k mod cycle length.
- Floyd's tortoise–hare finds a cycle in an implicit sequence x → f(x) with O(1) memory (e.g. pseudo-random generators, repeated states of a simulation: **detect the period, then skip ahead**).
- "Simulate 10^{18} steps of a deterministic process" ⇒ the state repeats; store first-seen time of each state in a map, jump by the period.

5.2.4 Permutations

- Decompose into cycles: `i → p[i]`. Minimum swaps (any two elements) to sort = n − #cycles. Adjacent swaps = #inversions (BIT, §3.2).
- p^k (apply k times): for each cycle of length L, shift by k mod L. Order of p = lcm of cycle lengths.
- Square root of a permutation, conjugacy: cycles of equal length pair up.
- Swaps restricted to graph edges (ACPC 2021 *Swapping Places*): within each connected component any permutation is achievable; construct by fixing leaves of a spanning tree first (route the right value to a leaf along the tree path, then delete the leaf) — $O(n^2)$ swaps.
- Rotations of a permutation maximising fixed points (ACPC 2022 *Maximum Fix*): element a_i is fixed by exactly one shift k = (i − a_i) mod n; count votes.

5.3 SCC, 2-SAT, Bridges, Articulation Points, Euler Paths

5.3.1 Strongly connected components (Tarjan)

```cpp
// [stress-tested] comp[v] in [0, ncomp). Components are numbered in REVERSE topological order
// (comp 0 is a sink): edge u->v between components implies comp[u] >= comp[v].
struct SCC {
    int n, timer = 0, ncomp = 0;
    vector<vector<int>> g; vector<int> tin, low, comp, st; vector<char> on;
    SCC(const vector<vector<int>>& g) : n(sz(g)), g(g), tin(n, -1), low(n), comp(n, -1), on(n) {
        rep(v, 0, n) if (tin[v] == -1) dfs(v);
    }
    void dfs(int v) {
        tin[v] = low[v] = timer++; st.push_back(v); on[v] = 1;
        for (int u : g[v]) {
            if (tin[u] == -1) { dfs(u); low[v] = min(low[v], low[u]); }
            else if (on[u]) low[v] = min(low[v], tin[u]);
        }
        if (low[v] == tin[v]) {
            while (true) {
                int u = st.back(); st.pop_back(); on[u] = 0; comp[u] = ncomp;
                if (u == v) break;
            }
            ++ncomp;
        }
    }
};
```

Condensation DAG: edge comp[u] → comp[v] for every edge with different comps. Typical follow-ups: min edges to add to make strongly connected = max(#sources, #sinks) (0 if one SCC); nodes reachable from all others = the unique sink SCC if exactly one sink.

Recursion depth = n; for $n \geq 5 \cdot 10^5$ on a path consider an iterative version or Kosaraju with explicit stacks.

5.3.2 2-SAT

```cpp
// [stress-tested] variables 0..n-1. lit(x, true) = x, lit(x, false) = not x.
struct TwoSat {
    int n; vector<vector<int>> g;
    TwoSat(int n) : n(n), g(2 * n) {}
    static int lit(int x, bool val) { return 2 * x + !val; }
    void implies(int a, int b) { g[a].push_back(b); g[b ^ 1].push_back(a ^ 1); }  // a -> b (and !b -> !a)
    void either(int a, int b) { implies(a ^ 1, b); }                              // a or b
    void setTrue(int a) { implies(a ^ 1, a); }
    void atMostOne(int a, int b) { implies(a, b ^ 1); }                           // not (a and b)
    void equal(int a, int b) { implies(a, b); implies(b, a); }
    bool solve(vector<bool>& val) {
        SCC s(g); val.assign(n, false);
        rep(x, 0, n) {
            if (s.comp[2 * x] == s.comp[2 * x + 1]) return false;
            val[x] = s.comp[2 * x] < s.comp[2 * x + 1];   // later in topo order = true
        }
        return true;
    }
};
```

Modelling: each boolean choice is a variable; constraints must be clauses of **two** literals. "Exactly one of a, b" = either(a, b) + atMostOne(a, b). "At most one among k literals" → prefix-variable encoding (O(k) extra variables). Often combined with binary search on a threshold (e.g. maximise min distance between chosen points: pairs closer than D can't both be chosen).

5.3.3 Bridges and articulation points

```cpp
// [stress-tested] undirected multigraph; edges given as (u, v). Handles parallel edges via edge ids.
struct BridgeCut {
    int n, timer = 0; vector<vector<pii>> g; vector<int> tin, low;
    vector<char> isBridge, isCut;
    BridgeCut(int n, const vector<pii>& edges) : n(n), g(n), tin(n, -1), low(n), isBridge(sz(edges)), isCut(n) {
        rep(i, 0, sz(edges)) { auto [u, v] = edges[i]; g[u].push_back({v, i}); g[v].push_back({u, i}); }
        rep(v, 0, n) if (tin[v] == -1) dfs(v, -1);
    }
    void dfs(int v, int pe) {
        tin[v] = low[v] = timer++; int children = 0;
        for (auto [u, id] : g[v]) {
            if (id == pe) continue;
            if (tin[u] != -1) { low[v] = min(low[v], tin[u]); continue; }
            dfs(u, id); low[v] = min(low[v], low[u]); ++children;
            if (low[u] > tin[v]) isBridge[id] = 1;
            if (low[u] >= tin[v] && pe != -1) isCut[v] = 1;
        }
        if (pe == -1 && children > 1) isCut[v] = 1;
    }
};
```

- **2-edge-connected components:** delete bridges, take components. Contracting them gives the **bridge tree** (a tree!) — then path questions become tree questions ("must every u–v path use edge e" ⇔ e is a bridge on the bridge-tree path).
- **Biconnected components / block-cut tree:** push edges on a stack in the DFS; when `low[u] >= tin[v]` pop edges up to (v, u) → one block. Block-cut tree: bipartite tree of blocks and cut vertices; "vertices that lie on every u–v path" = cut vertices on the block-cut tree path.
- Edge in a cycle ⇔ not a bridge. Graph stays connected after removing any one vertex ⇔ no articulation points.

5.3.4 Euler path / circuit (Hierholzer)

Existence: **undirected** connected (ignoring isolated vertices) with 0 odd-degree vertices (circuit) or exactly 2 (path, start at an odd one). **Directed**: every vertex in = out (circuit), or exactly one with out − in = 1 (start) and one with in − out = 1 (end).

```cpp
// [stress-tested] returns vertex sequence of length m+1 using every edge once, or {} if impossible
// from s. directed=false: each edge usable in both directions.
vector<int> eulerPath(int n, const vector<pii>& edges, int s, bool directed) {
    int m = sz(edges); vector<vector<pii>> g(n);
    rep(i, 0, m) { auto [u, v] = edges[i]; g[u].push_back({v, i}); if (!directed) g[v].push_back({u, i}); }
    vector<int> ptr(n, 0), path, st = {s}; vector<char> used(m, 0);
    while (!st.empty()) {
        int v = st.back();
        while (ptr[v] < sz(g[v]) && used[g[v][ptr[v]].second]) ++ptr[v];
        if (ptr[v] == sz(g[v])) { path.push_back(v); st.pop_back(); }
        else { auto [u, id] = g[v][ptr[v]]; used[id] = 1; st.push_back(u); }
    }
    if (sz(path) != m + 1) return {};
    reverse(all(path));
    return path;
}
```

Check the degree conditions yourself first (and pick s accordingly): if they fail, the routine can still return a sequence of length m+1 that is not a valid trail. The length check only catches edges unreachable from s. Uses: **De Bruijn sequence** (nodes = (k−1)-length strings, edges = k-length strings, Euler circuit), reconstructing a string from all its k-mers, "draw without lifting the pen", domino chains.

5.4 Trees: LCA, Euler Tour, HLD, Centroid Decomposition

5.4.1 LCA with binary lifting (+ distances, k-th ancestor, path aggregates)

```cpp
// [stress-tested] weighted tree, iterative DFS (safe for deep trees).
struct LCA {
    int n, LOG; vector<vector<int>> up; vector<int> dep, tin, tout; vector<ll> dist;
    LCA(const vector<vector<pair<int, ll>>>& g, int root = 0) : n(sz(g)) {
        LOG = 1; while ((1 << LOG) < n) ++LOG;
        up.assign(LOG, vector<int>(n, root)); dep.assign(n, 0); tin.assign(n, 0); tout.assign(n, 0);
        dist.assign(n, 0);
        int timer = 0; vector<pair<int, int>> st = {{root, 0}};   // (vertex, next child index)
        tin[root] = timer++;
        while (!st.empty()) {
            auto& [v, i] = st.back();
            if (i < sz(g[v])) {
```

```
                auto [u, w] = g[v][i++];
                if (u == up[0][v] && v != root) continue;
                up[0][u] = v; dep[u] = dep[v] + 1; dist[u] = dist[v] + w; tin[u] = timer++;
                st.push_back({u, 0});
            } else { tout[v] = timer++; st.pop_back(); }
        }
        rep(k, 1, LOG) rep(v, 0, n) up[k][v] = up[k - 1][up[k - 1][v]];
    }
    bool isAnc(int u, int v) const { return tin[u] <= tin[v] && tout[v] <= tout[u]; }
    int lca(int u, int v) const {
        if (isAnc(u, v)) return u;
        if (isAnc(v, u)) return v;
        for (int k = LOG - 1; k >= 0; --k) if (!isAnc(up[k][u], v)) u = up[k][u];
        return up[0][u];
    }
    ll distance(int u, int v) const { return dist[u] + dist[v] - 2 * dist[lca(u, v)]; }
    int kthAncestor(int v, int k) const {   // k <= dep[v]
        rep(j, 0, LOG) if (k >> j & 1) v = up[j][v];
        return v;
    }
    // k-th vertex on the path u -> v (0-indexed), assumes k <= #edges on the path
    int kthOnPath(int u, int v, int k) const {
        int w = lca(u, v), du = dep[u] - dep[w], dv = dep[v] - dep[w];
        return k <= du ? kthAncestor(u, k) : kthAncestor(v, du + dv - k);
    }
};
```

- **Path aggregates (max edge on path, min, sum):** add a table `mx[k][v]` = max edge weight on the 2^k jump; combine while lifting both endpoints to the LCA.
- **Path sums with point updates:** Euler tour + BIT: value on vertex v contributes to the whole subtree, so add at `tin[v]`, subtract at `tout[v]`; root-to-v sum = prefix query at `tin[v]`; path(u, v) = s(u) + s(v) − 2 s(lca) + val(lca).
- **Subtree queries:** with pre-order indices `tin` (counting only entries), subtree of v = `[tin[v], tin[v] + size[v] - 1]` → BIT/segment tree.
- **Distance between nodes in unweighted tree, many queries:** depth[u] + depth[v] − 2·depth[lca].
- **Ancestor check** without LCA: tin/tout as above (O(1)).
- **Diameter endpoints:** farthest from any node is an endpoint a; farthest from a is b. Every vertex's eccentricity = max(dist(v, a), dist(v, b)). **Centre** = middle of the diameter path.
- **Virtual (auxiliary) tree** for queries on k marked nodes: sort by tin, add LCAs of adjacent pairs, build with a stack → O(k log n) nodes; then run the DP only on that tree.

5.4.2 Heavy-light decomposition

Any root-to-leaf path crosses O(log n) heavy chains; positions on a chain are contiguous, so path queries become O(log n) segment-tree range queries (O(log² n) total).

```
// [stress-tested] call path(u, v, f): f(l, r) on inclusive position ranges. Use with any segment tree
// indexed by pos[]. Edge values: store the edge weight at pos[child] and pass skipLca = true.
struct HLD {
    int n; vector<int> par, heavy, dep, head, pos, sub;
    HLD(const vector<vector<int>>& g, int root = 0) : n(sz(g)), par(n, -1), heavy(n, -1), dep(n), head(n), pos(n), su
b(n, 1) {
        vector<int> order; order.reserve(n); order.push_back(root);
        rep(i, 0, n) { int v = order[i]; for (int u : g[v]) if (u != par[v]) { par[u] = v; dep[u] = dep[v] + 1; orde
r.push_back(u); } }
        for (int i = n - 1; i > 0; --i) { int v = order[i]; sub[par[v]] += sub[v]; }
        rep(i, 0, n) { int v = order[i]; int best = 0;
            for (int u : g[v]) if (u != par[v] && sub[u] > best) { best = sub[u]; heavy[v] = u; } }
        int cur = 0; vector<int> st = {root}; head[root] = root;
        while (!st.empty()) {                       // pre-order DFS, heavy child visited first
            int v = st.back(); st.pop_back(); pos[v] = cur++;
            for (int u : g[v]) if (u != par[v] && u != heavy[v]) { head[u] = u; st.push_back(u); }
            if (heavy[v] != -1) { head[heavy[v]] = head[v]; st.push_back(heavy[v]); }
        }
    }
    template <class F> void path(int u, int v, F f, bool skipLca = false) {
        for (; head[u] != head[v]; v = par[head[v]]) {
            if (dep[head[u]] > dep[head[v]]) swap(u, v);
            f(pos[head[v]], pos[v]);
        }
        if (dep[u] > dep[v]) swap(u, v);
        if (pos[u] + skipLca <= pos[v]) f(pos[u] + skipLca, pos[v]);
    }
```

```
    pair<int, int> subtree(int v) { return {pos[v], pos[v] + sub[v] - 1}; }
};
```

Positions are a DFS pre-order with the heavy child first, so both chains and subtrees are contiguous: `subtree(v)` and `path(u, v)` can be mixed on one segment tree.

5.4.3 Centroid decomposition

Centroid: vertex whose removal leaves components of size $\leq n/2$. Recursively decompose: depth O(log n); every path in the tree passes through the centroid of the smallest decomposition component containing both ends.

```
// cpar[v] = parent in the centroid tree (-1 for the root centroid)
struct Centroid {
    vector<vector<int>> g; vector<int> sub, cpar; vector<char> removed;
    Centroid(const vector<vector<int>>& g) : g(g), sub(sz(g)), cpar(sz(g), -1), removed(sz(g)) { build(0, -1); }
    int calcSize(int v, int p) { sub[v] = 1; for (int u : g[v]) if (u != p && !removed[u]) sub[v] += calcSize(u, v);
return sub[v]; }
    int find(int v, int p, int tot) {
        for (int u : g[v]) if (u != p && !removed[u] && sub[u] * 2 > tot) return find(u, v, tot);
        return v;
    }
    void build(int v, int p) {
        int c = find(v, -1, calcSize(v, -1));
        cpar[c] = p; removed[c] = 1;
        // ---- process all paths through c here (DFS from c over non-removed vertices) ----
        for (int u : g[c]) if (!removed[u]) build(u, c);
    }
};
```

Typical uses: - **Count paths with length $\leq$ K / exactly K:** at each centroid, collect depths from each child subtree; count pairs across different subtrees (sort + two pointers, subtract pairs within the same child). - **Nearest marked vertex queries:** `best[c]` = min distance from centroid c to a marked vertex in its component; update/query walks up `cpar` (O(log n) ancestors) using `dist(v, c)` (precomputed per level or via LCA). - Recursion depth of `calcSize` can be n on a path graph at the top level; fine for $n \leq 2 \cdot 10^5$ on Kattis, otherwise make it iterative.

6. Flows, Matching, Assignment

6.1 Max Flow, Min Cut, and Modelling

Signals: "each X used at most once/at most c times", "assign", "disjoint paths", "minimum set of things to remove to disconnect", "choose a subset with dependencies maximising profit", "minimum number of paths/chains covering …". Small n (≤ 500 nodes, ≤ 10^4–10^5 edges) is a strong hint.

6.1.1 Dinic

```cpp
// [stress-tested] O(V^2 E) worst, much faster in practice; unit-capacity graphs O(E sqrt V).
struct Dinic {
    struct E { int to; ll cap; };
    vector<E> e; vector<vector<int>> g; vector<int> level, it;
    Dinic(int n) : g(n), level(n), it(n) {}
    int addEdge(int u, int v, ll cap, ll rcap = 0) {        // rcap = cap for undirected edges
        g[u].push_back(sz(e)); e.push_back({v, cap});
        g[v].push_back(sz(e)); e.push_back({u, rcap});
        return sz(e) - 2;                                    // id; flow on it = original cap - e[id].cap
    }
    bool bfs(int s, int t) {
        fill(all(level), -1); level[s] = 0; queue<int> q; q.push(s);
        while (!q.empty()) {
            int v = q.front(); q.pop();
            for (int id : g[v]) if (e[id].cap > 0 && level[e[id].to] < 0) {
                level[e[id].to] = level[v] + 1; q.push(e[id].to);
            }
        }
        return level[t] >= 0;
    }
    ll dfs(int v, int t, ll f) {
        if (v == t) return f;
        ll pushed = 0;
        for (int& i = it[v]; i < sz(g[v]); ++i) {
            int id = g[v][i], to = e[id].to;
            if (e[id].cap <= 0 || level[to] != level[v] + 1) continue;
            ll d = dfs(to, t, min(f - pushed, e[id].cap));
            if (d > 0) {
                e[id].cap -= d; e[id ^ 1].cap += d; pushed += d;
                if (pushed == f) return pushed;
            }
        }
        return pushed;
    }
    ll maxflow(int s, int t, ll limit = INF) {
        ll flow = 0;
        while (flow < limit && bfs(s, t)) {
            fill(all(it), 0);
            flow += dfs(s, t, limit - flow);
        }
        return flow;
    }
    // after maxflow: S side of a minimum cut = vertices reachable from s in the residual graph
    bool inS(int v) const { return level[v] >= 0; }
};
```

- **Min cut edges:** edges u → v with `inS(u) && !inS(v)` (original edges only). Max flow = min cut capacity.
- Reuse: to rerun with modified capacities, rebuild the graph (or reset caps from a saved copy).
- Capacities as `ll`; use `INF` for "uncuttable" edges but make sure the answer can't involve them (otherwise "impossible").

6.1.2 Modelling cookbook

need	construction
several sources/sinks	super source S → each source (cap = supply), each sink → super sink T
vertex capacity c	split v into v_in → v_out with cap c; edges enter v_in, leave v_out
undirected edge cap c	one edge with cap c in both directions (`addEdge(u, v, c, c)`)
k edge-disjoint s–t paths	unit capacities, flow ≥ k (Menger); vertex-disjoint: split vertices with cap 1
bipartite matching	S → left (1), left → right (1), right → T (1)
each worker ≤ a_i tasks, task needs b_j workers	S → worker (a_i), worker → task (1), task → T (b_j); feasible iff flow = Σb_j

need	construction
minimise the maximum load	binary search the cap L on S → worker edges; feasible iff max flow = total
time steps (people moving over time)	time-expanded graph: node (v, t) for every t, edges (v,t) → (u,t+1)
pick at most one per row & column (grid)	bipartite rows ↔ columns, cell = edge
choose ≥/≤ constraints from both sides	lower bounds (below)

Project selection / maximum closure (min cut)

Projects with profit p_i (may be negative), dependencies "taking i requires taking j". Build: S → i with cap p_i for p_i > 0; i → T with cap −p_i for p_i < 0; i → j with cap INF for each dependency. **Max profit = Σ positive p_i − maxflow.** Chosen set = S side.

General "two-label" min cut (each item either A or B): cost a_i if item is in B (edge S → i, cap a_i), cost b_i if in A (edge i → T, cap b_i), penalty c for i in A and j in B (edge i → j, cap c). Works when penalties are non-negative and pairwise ("submodular"). Image segmentation, "assign each student to one of two buildings with friendship penalties".

Lower bounds (circulations with demands)

Edge u → v with flow in [l, c]: add edge with cap c − l, and record `ex[v] += l`, `ex[u] -= l`. Then S' → v (cap ex[v]) for ex[v] > 0, v → T' (cap −ex[v]) for ex[v] < 0. A feasible circulation exists iff maxflow(S', T') saturates all S' edges. For an s–t flow add edge t → s with cap INF first.

- **Max flow with lower bounds:** after feasibility, F = flow on the t → s edge; remove that edge (set both caps to 0); answer = F + maxflow(s, t) on the same residual graph (S', T' edges are saturated and can't be used).
- **Min flow with lower bounds:** feasible value F = flow on the t → s edge; remove that edge (set both caps to 0); answer = F − maxflow(t, s).

```
// [stress-tested] Minimum number of s->t paths in a DAG covering every edge at least once
// (ACPC 2022 Escape Routes). Every edge: lower bound 1, capacity INF. Assumes each edge lies on
// some s->t path. Generalises: give each edge its own [l, c].
ll minFlowCoverEdges(int n, const vector<pii>& edges, int s, int t) {
    int S = n, T = n + 1; Dinic d(n + 2); vector<ll> ex(n, 0);
    const ll BIG = 1e12;
    for (auto [u, v] : edges) { d.addEdge(u, v, BIG - 1); ex[v] += 1; ex[u] -= 1; }
    int ts = d.addEdge(t, s, BIG);
    ll need = 0;
    rep(v, 0, n) { if (ex[v] > 0) { d.addEdge(S, v, ex[v]); need += ex[v]; } else if (ex[v] < 0) d.addEdge(v, T, -ex
[v]); }
    if (d.maxflow(S, T) != need) return -1;                // infeasible
    ll F = BIG - d.e[ts].cap;                                // current s->t flow value
    d.e[ts].cap = d.e[ts ^ 1].cap = 0;
    return F - d.maxflow(t, s);
}
```

6.1.3 Flow theorems worth knowing

- **Max-flow = min-cut.** Integral capacities ⇒ integral max flow (so flows decompose into paths: each unit = one path).
- **Menger:** max #edge-disjoint s–t paths = min #edges to delete to disconnect s from t (same for vertices with splitting).
- **Hall:** a bipartite graph has a matching saturating the left side iff every left subset X has |N(X)| ≥ |X|. Deficiency version: max matching = |L| − max_X (|X| − |N(X)|).
- Max flow in a graph where all capacities are multiplied by k is k times — so "fractional" answers often come from binary search on a ratio, not from fractional flows.
- **Gomory–Hu tree** (all-pairs min cut, n − 1 max flows) exists if you need min cut between many pairs in an undirected graph.

6.2 Bipartite Matching and Covers

6.2.1 Hopcroft–Karp (O(E √V))

```
// [stress-tested] left 0..n-1, right 0..m-1
struct HopcroftKarp {
    int n, m; vector<vector<int>> g; vector<int> mL, mR, dist;
    HopcroftKarp(int n, int m) : n(n), m(m), g(n), mL(n, -1), mR(m, -1), dist(n) {}
    void addEdge(int u, int v) { g[u].push_back(v); }
    bool bfs() {
        queue<int> q; bool found = false;
        rep(u, 0, n) { if (mL[u] < 0) { dist[u] = 0; q.push(u); } else dist[u] = -1; }
        while (!q.empty()) {
            int u = q.front(); q.pop();
            for (int v : g[u]) {
                int w = mR[v];
```

```
                if (w < 0) found = true;
                else if (dist[w] < 0) { dist[w] = dist[u] + 1; q.push(w); }
            }
        }
        return found;
    }
    bool dfs(int u) {
        for (int v : g[u]) {
            int w = mR[v];
            if (w < 0 || (dist[w] == dist[u] + 1 && dfs(w))) { mL[u] = v; mR[v] = u; return true; }
        }
        dist[u] = -1;
        return false;
    }
    int maxMatching() {
        int res = 0;
        while (bfs()) rep(u, 0, n) if (mL[u] < 0 && dfs(u)) ++res;
        return res;
    }
    // König: minimum vertex cover (call after maxMatching). Returns (left cover, right cover).
    pair<vector<int>, vector<int>> minVertexCover() {
        vector<char> visL(n, 0), visR(m, 0); queue<int> q;
        rep(u, 0, n) if (mL[u] < 0) { visL[u] = 1; q.push(u); }
        while (!q.empty()) {                                    // alternating BFS from free left
            int u = q.front(); q.pop();
            for (int v : g[u]) if (!visR[v] && mL[u] != v) {
                visR[v] = 1;
                if (mR[v] >= 0 && !visL[mR[v]]) { visL[mR[v]] = 1; q.push(mR[v]); }
            }
        }
        vector<int> L, R;
        rep(u, 0, n) if (!visL[u]) L.push_back(u);
        rep(v, 0, m) if (visR[v]) R.push_back(v);
        return {L, R};
    }
};
```

For tiny graphs, Kuhn's algorithm (DFS augment from each left vertex, O(VE)) is shorter; Hopcroft–Karp is safe up to $\sim 10^5$ vertices / 10^6 edges.

6.2.2 Facts (bipartite graphs, V = L ∪ R)

quantity	equals
min vertex cover	max matching (König)
max independent set	V − max matching (complement of a min vertex cover)
min edge cover (no isolated vertices)	V − max matching
min path cover of a DAG (vertex-disjoint)	n − max matching in split graph (u_out → v_in for each edge)
min path cover of a DAG (paths may share vertices)	same on the transitive closure
min chains covering a poset	max antichain (Dilworth) = min path cover on the closure
perfect matching exists	Hall's condition / max matching = n

- **Grid problems** (dominoes, placing non-attacking pieces, covering cells): colour cells like a chessboard → bipartite. Max non-attacking rooks on a board with holes: rows ↔ columns.
- **Max independent set in a bipartite conflict graph** = choose max items with no conflicting pair (e.g. points where conflicts only between "even" and "odd" sums).
- **Min vertex cover in a bipartite graph** = "minimum number of lines (rows/cols) to cover all marked cells".
- **Maximum weight** bipartite matching → Hungarian / min cost flow (§6.3).
- **General (non-bipartite) graphs:** max matching needs Edmonds' blossom ($O(V^3)$); for $n \le 20$–25, bitmask DP over matched vertices; for trees, greedy from leaves.
- **Path cover reconstruction**: matched edge u_out → v_in means "v follows u in its path"; paths start at vertices whose v_in is unmatched.

6.2.3 Stable marriage (Gale–Shapley)

```
// prefM[i] = men's preference lists (women ids, best first), prefW[j][i] = rank of man i for woman j
vector<int> galeShapley(const vector<vector<int>>& prefM, const vector<vector<int>>& rankW) {
    int n = sz(prefM); vector<int> wife(n, -1), husband(n, -1), nxt(n, 0); queue<int> freeMen;
    rep(i, 0, n) freeMen.push(i);
    while (!freeMen.empty()) {
```

```
        int m = freeMen.front(); freeMen.pop();
        int w = prefM[m][nxt[m]++];
        if (husband[w] < 0) { husband[w] = m; wife[m] = w; }
        else if (rankW[w][m] < rankW[w][husband[w]]) {
            wife[husband[w]] = -1; freeMen.push(husband[w]); husband[w] = m; wife[m] = w;
        } else freeMen.push(m);
    }
    return wife;   // man-optimal stable matching
}
```

6.3 Min-Cost Flow and the Assignment Problem

6.3.1 Hungarian algorithm (n ≤ m, O(n²m))

```
// [stress-tested] a = n x m cost matrix (n <= m). Minimum total cost assigning every row to a
// distinct column. Returns {cost, match} with match[row] = column. Maximise: negate costs.
pair<ll, vector<int>> hungarian(const vector<vector<ll>>& a) {
    int n = sz(a), m = sz(a[0]);
    const ll BIG = LLONG_MAX / 4;
    vector<ll> u(n + 1), v(m + 1); vector<int> p(m + 1), way(m + 1);
    rep(i, 1, n + 1) {
        p[0] = i; int j0 = 0;
        vector<ll> minv(m + 1, BIG); vector<char> used(m + 1, 0);
        do {
            used[j0] = 1; int i0 = p[j0], j1 = 0; ll delta = BIG;
            rep(j, 1, m + 1) if (!used[j]) {
                ll cur = a[i0 - 1][j - 1] - u[i0] - v[j];
                if (cur < minv[j]) { minv[j] = cur; way[j] = j0; }
                if (minv[j] < delta) { delta = minv[j]; j1 = j; }
            }
            rep(j, 0, m + 1) if (used[j]) { u[p[j]] += delta; v[j] -= delta; } else minv[j] -= delta;
            j0 = j1;
        } while (p[j0] != 0);
        do { int j1 = way[j0]; p[j0] = p[j1]; j0 = j1; } while (j0);
    }
    vector<int> match(n);
    rep(j, 1, m + 1) if (p[j]) match[p[j] - 1] = j - 1;
    return {-v[0], match};
}
```

- Forbidden pairs: cost = large constant `C` (e.g. 1e12, but ≤ BIG/n); if the result ≥ C, no valid assignment exists.
- n > m: transpose. "Not everyone must be matched": add dummy columns with cost 0 (or the cost of staying unmatched).
- 500 × 500 runs in well under a second.

Worked pattern: enumerate the maximum, then assign (ACPC 2018 *Dog Trouble*)

Minimise `Σ (t - t_i)` where t = max assigned time ⇒ `N·t - Σ t_i`. Enumerate t over all distinct values (≤ 200); allow only pairs with t_ij ≤ t; maximise Σ t_ij with Hungarian (cost −t_ij, forbidden = C). Answer = min over t of `N·t - maxSum(t)`. Using a t larger than the true max only overestimates, so taking the min over all t is exact. 200 × O(50³) is tiny.

General lesson: **an objective mixing "max" and "sum" → fix the max by enumeration or binary search, then optimise the sum.**

6.3.2 Min-cost max-flow (successive shortest paths with SPFA)

```
// [stress-tested] handles negative edge costs (no negative cycles). O(F · V · E) worst case.
struct MCMF {
    struct E { int to; ll cap, cost; };
    vector<E> e; vector<vector<int>> g; vector<ll> dist; vector<int> pe; vector<char> inq;
    MCMF(int n) : g(n), dist(n), pe(n), inq(n) {}
    int addEdge(int u, int v, ll cap, ll cost) {
        g[u].push_back(sz(e)); e.push_back({v, cap, cost});
        g[v].push_back(sz(e)); e.push_back({u, 0, -cost});
        return sz(e) - 2;
    }
    // returns {flow, cost}. stopIfPositive: stop when the cheapest path costs >= 0
    // (gives min-cost *any-amount* flow, i.e. max profit when costs are negated profits)
    pair<ll, ll> run(int s, int t, ll maxf = INF, bool stopIfPositive = false) {
        ll flow = 0, cost = 0;
        while (flow < maxf) {
            fill(all(dist), INF); dist[s] = 0; queue<int> q; q.push(s); inq[s] = 1;
            while (!q.empty()) {
                int u = q.front(); q.pop(); inq[u] = 0;
                for (int id : g[u]) {
                    const E& ed = e[id];
```

```
                    if (ed.cap > 0 && dist[u] + ed.cost < dist[ed.to]) {
                        dist[ed.to] = dist[u] + ed.cost; pe[ed.to] = id;
                        if (!inq[ed.to]) { inq[ed.to] = 1; q.push(ed.to); }
                    }
                }
            }
            if (dist[t] == INF || (stopIfPositive && dist[t] >= 0)) break;
            ll f = maxf - flow;
            for (int v = t; v != s; v = e[pe[v] ^ 1].to) f = min(f, e[pe[v]].cap);
            for (int v = t; v != s; v = e[pe[v] ^ 1].to) { e[pe[v]].cap -= f; e[pe[v] ^ 1].cap += f; }
            flow += f; cost += f * dist[t];
        }
        return {flow, cost};
    }
};
```

Modelling with costs: - **Assignment with capacities** (worker i can do up to c_i jobs): S → worker (c_i, 0), worker → job (1, cost), job → T (1, 0). - **k disjoint paths of minimum total length**: unit capacities, run with `maxf = k`. - **Choose at most k items with conflicts along a line** (k non-overlapping interval families, max weight): chain i → i+1 with (k, 0), interval [l, r) as edge l → r with (1, −w). - **Max-weight matching where not all must be matched:** costs = −weight, `stopIfPositive = true`. - Convex costs (cost of the j-th unit is increasing): parallel edges with capacity 1 and costs c_1 ≤ c_2 ≤ …. - If it's too slow (large flow), use Dijkstra with Johnson potentials instead of SPFA, or capacity scaling — rarely needed at ACPC sizes.

7. Math

7.1 Modular Arithmetic, GCD, CRT, Binomials

7.1.1 Basics

```
// [stress-tested]
ll power(ll b, ll e, ll m) {                    // b^e mod m, m < 2^62 safe via __int128
    ll r = 1 % m; b %= m; if (b < 0) b += m;
    for (; e > 0; e >>= 1, b = (__int128)b * b % m) if (e & 1) r = (__int128)r * b % m;
    return r;
}
ll inv(ll a, ll p) { return power(a, p - 2, p); } // p prime, a not divisible by p
ll extgcd(ll a, ll b, ll& x, ll& y) {           // a*x + b*y = g = gcd(a, b)
    if (b == 0) { x = 1; y = 0; return a; }
    ll x1, y1, g = extgcd(b, a % b, x1, y1);
    x = y1; y = x1 - (a / b) * y1;
    return g;
}
ll invGeneral(ll a, ll m) {                     // inverse mod any m, or -1 if gcd(a, m) != 1
    ll x, y; ll g = extgcd(((a % m) + m) % m, m, x, y);
    return g != 1 ? -1 : ((x % m) + m) % m;
}
```

- Always reduce after every `+`, `-` (`((a - b + MOD) % MOD)`), `*` (cast to `ll` first: `1LL * a * b % MOD`).
- Division by d ⇒ multiply by `inv(d)`; valid only if gcd(d, MOD) = 1. **Don't divide before taking mod in a formula unless d divides exactly** (e.g. n(n+1)/2 mod p: compute with the inverse of 2, or divide n or n+1 first).
- **Exponent reduction** (Fermat/Euler): a^e mod p = a ^ (e mod (p−1)) mod p when gcd(a, p) = 1. For towers `a^(b^c) mod m` use φ(m): $a^e \equiv$ a ^ (e mod φ(m) + φ(m)) for e ≥ log2 m (generalised Euler).
- Inverses of 1..n in O(n): `inv[1] = 1; inv[i] = (MOD - MOD/i) * inv[MOD % i] % MOD`.
- `mod 10^9 + 9`, `998244353`, `10^6` (not prime! no inverses) — read the statement carefully.

7.1.2 Factorials and binomials

```
struct Comb {                                   // needs n < p (else n! = 0 mod p)
    vector<ll> f, fi; ll p;
    Comb(int n, ll p) : f(n + 1), fi(n + 1), p(p) {
        f[0] = 1; rep(i, 1, n + 1) f[i] = f[i - 1] * i % p;
        fi[n] = power(f[n], p - 2, p);
        for (int i = n; i > 0; --i) fi[i - 1] = fi[i] * i % p;
    }
    ll C(int n, int k) const { return k < 0 || k > n ? 0 : f[n] * fi[k] % p * fi[n - k] % p; }
};
// C(n, k) for huge n (up to 1e18) and small k: product formula, p prime > k
ll Cbig(ll n, int k, ll p) {
    if (k < 0 || k > n) return 0;
    ll num = 1, den = 1;
    rep(i, 0, k) { num = (__int128)num * ((n - i) % p) % p; den = den * ((i + 1) % p) % p; }
    return num * power(den, p - 2, p) % p;
}
// Lucas: C(n, k) mod small prime p, with c = Comb(p - 1, p)
ll lucas(ll n, ll k, const Comb& c) {
    if (k < 0 || k > n) return 0;
    ll res = 1;
    while (n || k) { res = res * c.C(n % c.p, k % c.p) % c.p; n /= c.p; k /= c.p; }
    return res;
}
```

- Without a prime modulus (or n ≤ 60 exact): Pascal's triangle `C[n][k] = C[n-1][k-1] + C[n-1][k]` (fits `ll` up to n = 66 for all k; `__int128` further).
- C(n, k) mod prime power / composite: factor m, Lucas–Granville or CRT; rarely needed.

7.1.3 Linear Diophantine equations and CRT

a·x + b·y = c solvable iff g = gcd(a, b) divides c. One solution: `x0 = x·(c/g)`, `y0 = y·(c/g)` from extgcd. All solutions: `x = x0 + k·(b/g)`, `y = y0 - k·(a/g)`. To find the smallest non-negative x: `x = ((x0 % (b/g)) + b/g) % (b/g)` (use `__int128` for the multiplication).

```cpp
// [stress-tested] x ≡ a1 (mod m1), x ≡ a2 (mod m2), moduli need not be coprime.
// Returns {x, lcm} with 0 <= x < lcm, or {-1, -1} if no solution. Keep lcm < 2^62.
pair<ll, ll> crt(ll a1, ll m1, ll a2, ll m2) {
    ll x, y, g = extgcd(m1, m2, x, y);
    if ((a2 - a1) % g != 0) return {-1, -1};
    ll l = m1 / g * m2;
    __int128 t = (__int128)(a2 - a1) / g * x % (m2 / g);
    __int128 res = (a1 + t * m1) % l;
    if (res < 0) res += l;
    return {(ll)res, l};
}
```

Fold over many congruences pairwise. Uses: combine answers mod several primes, "the clock shows x mod a and y mod b" problems, periodic events that must coincide (first time t where several cyclic processes are simultaneously in given states).

7.1.4 GCD / LCM facts

- gcd(a, b) = gcd(a, b − a) = gcd(a, b mod a); gcd of an array with one changing element → prefix/suffix gcds.
- gcd(a + k, b + k) as k varies: gcd(a + k, b − a) → only divisors of |b − a| matter.
- Prefix gcds of an array take ≤ log2(max) distinct values (each change at least halves) — "all subarray gcds" has O(n log V) distinct (value, start) pairs.
- lcm can overflow: `a / gcd(a, b) * b`, check with `__int128` or cap.
- Number of pairs (i, j) ≤ n with gcd = 1 → Möbius / φ sums (§7.2). Σ_{d|n} φ(d) = n.
- `__gcd`, `std::gcd`, `std::lcm` (C++17) work on `ll`.
- "**Find p given several values s_i² mod p**" (ACPC 2024 *Four Questions*): p divides `s_i² - r_i` for each query, so p divides `gcd` of these numbers. Choose queries so the gcd is p (use s > √p so s² actually wraps; several queries remove spurious factors).

7.1.5 Useful closed forms

- Σi = n(n+1)/2, Σi² = n(n+1)(2n+1)/6, Σi³ = (n(n+1)/2)², geometric Σ_{i<n} $r^i = (r^n - 1)/(r - 1)$.
- Σ_{i=1}^{n} ⌊n/i⌋ = O(n log n) to iterate directly; computing it fast uses the O(√n) block trick (§2.5).
- Harmonic numbers H_n ≈ ln n + 0.5772. Number of primes ≤ n ≈ n / ln n (78498 below 10^6, 664579 below 10^7).

7.2 Primes, Factorisation, Divisors, Multiplicative Functions

7.2.1 Sieves

```cpp
// [stress-tested] smallest prime factor for all n <= N (linear sieve) + list of primes. N = 1e7 ~ 0.1 s, 40 MB.
struct Sieve {
    vector<int> spf, primes;
    Sieve(int N) : spf(N + 1, 0) {
        rep(i, 2, N + 1) {
            if (!spf[i]) { spf[i] = i; primes.push_back(i); }
            for (int p : primes) {
                if (p > spf[i] || (ll)i * p > N) break;
                spf[i * p] = p;
            }
        }
    }
    vector<pair<int, int>> factor(int x) const {      // x <= N, O(log x)
        vector<pair<int, int>> f;
        while (x > 1) { int p = spf[x], e = 0; while (x % p == 0) x /= p, ++e; f.push_back({p, e}); }
        return f;
    }
};
```

- Plain Eratosthenes with `vector<bool>` handles N = 10^8 in ~1 s.
- **Segmented sieve** for primes in [L, R] with R ≤ 10^{12}, R − L ≤ 10^6: sieve primes up to √R, then cross out multiples in the window.
- The linear sieve also computes multiplicative functions: φ, μ, number of divisors — handle the `i % p == 0` case separately.

7.2.2 Trial division (n ≤ 10^{12}–10^{14}) and factoring big numbers

```cpp
vector<pair<ll, int>> factorTrial(ll n) {      // O(sqrt n); with a primes list, O(pi(sqrt n))
    vector<pair<ll, int>> f;
    for (ll p = 2; p * p <= n; ++p) if (n % p == 0) {
        int e = 0; while (n % p == 0) n /= p, ++e; f.push_back({p, e});
    }
    if (n > 1) f.push_back({n, 1});
```

```
    return f;
}
```

"All prime factors $< 10^6$" (ACPC 2022 *Tricky Factoring*) ⇒ divide out primes up to 10^6 from a sieve, even if n is 10^{16}.

```
// [stress-tested] deterministic Miller-Rabin for n < 2^64, Pollard rho. ~1 ms per 64-bit number.
ull mulmod(ull a, ull b, ull m) { return (unsigned __int128)a * b % m; }
ull powmod(ull b, ull e, ull m) { ull r = 1; b %= m; for (; e; e >>= 1, b = mulmod(b, b, m)) if (e & 1) r = mulmod(r,
b, m); return r; }
bool isPrime(ull n) {
    if (n < 2) return false;
    for (ull p : {2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37}) if (n % p == 0) return n == p;
    ull d = n - 1; int s = 0; while (!(d & 1)) d >>= 1, ++s;
    for (ull a : {2, 325, 9375, 28178, 450775, 9780504, 1795265022}) {
        ull x = powmod(a % n, d, n);
        if (x == 0 || x == 1 || x == n - 1) continue;
        bool comp = true;
        rep(i, 1, s) { x = mulmod(x, x, n); if (x == n - 1) { comp = false; break; } }
        if (comp) return false;
    }
    return true;
}
ull pollard(ull n) {
    if (n % 2 == 0) return 2;
    static mt19937_64 rr(1234);
    while (true) {
        ull x = rr() % n, y = x, c = rr() % (n - 1) + 1, d = 1;
        auto f = [&](ull v) { return (mulmod(v, v, n) + c) % n; };
        while (d == 1) {
            ull q = 1, xs = x;
            rep(i, 0, 64) { x = f(x); y = f(f(y)); q = mulmod(q, x > y ? x - y : y - x, n); if (!q) break; }
            d = __gcd(q, n);
            if (d == 1) continue;
            if (d == n) {                       // batch overshot: redo step by step
                x = xs; d = 1;
                while (d == 1) { x = f(x); y = f(f(y)); d = __gcd(x > y ? x - y : y - x, n); }
            }
        }
        if (d != n) return d;
    }
}
void factorRho(ull n, map<ull, int>& f) {
    if (n == 1) return;
    if (isPrime(n)) { f[n]++; return; }
    ull d = pollard(n); factorRho(d, f); factorRho(n / d, f);
}
```

7.2.3 Divisors

- Enumerate all divisors from the factorisation (recursive product over prime powers). Count = Π(e_i + 1), sum σ = Π $(p^{e+1} - 1)/(p - 1)$.
- Max divisor count: ≤ 10^6 → 240; ≤ 10^9 → 1344; ≤ 10^{12} → 6720; ≤ 10^{16} → 41472; ≤ 10^{18} → 103680. So "iterate over all divisors" or even "all pairs of divisors" of a single number is often fine.
- **Harmonic enumeration:** for every d ≤ N, iterate multiples of d: Σ N/d = O(N log N). Use for "count pairs with gcd = d" (cnt[d] = #multiples choose 2, then subtract cnt of multiples of d from larger to smaller).
- Sum of proper divisors (aliquot, ACPC 2023 *Outlaw Numbers*) = σ(n) − n.
- Counting divisors of all numbers ≤ N: `for d: for m = d; m <= N; m += d: cnt[m]++`.

7.2.4 Euler's totient and Möbius

```
ll phi(ll n) {                    // O(sqrt n)
    ll r = n;
    for (ll p = 2; p * p <= n; ++p) if (n % p == 0) { while (n % p == 0) n /= p; r -= r / p; }
    if (n > 1) r -= r / n;
    return r;
}
// sieve: phi[i] = i; for each prime p: for multiples m: phi[m] -= phi[m] / p
```

- φ(n) = #{1 ≤ k ≤ n : gcd(k, n) = 1}; Σ_{d|n} φ(d) = n; a^φ(m) ≡ 1 (mod m) when gcd(a, m) = 1.
- **Möbius** μ(n) = 0 if a square divides n, else $(-1)^{\#\text{primes}}$. Σ_{d|n} μ(d) = [n = 1].
- **Möbius inversion:** if g(n) = Σ_{d|n} f(d) then f(n) = Σ_{d|n} μ(d) g(n/d). Same for "multiples": if G(d) = Σ_{d|m} F(m) then F(d) = Σ_{d|m} μ(m/d) G(m).

- **Count coprime pairs** ($i \le A$, $j \le B$): $\Sigma_d \mu(d) \cdot \lfloor A/d \rfloor \cdot \lfloor B/d \rfloor$ — with the $\sqrt{}$ block trick $O(\sqrt{n})$ per query.
- $\Sigma_{i=1}^{n} \gcd(i, n) = \Sigma_{d \mid n} d \cdot \varphi(n/d)$.

7.2.5 Other facts

- **Legendre:** exponent of prime p in $n! = \Sigma \lfloor n/p^k \rfloor$. Trailing zeros of $n!$ = exponent of 5.
- **Kummer:** exponent of p in $C(n, k)$ = number of carries when adding k and $n - k$ in base p. $C(n, k)$ odd $\Leftrightarrow$ $(k \,\&\, n) == k$.
- **Wilson:** $(p - 1)! \equiv -1 \pmod p$. **Fermat:** $a^{p-1} \equiv 1$.
- Every integer > 1 is a product of primes; there are infinitely many primes in every coprime residue class; prime gaps below 2^{64} are ≤ 1550 (so "next prime after n" by testing successive numbers is fast).
- **Goldbach** (verified far beyond contest ranges): even $n > 2$ is a sum of two primes; odd $n > 5$ is a sum of three primes.
- **Quadratic residues:** for odd prime p, exactly $(p-1)/2$ non-zero squares; Euler's criterion $a^{(p-1)/2} \equiv \pm 1$.
- Integer `a^b` exact comparison without overflow: compare `b·log(a)` with a margin, then verify exactly with `__int128`/capped multiplication (ACPC 2021 *Associative Exponents*).

7.3 Counting

7.3.1 The standard toolbox

objects	count
sequences of length k from n symbols	n^k
arrangements of k distinct out of n	$n!/(n-k)!$
subsets of size k	$C(n, k)$
multisets of size k from n types (stars & bars)	$C(n + k - 1, k)$
solutions of $x_1 + \dots + x_n = k$, $x_i \ge 0$	$C(n + k - 1, n - 1)$
same with $x_i \ge 1$	$C(k - 1, n - 1)$
same with $x_i \le c$	inclusion–exclusion: $\Sigma_j (-1)^j C(n, j) C(k - j(c+1) + n - 1, n - 1)$
permutations of a multiset (counts c_i)	$n! / \Pi c_i!$
lattice paths $(0,0) \to (a,b)$ with steps R/U	$C(a + b, a)$
paths that never go below the diagonal	Catalan-type (reflection, below)
surjections $n \to k$	$\Sigma_j (-1)^j C(k, j)(k - j)^n = k! \cdot S(n, k)$
derangements D_n	$(n - 1)(D_{n-1} + D_{n-2})$, $D_n \approx n!/e$
labelled trees on n vertices	n^{n-2} (Cayley); forests with k given roots: $k \cdot n^{n-k-1}$
binary strings of length n with no two adjacent 1s	Fibonacci F_{n+2}

Catalan $C_n = C(2n, n)/(n+1) = C(2n, n) - C(2n, n+1)$: 1, 1, 2, 5, 14, 42, 132, 429, 1430, 4862, 16796. Counts balanced bracket sequences, full binary trees with $n+1$ leaves, triangulations of an $(n+2)$-gon, monotone lattice paths not crossing the diagonal, non-crossing matchings of $2n$ points, stack-sortable permutations. Recurrence $C_{n+1} = \Sigma\, C_i\, C_{n-i}$.

Reflection principle: a path that touches the forbidden line can be reflected (from its first touch) to a path starting at the mirrored start point; so #bad = #paths from the mirrored start. **Ballot:** sequences of n R-steps and m U-steps ($m \le n$) in which every prefix has #U ≤ #R: $C(n + m, m) - C(n + m, m - 1)$ ($n = m$ gives Catalan). Strictly more R than U in every non-empty prefix: $(n - m)/(n + m) \cdot C(n + m, m)$.

Stirling numbers of the 2nd kind $S(n, k)$ (partitions of n labelled items into k non-empty unlabelled groups): $S(n, k) = k \cdot S(n-1, k) + S(n-1, k-1)$. **Bell numbers** $B_n = \Sigma_k S(n, k)$: 1, 1, 2, 5, 15, 52, 203, 877, 4140. **Stirling 1st kind** (permutations with k cycles): $c(n, k) = (n-1)c(n-1, k) + c(n-1, k-1)$.

Integer partitions $p(n)$ (unordered sums): $p(n) = 1, 1, 2, 3, 5, 7, 11, 15, 22, 30, 42 \dots$, $p(100) \approx 1.9 \cdot 10^8$. DP: `for part in 1..n: for s = part..n: p[s] += p[s - part]` (coin-change counting).

7.3.2 Identities

- Pascal $C(n, k) = C(n-1, k-1) + C(n-1, k)$; symmetry $C(n, k) = C(n, n-k)$; $\Sigma_k C(n, k) = 2^n$; $\Sigma_k (-1)^k C(n, k) = 0$.
- **Hockey stick:** $\Sigma_{i=k}^{n} C(i, k) = C(n+1, k+1)$. **Vandermonde:** $\Sigma_j C(a, j)\, C(b, k-j) = C(a + b, k)$.
- $k \cdot C(n, k) = n \cdot C(n-1, k-1)$. $\Sigma_k\, k \cdot C(n, k) = n \cdot 2^{n-1}$.
- $C(n, k)$ is maximal at $k = n/2$; $C(60, 30) \approx 1.2 \cdot 10^{17}$ (fits in `ll`), $C(67, 33)$ overflows `ll`.

- Number of monotone paths in a grid avoiding forbidden cells: sort cells, dp[i] = paths to cell i avoiding earlier forbidden ones = C(to i) − Σ_{j before i} dp[j]·C(j → i).

7.3.3 Inclusion–exclusion

|A_1 ∪ … ∪ A_n| = Σ_{nonempty S} (−1)^{|S|+1} |∩{i∈S} A_i|. *Typical shape: count objects violating **none** of n conditions = Σ (#objects violating at least all in S). With symmetric conditions this collapses to* $\Sigma_j (-1)^j C(n, j)\cdot f(j)$.} (−1)^{|S|}

- Count integers ≤ N divisible by none of primes p_1..p_k: Σ over subsets (−1)^{|S|} ⌊N / Π p⌋ ($k \le 15 \to 2^k$ terms; prune when the product exceeds N).
- Count arrangements where at least one of p properties holds (ACPC 2024 *A Little to the Right*): either inclusion–exclusion over subsets of properties, or observe structure (a property with all-distinct values defines exactly one order; answer = number of distinct such orders).
- Inclusion–exclusion over subsets of n ≤ 20 with SOS (§4.2) when the "intersection" depends on a mask.

7.3.4 Burnside / Pólya (counting up to symmetry)

7.3 distinct objects = (1/|G|) Σ_{g ∈ G} fix(g). For colourings, fix(g) = k^{c(g)} with c(g) = number of cycles of g on positions.

- **Necklaces** (rotations only), n beads, k colours: (1/n) Σ_{d|n} φ(d) k^{n/d}.
- **Bracelets** (rotations + reflections): necklaces/2 + (n odd: k^{(n+1)/2}/2; n even: (k^{n/2} + k^{n/2+1})/4).
- Cube faces with k colours (rotations): $(k^6 + 3k^4 + 12k^3 + 8k^2)/24$.
- Modulo p: divide by |G| using an inverse (p must not divide |G|).

7.3.5 Probability-flavoured counting

- Expected number of distinct values when drawing k times uniformly from n: $n(1 - (1 - 1/n)^k)$.
- Random permutation: expected cycles H_n; probability that 1 and 2 are in the same cycle 1/2.
- Birthday: collision among k random values in [0, N) likely once $k \approx \sqrt{N}$ (so 64-bit hashes of 10^6 strings are safe, 32-bit are not).

7.4 Matrices, Linear Recurrences, Gaussian Elimination, XOR Basis

7.4.1 Matrix exponentiation

```
// [stress-tested]
using Mat = vector<vector<ll>>;
Mat mul(const Mat& A, const Mat& B, ll mod) {
    int n = sz(A), m = sz(B[0]), k = sz(B);
    Mat C(n, vector<ll>(m, 0));
    rep(i, 0, n) rep(t, 0, k) if (A[i][t]) rep(j, 0, m) C[i][j] = (C[i][j] + A[i][t] * B[t][j]) % mod;
    return C;
}
Mat matpow(Mat A, ll e, ll mod) {
    int n = sz(A); Mat R(n, vector<ll>(n, 0)); rep(i, 0, n) R[i][i] = 1 % mod;
    for (; e > 0; e >>= 1, A = mul(A, A, mod)) if (e & 1) R = mul(R, A, mod);
    return R;
}
```

- **Linear recurrence** f(n) = c_1 f(n−1) + … + c_k f(n−k): companion matrix (first row = c, sub-diagonal ones), $O(k^3 \log n)$. Add a constant/polynomial term by adding extra state rows (e.g. a row that stays 1).
- **Number of walks of length L** between all pairs = A^L (adjacency matrix). Shortest walk with exactly L edges: (min, +) matrix "multiplication" with the same `matpow` skeleton.
- **DP with a fixed linear transition repeated 10^{18} times** (e.g. tiling a 4 × n strip, states = profiles) → matrix power over the state vector.
- Cost is $k^3 \cdot \log n$: k = 100 → $10^6 \cdot 60$ = fine; k = 200 is the practical limit.
- Fibonacci: $[[1,1],[1,0]]^n = [[F_{n+1}, F_n],[F_n, F_{n-1}]]$. Identities: $F_{2n} = F_n(2F_{n+1} - F_n)$, $F_{2n+1} = F_n^2 + F_{n+1}^2$.

7.4.2 Berlekamp–Massey: guess the recurrence from terms

Compute the first ~2k terms of any sequence that satisfies a linear recurrence of order ≤ k (by brute force / DP), feed them to BM, get the recurrence; then jump to the n-th term in O(k² log n). Great for "count tilings/paths of size n ≤ 10^{18}" when the DP is linear but you don't want to derive the matrix.

```
// [stress-tested] mod must be prime. Returns tr with s[i] = sum_j tr[j] * s[i-1-j].
vector<ll> berlekampMassey(const vector<ll>& s, ll mod) {
    auto pw = [&](ll x, ll e) { ll r = 1; for (x %= mod; e; e >>= 1, x = x * x % mod) if (e & 1) r = r * x % mod; ret
urn r; };
    int n = sz(s), L = 0, m = 0; vector<ll> C(n, 0), B(n, 0), T; C[0] = B[0] = 1; ll b = 1;
    rep(i, 0, n) {
        ++m; ll d = s[i] % mod;
        rep(j, 1, L + 1) d = (d + C[j] * s[i - j]) % mod;
        if (!d) continue;
        T = C; ll coef = d * pw(b, mod - 2) % mod;
        rep(j, m, n) C[j] = ((C[j] - coef * B[j - m]) % mod + mod) % mod;
        if (2 * L > i) continue;
        L = i + 1 - L; B = T; b = d; m = 0;
    }
    C.resize(L + 1); C.erase(C.begin());
    for (ll& x : C) x = (mod - x) % mod;
    return C;
}
// k-th term (0-indexed) given first terms S (size >= sz(tr)) and recurrence tr
ll linearRec(const vector<ll>& S, const vector<ll>& tr, ll k, ll mod) {
    int n = sz(tr);
    if (n == 0) return 0;
    auto combine = [&](const vector<ll>& a, const vector<ll>& b) {
        vector<ll> res(2 * n + 1, 0);
        rep(i, 0, n + 1) rep(j, 0, n + 1) res[i + j] = (res[i + j] + a[i] * b[j]) % mod;
        for (int i = 2 * n; i > n; --i) rep(j, 0, n) res[i - 1 - j] = (res[i - 1 - j] + res[i] * tr[j]) % mod;
        res.resize(n + 1); return res;
    };
    vector<ll> pol(n + 1, 0), e(n + 1, 0); pol[0] = e[1] = 1;
    for (++k; k; k /= 2) { if (k % 2) pol = combine(pol, e); e = combine(e, e); }
    ll res = 0;
    rep(i, 0, n) res = (res + pol[i + 1] * S[i]) % mod;
    return res;
}
```

Give BM at least 2·(order) terms, preferably more (it returns the shortest recurrence consistent with what it saw).

7.4.3 Gaussian elimination (reals)

```
// [stress-tested] a = n rows of (m coefficients + rhs). Returns 0: no solution, 1: unique, 2: infinitely many.
// ans = one solution (free variables = 0).
int gauss(vector<vector<double>> a, vector<double>& ans) {
    const double EPS = 1e-9;
    int n = sz(a), m = sz(a[0]) - 1; vector<int> where(m, -1);
    for (int col = 0, row = 0; col < m && row < n; ++col) {
        int sel = row;
        rep(i, row, n) if (fabs(a[i][col]) > fabs(a[sel][col])) sel = i;
        if (fabs(a[sel][col]) < EPS) continue;
        swap(a[sel], a[row]); where[col] = row;
        rep(i, 0, n) if (i != row) {
            double c = a[i][col] / a[row][col];
            rep(j, col, m + 1) a[i][j] -= a[row][j] * c;
        }
        ++row;
    }
    ans.assign(m, 0);
    rep(i, 0, m) if (where[i] != -1) ans[i] = a[where[i]][m] / a[where[i]][i];
    rep(i, 0, n) {
        double s = 0; rep(j, 0, m) s += ans[j] * a[i][j];
        if (fabs(s - a[i][m]) > EPS) return 0;
    }
    rep(i, 0, m) if (where[i] == -1) return 2;
    return 1;
}
```

- **Expected values with cycles** (random walk on a graph, Markov chains): one equation per state `E_v - Σ p_vu E_u = c_v`, absorbing states `E = 0`. n ≤ 500.
- Mod p version: same code with modular inverse instead of division (exact, no EPS). **Determinant** = product of pivots × $(-1)^{\#\text{swaps}}$; **rank** = number of pivots.

- **Kirchhoff (matrix-tree):** #spanning trees = any cofactor of the Laplacian L = D − A (delete one row and column, take determinant; mod p or doubles for small n).

7.4.4 XOR basis (Gaussian elimination over GF(2))

```
// [stress-tested]
struct XorBasis {
    ll b[62] = {}; int cnt = 0;
    bool insert(ll x) {                              // returns false if x already representable
        for (int i = 61; i >= 0; --i) if (x >> i & 1) {
            if (!b[i]) { b[i] = x; ++cnt; return true; }
            x ^= b[i];
        }
        return false;
    }
    bool canMake(ll x) const { for (int i = 61; i >= 0; --i) if (x >> i & 1) x ^= b[i]; return x == 0; }
    ll maxXor(ll x = 0) const { for (int i = 61; i >= 0; --i) x = max(x, x ^ b[i]); return x; }
};
```

- Number of distinct subset-xors = 2^{cnt}; each achievable value is hit by $2^{n - \text{cnt}}$ subsets.
- **Max xor path** in an undirected graph between u and v: xor of any path + any subset of cycle xors → DFS xor-to-root `d`, insert `d[a] ^ d[b] ^ w` for every non-tree edge, answer `maxXor(d[u] ^ d[v])`.
- Systems of equations mod 2 with up to ~2000 variables: use `bitset<N>` rows, elimination O(n³/64).
- Lights-out style puzzles (press toggles neighbours): linear system over GF(2).

7.5 FFT / NTT (Convolution)

Convolution c_k = Σ_{i+j=k} a_i b_j in O(n log n). Recognise it: "**for every shift/offset/sum k, aggregate over pairs (i, j) with i + j = k (or i − j = k)**". Reverse one array to turn differences into sums.

7.5.1 FFT with doubles (exact for integer results up to ~10^{14})

```
// [stress-tested]
using cd = complex<double>;
void fft(vector<cd>& a, bool invert) {
    int n = sz(a);
    for (int i = 1, j = 0; i < n; ++i) {                // bit reversal
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    vector<cd> rt(max(1, n / 2));                        // accurate roots, computed in long double
    rep(k, 0, n / 2) { long double ang = 2 * acosl(-1) * k / n; rt[k] = cd(cosl(ang), sinl(ang)); }
    for (int len = 2; len <= n; len <<= 1) {
        int step = n / len;
        for (int i = 0; i < n; i += len) rep(j, 0, len / 2) {
            cd w = rt[j * step]; if (invert) w = conj(w);
            cd u = a[i + j], v = a[i + j + len / 2] * w;
            a[i + j] = u + v; a[i + j + len / 2] = u - v;
        }
    }
    if (invert) for (cd& x : a) x /= n;
}
vector<ll> multiply(const vector<ll>& a, const vector<ll>& b) {
    if (a.empty() || b.empty()) return {};
    int n = 1; while (n < sz(a) + sz(b)) n <<= 1;
    vector<cd> fa(all(a)), fb(all(b)); fa.resize(n); fb.resize(n);
    fft(fa, false); fft(fb, false);
    rep(i, 0, n) fa[i] *= fb[i];
    fft(fa, true);
    vector<ll> res(sz(a) + sz(b) - 1);
    rep(i, 0, sz(res)) res[i] = llround(fa[i].real());
    return res;
}
```

7.5.2 NTT modulo 998244353 (exact)

```
// [stress-tested] mod = 998244353 = 119 * 2^23 + 1, primitive root 3. Max length 2^23.
const ll NMOD = 998244353;
ll npow(ll b, ll e) { ll r = 1; b %= NMOD; for (; e; e >>= 1, b = b * b % NMOD) if (e & 1) r = r * b % NMOD; return
r; }
void ntt(vector<ll>& a, bool invert) {
```

```
    int n = sz(a);
    for (int i = 1, j = 0; i < n; ++i) {
        int bit = n >> 1;
        for (; j & bit; bit >>= 1) j ^= bit;
        j ^= bit;
        if (i < j) swap(a[i], a[j]);
    }
    for (int len = 2; len <= n; len <<= 1) {
        ll w = npow(3, (NMOD - 1) / len); if (invert) w = npow(w, NMOD - 2);
        for (int i = 0; i < n; i += len) {
            ll wn = 1;
            rep(j, 0, len / 2) {
                ll u = a[i + j], v = a[i + j + len / 2] * wn % NMOD;
                a[i + j] = u + v < NMOD ? u + v : u + v - NMOD;
                a[i + j + len / 2] = u - v >= 0 ? u - v : u - v + NMOD;
                wn = wn * w % NMOD;
            }
        }
    }
    if (invert) { ll ni = npow(n, NMOD - 2); for (ll& x : a) x = x * ni % NMOD; }
}
vector<ll> multiplyMod(vector<ll> a, vector<ll> b) {
    if (a.empty() || b.empty()) return {};
    int need = sz(a) + sz(b) - 1, n = 1; while (n < need) n <<= 1;
    a.resize(n); b.resize(n); ntt(a, false); ntt(b, false);
    rep(i, 0, n) a[i] = a[i] * b[i] % NMOD;
    ntt(a, true); a.resize(need); return a;
}
```

For mod 10^9+7 (not NTT-friendly): split each coefficient as `x = hi·2^15 + lo` and do 3–4 double FFTs, or three NTT primes + CRT. Usually the problem uses 998244353 if it intends NTT.

7.5.3 Applications

- **Count pairs by sum:** `cnt = multiply(freq, freq)`; pairs with i < j: subtract the diagonal (`freq[x]` at index 2x) and halve.
- **Pattern matching with mismatches / count matching positions** (for every alignment i, how many j have text[i+j] == pat[j]): for each letter c, `multiply(isC(text), reverse(isC(pat)))`; entry at index `i + m - 1` counts matches for alignment i. Sum over letters (or over the 2 colours of a binary image). O(σ·n log n).
- **Wildcards** (`?` matches anything): with p_j = 0 for `?`, alignment i matches iff Σ_j p_j t_{i+j}(p_j − t_{i+j})² = 0 → expands to 3 convolutions. Use random values for letters to avoid accidental zeros when working mod p.
- **2-D matching** (ACPC 2017 *Matchings*: count equal pixels for every placement of a small image inside a big one): flatten both images with the **big image's width W**: index r·W + c. Place the small image's pixel (r, c) at r·W + c too. Then one 1-D correlation gives all placements; only use top-left positions with c + w ≤ W and r + h ≤ H.
- **Big integer multiplication:** digits as coefficients, then carry.
- **Subset sums with multiplicities / knapsack counting** with many items of small weight: multiply generating polynomials (divide & conquer on the product to keep sizes balanced).
- **Polynomial with sums over all pairs of differences** |a_i − a_j| = d for every d: multiply freq by reversed freq.
- Sizes: n up to ~10^6 per multiplication is fine (a few hundred ms).

8. Strings

8.1 Hashing, Prefix Function (KMP), Z-Function

8.1.1 Polynomial hashing (mod $2^{61} - 1$, random base)

```
// [stress-tested] O(1) substring hash after O(n) preprocessing. Collision prob ~ n^2 / 2^61.
struct Hasher {
    static const ull M = (1ULL << 61) - 1;
    static ull mul(ull a, ull b) {
        __uint128_t c = (__uint128_t)a * b;
        ull r = (ull)(c & M) + (ull)(c >> 61);
        return r >= M ? r - M : r;
    }
    static ull base() { static ull B = mt19937_64(chrono::steady_clock::now().time_since_epoch().count())() % (M - 10
00) + 500; return B; }
    vector<ull> h, p;
    Hasher(const string& s) : h(sz(s) + 1, 0), p(sz(s) + 1, 1) {
        ull B = base();
        rep(i, 0, sz(s)) { h[i + 1] = (mul(h[i], B) + (unsigned char)s[i]) % M; p[i + 1] = mul(p[i], B); }
    }
    ull get(int l, int r) const {                   // hash of s[l, r)
        ull v = h[r] + M - mul(h[l], p[r - l]);
        return v >= M ? v - M : v;
    }
    // hash of concatenation A + B given hash(A), hash(B), len(B): mul(hA, p^lenB) + hB  (mod M)
};
```

Uses: - **Compare substrings** in O(1); **LCP of two suffixes** in O(log n) by binary search on length → suffix comparison → sort suffixes/rotations with a custom comparator in O(n $\log^2$ n). - **Longest repeated substring / longest common substring of k strings**: binary search the length, put all hashes of that length in a hash set. - **Palindrome check** of s[l, r): hash of s vs hash of reverse(s) on the mirrored range. - **Canonical forms** for sets of strings: hash of the minimal rotation (§8.2) for "equal up to rotation" (ACPC 2020 *Password Rotation*: insert min-rotation of s and of reverse(s)). - Hashing sequences of ints: use values + 1 (avoid 0) and the same formula; for multisets use sum/xor of random 64-bit values per element (§2.5). - Anti-hack is irrelevant on Kattis, but **never use base 31/mod 2^{64} with natural overflow** on adversarial tests (Thue–Morse strings break it).

8.1.2 Prefix function and KMP

`pi[i]` = length of the longest proper border (prefix = suffix) of s[0..i].

```
// [stress-tested]
vector<int> prefixFunction(const string& s) {
    int n = sz(s); vector<int> pi(n, 0);
    rep(i, 1, n) {
        int j = pi[i - 1];
        while (j > 0 && s[i] != s[j]) j = pi[j - 1];
        if (s[i] == s[j]) ++j;
        pi[i] = j;
    }
    return pi;
}
vector<int> kmpFind(const string& text, const string& pat) {    // start indices of occurrences
    vector<int> pi = prefixFunction(pat), res; int j = 0, m = sz(pat);
    if (m == 0) return res;
    rep(i, 0, sz(text)) {
        while (j > 0 && text[i] != pat[j]) j = pi[j - 1];
        if (text[i] == pat[j]) ++j;
        if (j == m) { res.push_back(i - m + 1); j = pi[j - 1]; }
    }
    return res;
}
```

- **Smallest period** of s: p = n − pi[n−1]; s is a repetition of its prefix of length p iff n % p == 0.
- All borders: n, pi[n−1], pi[pi[n−1]−1], … (chain).
- **Count occurrences of each prefix** in s: `cnt[pi[i]]++`, then propagate `cnt[pi[i-1]] += cnt[i]` from long to short, plus 1 each.
- **KMP automaton**: `go[state][c]` for building DP over strings that must avoid/contain a pattern ("count strings of length n not containing P": dp over (length, automaton state) — matrix power for huge n).
- Rotation check: t is a rotation of s ⇔ |s| = |t| and t occurs in s + s.

8.1.3 Z-function

`z[i]` = length of the longest common prefix of s and s[i..].

```
// [stress-tested]
vector<int> zFunction(const string& s) {
    int n = sz(s); vector<int> z(n, 0);
    if (n) z[0] = n;
    for (int i = 1, l = 0, r = 0; i < n; ++i) {
        if (i < r) z[i] = min(r - i, z[i - l]);
        while (i + z[i] < n && s[z[i]] == s[i + z[i]]) ++z[i];
        if (i + z[i] > r) { l = i; r = i + z[i]; }
    }
    return z;
}
```

- Occurrences of P in T: z-function of `P + '#' + T`, positions with $z \geq |P|$.
- For each suffix of T, the longest prefix of P that starts there: same concatenation trick.
- Number of distinct substrings in $O(n^2)$ by appending characters and using z on the reversed prefix.
- z and pi are interconvertible; use whichever is easier to reason about.

8.2 Palindromes, Rotations, Tries, Aho–Corasick, Suffix Structures

8.2.1 Manacher (all maximal palindromes in O(n))

```
// [stress-tested] t = "#s0#s1#...#"; p[i] = radius in t (including centre).
// The palindrome centred at t[i] has length p[i] - 1 in s and starts at s index (i - p[i] + 1) / 2.
vector<int> manacher(const string& s) {
    string t = "#"; for (char c : s) { t += c; t += '#'; }
    int n = sz(t); vector<int> p(n, 0);
    for (int i = 0, l = 0, r = -1; i < n; ++i) {
        int k = i > r ? 1 : min(p[l + r - i], r - i + 1);
        while (i - k >= 0 && i + k < n && t[i - k] == t[i + k]) ++k;
        p[i] = k--;
        if (i + k > r) { l = i - k; r = i + k; }
    }
    return p;
}
```

- Longest palindromic substring: max over i of `p[i] - 1`.
- Number of palindromic substrings: $\Sigma_i \lfloor p[i] / 2 \rfloor$.
- Is s[l..r] a palindrome: centre $i = l + r + 1$ in t; check `p[i] - 1 ≥ r - l + 1`.
- Palindromic **subsequences** are a different problem: interval DP (§4.1).
- Minimum insertions to make a palindrome = n − LPS (longest palindromic subsequence = LCS(s, reverse s)).

8.2.2 Minimal rotation / canonical cyclic form

```
// [stress-tested] index of the lexicographically smallest rotation, O(n)
int minRotation(const string& s) {
    int n = sz(s), i = 0, j = 1, k = 0;
    while (i < n && j < n && k < n) {
        char a = s[(i + k) % n], b = s[(j + k) % n];
        if (a == b) { ++k; continue; }
        if (a > b) i += k + 1; else j += k + 1;
        if (i == j) ++j;
        k = 0;
    }
    return min(i, j);
}
// canonical form up to rotation:      rotate(s, minRotation(s))
// up to rotation AND reflection:       min(canon(s), canon(reverse(s)))
```

Works for any sequence (vector of ints) — useful for necklaces, cyclic polygons (canonical edge/angle sequence), cycle comparison in graphs.

8.2.3 Trie

```
struct Trie {
    vector<array<int, 26>> nxt; vector<int> endCnt, passCnt;
    Trie() { newNode(); }
    int newNode() { nxt.push_back({}); nxt.back().fill(-1); endCnt.push_back(0); passCnt.push_back(0); return sz(nxt)
- 1; }
    void insert(const string& s) {
```

```
        int v = 0; passCnt[0]++;
        for (char ch : s) { int c = ch - 'a'; if (nxt[v][c] < 0) { int u = newNode(); nxt[v][c] = u; } v = nxt[v][c];
passCnt[v]++; }
        endCnt[v]++;
    }
    int countPrefix(const string& s) {              // # inserted strings with prefix s
        int v = 0; for (char ch : s) { v = nxt[v][ch - 'a']; if (v < 0) return 0; } return passCnt[v];
    }
};
```

Binary trie over bits (high to low) for max xor with a set: at each level go to the child with the opposite bit if it exists. Store counts to support deletion and "count values with x ^ v < k".

8.2.4 Aho–Corasick (many patterns in one text)

```
// [stress-tested] lowercase. After build(): cnt[state] = #patterns that end at state (incl. via links).
struct AhoCorasick {
    vector<array<int, 26>> go; vector<int> link, cnt, order;
    AhoCorasick() { add(); }
    int add() { go.push_back({}); go.back().fill(-1); link.push_back(0); cnt.push_back(0); return sz(go) - 1; }
    int insert(const string& s) {                   // returns terminal state of s
        int v = 0;
        for (char ch : s) { int c = ch - 'a'; if (go[v][c] < 0) { int u = add(); go[v][c] = u; } v = go[v][c]; }
        cnt[v]++; return v;
    }
    void build() {
        queue<int> q; order.clear();
        rep(c, 0, 26) { if (go[0][c] < 0) go[0][c] = 0; else { link[go[0][c]] = 0; q.push(go[0][c]); } }
        while (!q.empty()) {
            int v = q.front(); q.pop(); order.push_back(v);
            cnt[v] += cnt[link[v]];
            rep(c, 0, 26) {
                int u = go[v][c];
                if (u < 0) go[v][c] = go[link[v]][c];
                else { link[u] = go[link[v]][c]; q.push(u); }
            }
        }
    }
    ll countMatches(const string& text) {           // total pattern occurrences in text
        ll res = 0; int v = 0;
        for (char ch : text) { v = go[v][ch - 'a']; res += cnt[v]; }
        return res;
    }
};
```

- Occurrences **per pattern**: count visits per state while scanning, then push counts up the suffix-link tree in reverse BFS `order` (`visits[link[v]] += visits[v]`), read at each pattern's terminal state.
- DP over the automaton: "count strings of length n containing none of the patterns" = dp over (length, state) avoiding states with cnt > 0.

8.2.5 Suffix array + LCP

```
// [stress-tested] sa[0] = n (empty suffix), sa[1..n] = suffixes in sorted order.
// lcp[i] = LCP(suffix sa[i-1], suffix sa[i]) for i >= 1. Characters must be > 0.
struct SuffixArray {
    vector<int> sa, lcp;
    SuffixArray(string s, int lim = 256) {
        s.push_back(0); int n = sz(s), k = 0, a, b;
        vector<int> x(all(s)), y(n), ws(max(n, lim));
        sa = lcp = y; iota(all(sa), 0);
        for (int j = 0, p = 0; p < n; j = max(1, j * 2), lim = p) {
            p = j; iota(all(y), n - j);
            rep(i, 0, n) if (sa[i] >= j) y[p++] = sa[i] - j;
            fill(all(ws), 0);
            rep(i, 0, n) ws[x[i]]++;
            rep(i, 1, lim) ws[i] += ws[i - 1];
            for (int i = n; i--;) sa[--ws[x[y[i]]]] = y[i];
            swap(x, y); p = 1; x[sa[0]] = 0;
            rep(i, 1, n) { a = sa[i - 1]; b = sa[i]; x[b] = (y[a] == y[b] && y[a + j] == y[b + j]) ? p - 1 : p++; }
        }
        for (int i = 0, j; i < n - 1; lcp[x[i++]] = k)
            for (k && k--, j = sa[x[i] - 1]; s[i + k] == s[j + k]; k++);
    }
};
```

- **Distinct substrings** = $n(n+1)/2 - \Sigma$ lcp.

- **LCP of any two suffixes** = range-min of lcp between their ranks (sparse table, §3.4).
- **Longest repeated substring** = max lcp. **Longest common substring of A and B**: SA of `A + '#' + B` (separator smaller than letters but > 0, e.g. '#'), max lcp over adjacent pairs from different sides.
- **k-th distinct substring**: walk sa in order, each suffix contributes (n − sa[i] − lcp[i]) new substrings.
- Pattern search: binary search over sa comparing with the pattern, O(m log n).

8.2.6 Suffix automaton (when you need counts of occurrences of every substring)

```
// [stress-tested] states <= 2n. Distinct substrings = sum over v != 0 of len[v] - len[link[v]].
struct SAM {
    struct St { int len, link; map<char, int> next; };
    vector<St> st; int last = 0;
    SAM() { st.push_back({0, -1, {}}); }
    void extend(char c) {
        int cur = sz(st); st.push_back({st[last].len + 1, 0, {}});
        int p = last;
        while (p != -1 && !st[p].next.count(c)) { st[p].next[c] = cur; p = st[p].link; }
        if (p != -1) {
            int q = st[p].next[c];
            if (st[p].len + 1 == st[q].len) st[cur].link = q;
            else {
                int clone = sz(st); St cl = st[q]; cl.len = st[p].len + 1; st.push_back(cl);
                while (p != -1 && st[p].next.count(c) && st[p].next[c] == q) { st[p].next[c] = clone; p = st[p].link;
}
                st[q].link = st[cur].link = clone;
            }
        }
        last = cur;
    }
};
```

Occurrences of each state's substrings: `occ[cur] = 1` for non-clone states, then add occ[v] to occ[link[v]] in decreasing order of len. A string P occurs in s iff walking `next` from state 0 along P never fails.

9. Geometry

9.1 Geometry Primitives

Rule 1: stay in integers as long as possible. Inputs are usually integers; cross/dot products of integer points are exact in `ll` ($|\text{coords}| \le 10^9 \Rightarrow$ products $\le 4 \cdot 10^{18}$: borderline, use `__int128` for cross products of differences if coordinates reach 10^9). Only divide/sqrt at the very end. **Rule 2:** with doubles, compare with `EPS` (1e-9 relative to the scale of the input).

9.1.1 Point / vector

```
// [stress-tested]
template <class T> struct P {
    T x, y;
    P(T x = 0, T y = 0) : x(x), y(y) {}
    P operator+(P o) const { return {x + o.x, y + o.y}; }
    P operator-(P o) const { return {x - o.x, y - o.y}; }
    P operator*(T k) const { return {x * k, y * k}; }
    P operator/(T k) const { return {x / k, y / k}; }
    bool operator<(P o) const { return tie(x, y) < tie(o.x, o.y); }
    bool operator==(P o) const { return x == o.x && y == o.y; }
    T dot(P o) const { return x * o.x + y * o.y; }
    T cross(P o) const { return x * o.y - y * o.x; }
    T cross(P a, P b) const { return (a - *this).cross(b - *this); }   // >0: a->b turns left around *this
    T norm2() const { return x * x + y * y; }
    double len() const { return sqrt((double)norm2()); }
    double angle() const { return atan2((double)y, (double)x); }        // (-pi, pi]
    P perp() const { return {-y, x}; }                                  // rotate +90°
    P<double> rotate(double a) const { return {x * cos(a) - y * sin(a), x * sin(a) + y * cos(a)}; }
};
using Pt = P<ll>; using Pd = P<double>;
template <class T> int sgn(T v) { return (v > 0) - (v < 0); }
// orientation of c relative to line a->b: +1 left (CCW), -1 right (CW), 0 collinear
template <class T> int orient(P<T> a, P<T> b, P<T> c) { return sgn(a.cross(b, c)); }
```

9.1.2 Segments and lines

```
// [stress-tested] is p on segment [a, b] (integers, exact)
bool onSegment(Pt a, Pt b, Pt p) {
    return a.cross(b, p) == 0 && min(a.x, b.x) <= p.x && p.x <= max(a.x, b.x)
                              && min(a.y, b.y) <= p.y && p.y <= max(a.y, b.y);
}
// do closed segments [a,b] and [c,d] intersect (touching counts)
bool segIntersect(Pt a, Pt b, Pt c, Pt d) {
    int o1 = orient(a, b, c), o2 = orient(a, b, d), o3 = orient(c, d, a), o4 = orient(c, d, b);
    if (o1 * o2 < 0 && o3 * o4 < 0) return true;                        // proper crossing
    return onSegment(a, b, c) || onSegment(a, b, d) || onSegment(c, d, a) || onSegment(c, d, b);
}
// intersection point of infinite lines (a,b) and (c,d); requires non-parallel
Pd lineIntersection(Pd a, Pd b, Pd c, Pd d) {
    double t = (c - a).cross(d - c) / (b - a).cross(d - c);
    return a + (b - a) * t;
}
// distance from p to segment [a, b]
double distPointSegment(Pd a, Pd b, Pd p) {
    Pd ab = b - a; double L = ab.norm2();
    if (L == 0) return (p - a).len();
    double t = max(0.0, min(1.0, (p - a).dot(ab) / L));
    return (p - (a + ab * t)).len();
}
// projection of p onto the line (a, b); reflection = 2 * proj - p
Pd projectLine(Pd a, Pd b, Pd p) { Pd ab = b - a; return a + ab * ((p - a).dot(ab) / ab.norm2()); }
```

- Distance from p to the infinite line (a, b): `|cross(b - a, p - a)| / |b - a|`.
- Segment–segment distance: 0 if they intersect, else min of the 4 point–segment distances.
- **Lattice points** on segment (a, b) inclusive: `gcd(|dx|, |dy|) + 1`.
- Line through two points as `A x + B y = C`: A = y2 − y1, B = x1 − x2, C = A·x1 + B·y1 (integers).
- Parallel ⇔ cross(b − a, d − c) == 0; perpendicular ⇔ dot == 0.

9.1.3 Angles and sorting by angle

```
// [stress-tested] sort vectors by polar angle in [0, 2pi), exact for integers; ties by distance not handled
bool half(Pt p) { return p.y < 0 || (p.y == 0 && p.x < 0); }
void polarSort(vector<Pt>& v) {
    sort(all(v), [](Pt a, Pt b) {
        if (half(a) != half(b)) return half(a) < half(b);
        return a.cross(b) > 0;
    });
}
```

- Angle between vectors: `atan2(cross, dot)` (signed, robust) rather than `acos(dot/(|a||b|))` (inaccurate near 0, π).
- Rotating sweep around a point (max points visible in an angle window): polar sort, double the array with angles + 2π, two pointers.

9.1.4 Polygons

```
// [stress-tested] twice the signed area (positive if counter-clockwise). Exact in integers.
ll area2(const vector<Pt>& poly) {
    ll a = 0; int n = sz(poly);
    rep(i, 0, n) a += poly[i].cross(poly[(i + 1) % n]);
    return a;
}
// point in simple polygon: 1 inside, 0 outside, -1 on boundary. O(n), exact.
int inPolygon(const vector<Pt>& poly, Pt p) {
    int n = sz(poly); bool in = false;
    rep(i, 0, n) {
        Pt a = poly[i], b = poly[(i + 1) % n];
        if (onSegment(a, b, p)) return -1;
        if ((a.y > p.y) != (b.y > p.y)) {               // edge crosses the horizontal line y = p.y
            __int128 c = (__int128)(b - a).cross(p - a);
            if ((c > 0) == (b.y > a.y)) in = !in;       // p is strictly left of an upward edge, etc.
        }
    }
    return in ? 1 : 0;
}
```

- **Pick's theorem** (lattice polygon): Area = I + B/2 − 1 ⇒ interior points I = (area2 − B + 2) / 2 with B = Σ gcd(|dx|, |dy|) over edges.
- Centroid of polygon: C = Σ (p_i + p_{i+1})·cross(p_i, p_{i+1}) / (3·area2).
- **Point in convex polygon in O(log n):** binary search the fan triangle from vertex 0 (`orient` with poly[0], poly[mid]).
- Polygon is convex ⇔ all consecutive turns have the same sign (ignore zero for collinear).
- Area of a polygon cut by a half-plane: walk edges, keep inside vertices + intersection points (Sutherland–Hodgman), O(n).
- **Polar area** (curve r(θ)): area = ∫ r(θ)²/2 dθ (ACPC 2021 *Wonky Pizza*: r linear in θ → closed-form cubic; then binary search the cut angle).

9.1.5 Precision checklist

- Use `long double` if values ~10^9 and you need 1e-6 after subtractive operations.
- Never test `==` on doubles; `fabs(a - b) < EPS`. Choose EPS by scale: coordinates 10^4 → EPS 1e-9 fine; 10^9 → use relative error.
- `sqrt` of a slightly negative number → NaN: `sqrt(max(0.0, x))`. Same for `acos(min(1.0, max(-1.0, x)))`.
- Print `-0.000` as `0.000`: add `0.0` or check `fabs(x) < 5e-10`.
- Integer geometry: test collinearity / orientation exactly; compare distances by squared lengths.

9.2 Convex Hull, Calipers, Closest Pair, Circles, Integration

Needs `P`, `Pt`, `Pd` from §9.1.

9.2.1 Convex hull (monotone chain)

```
// [stress-tested] counter-clockwise, no collinear points, no duplicate first/last. Handles n <= 2.
vector<Pt> convexHull(vector<Pt> pts) {
    if (sz(pts) <= 1) return pts;
    sort(all(pts));
    vector<Pt> h(sz(pts) + 1);
    int s = 0, t = 0;
    for (int it = 2; it--; s = --t, reverse(all(pts)))
        for (Pt p : pts) {
            while (t >= s + 2 && h[t - 2].cross(h[t - 1], p) <= 0) t--;   // < 0 keeps collinear points
```

```
            h[t++] = p;
        }
    return {h.begin(), h.begin() + t - (t == 2 && h[0] == h[1])};
}
```

- Hull perimeter/area: sum edge lengths / `area2` of the hull.
- Keeping collinear boundary points (`< 0`) breaks when *all* points are collinear (returns duplicates) — handle that case separately.
- **Online upper hull** for CHT-like problems → §4.4. **Dynamic hull** is rarely needed.
- Fractal/self-similar shapes (ACPC 2020 *Wrapping Trees*): the hull of the limit set is determined by extreme points of the generator; reason about which corners survive.

9.2.2 Rotating calipers

```
// [stress-tested] squared diameter (farthest pair) of a convex polygon given CCW (output of convexHull)
ll diameter2(const vector<Pt>& h) {
    int n = sz(h);
    if (n == 1) return 0;
    if (n == 2) return (h[0] - h[1]).norm2();
    ll best = 0;
    for (int i = 0, j = 1; i < n; ++i) {
        while ((h[(i + 1) % n] - h[i]).cross(h[(j + 1) % n] - h[j]) > 0) j = (j + 1) % n;
        best = max({best, (h[i] - h[j]).norm2(), (h[(i + 1) % n] - h[j]).norm2()});
    }
    return best;
}
```

Same two-pointer skeleton gives: minimum width (min over edges of max distance to the antipodal vertex), smallest enclosing rectangle, max distance between two convex polygons.

9.2.3 Closest pair of points

```
// [stress-tested] squared distance of the closest pair, O(n log n)
ll closestPair2(vector<Pt> p) {
    sort(all(p), [](Pt a, Pt b) { return a.x < b.x; });
    set<pair<ll, ll>> S; ll best = LLONG_MAX; int j = 0;
    rep(i, 0, sz(p)) {
        ll d = (ll)ceil(sqrt((long double)best));
        while (p[i].x - p[j].x > d) { S.erase({p[j].y, p[j].x}); ++j; }
        auto lo = S.lower_bound({p[i].y - d, LLONG_MIN}), hi = S.upper_bound({p[i].y + d, LLONG_MAX});
        for (auto it = lo; it != hi; ++it) {
            ll dx = p[i].x - it->second, dy = p[i].y - it->first;
            best = min(best, dx * dx + dy * dy);
        }
        S.insert({p[i].y, p[i].x});
    }
    return best;
}
```

9.2.4 Circles

```
// circumcentre of triangle ABC (non-collinear)
Pd ccCenter(Pd A, Pd B, Pd C) {
    Pd b = C - A, c = B - A;
    return A + (b * c.norm2() - c * b.norm2()).perp() / b.cross(c) / 2;
}
// [stress-tested] minimum enclosing circle, expected O(n) (Welzl, randomised)
pair<Pd, double> minEnclosingCircle(vector<Pd> ps) {
    shuffle(all(ps), mt19937(1));
    Pd o = ps[0]; double r = 0, EPS = 1 + 1e-9;
    rep(i, 0, sz(ps)) if ((o - ps[i]).len() > r * EPS) {
        o = ps[i]; r = 0;
        rep(j, 0, i) if ((o - ps[j]).len() > r * EPS) {
            o = (ps[i] + ps[j]) / 2; r = (o - ps[i]).len();
            rep(k, 0, j) if ((o - ps[k]).len() > r * EPS) { o = ccCenter(ps[i], ps[j], ps[k]); r = (o - ps[i]).len();
}
        }
    }
    return {o, r};
}
// intersections of circle (c, r) with the infinite line (a, b)
vector<Pd> circleLine(Pd c, double r, Pd a, Pd b) {
    Pd ab = b - a, p = a + ab * ((c - a).dot(ab) / ab.norm2());
    double s = a.cross(b, c), h2 = r * r - s * s / ab.norm2();
    if (h2 < 0) return {};
```

```
    if (h2 == 0) return {p};
    Pd h = ab / ab.len() * sqrt(h2);
    return {p - h, p + h};
}
// intersections of two circles (0, 1 or 2 points); identical circles -> {}
vector<Pd> circleCircle(Pd a, double r1, Pd b, double r2) {
    Pd v = b - a; double d2 = v.norm2();
    if (d2 == 0) return {};
    double sum = r1 + r2, dif = r1 - r2;
    if (sum * sum < d2 || dif * dif > d2) return {};
    double p = (d2 + r1 * r1 - r2 * r2) / (2 * d2), h2 = r1 * r1 - p * p * d2;
    Pd mid = a + v * p, per = v.perp() * sqrt(max(0.0, h2) / d2);
    if (h2 <= 0) return {mid};
    return {mid + per, mid - per};
}
```

- Tangent length from external point P: $\sqrt{(|PC|^2 - r^2)}$. Angle subtended: $2\cdot\text{asin}(r / |PC|)$.
- Area of intersection of two circles with distance d: `r1²·acos((d² + r1² − r2²)/(2 d r1)) + r2²·acos((d² + r2² − r1²)/(2 d r2)) − ½√((−d+r1+r2)(d+r1−r2)(d−r1+r2)(d+r1+r2))` (handle containment / disjoint first).
- Circle through 3 points: `ccCenter`. Point inside circle: compare squared distances.
- Moving circle / point vs shrinking zone (ACPC 2018 *PUBNite*): write distance as a function of time, binary/ternary search the time, or solve the quadratic directly.

9.2.5 Sphere / 3-D quick facts

- Great-circle distance between points with unit vectors u, v on a sphere radius R: `R·acos(u·v)` (clamp the dot). Chord length `2R·sin(θ/2)`.
- Lat/long → xyz: $(\cos\varphi\cos\lambda, \cos\varphi\sin\lambda, \sin\varphi)$.
- Moving along a tangent line a distance s from the surface point: distance to centre $\sqrt{(R^2 + s^2)}$ (ACPC 2023 *Flying Straight* compares arc vs tangent).
- 3-D cross product `(a.y b.z − a.z b.y, a.z b.x − a.x b.z, a.x b.y − a.y b.x)`; volume of tetrahedron $= |a\cdot(b\times c)|/6$.

9.2.6 Numerical integration and root finding

```
// composite Simpson on [a, b] with n (even) intervals; error O(h^4)
template <class F> double simpson(F f, double a, double b, int n = 2000) {
    double h = (b - a) / n, s = f(a) + f(b);
    rep(i, 1, n) s += f(a + i * h) * (i & 1 ? 4 : 2);
    return s * h / 3;
}
```

- Area of a union of shapes: integrate the covered length of the vertical line $x = t$ (sweep events at all intersection x-coordinates, Simpson within each slab).
- Roots of a monotone function: binary search (§2.1). Polynomial root isolation: roots of the derivative split the axis into monotone pieces.